CHAPTER 1

Biomolecules : Constituents of Life

The living matter is composed of mainly six elements-*carbon, hydrogen, oxygen, nitrogen, phosphorus and sulfur*. These elements together constitute about 90% of the dry weight of the human body. Several other functionally important elements are also found in the cells. These include Ca, K, Na, Cl, Mg, Fe, Cu, Co, I, Zn, F, Mo and Se.

Complex Biomolecules

The organic compounds such as amino acids, nucleotides and monosaccharides serve as the *monomeric units* or building blocks of complex biomolecules-proteins, nucleic acids (DNA and RNA) and polysaccharides, respectively.

Table. The major complex biomolecules of cells

Biomolecule	Building block (repeating unit)	Major functions
1. Protein	Amino acids	Fundamental basis of structure and function of cell (static and dynamic functions).
2. Deoxyribonucleic acid (DNA)	Deoxyribonucleotides	Repository of hereditary information.
3. Ribonucleic acid (RNA)	Ribonucleotides	Essentially required for protein biosynthesis.
4. Polysaccharide (glycogen)	Monosaccharides (glucose)	Storage form of energy to meet short term demands.
5. Lipids	Fatty acids, glycerol	Storage form of energy to meet long term demands; structural components of membrances.

Carbohydrate

The term '*carbohydrate*' was originally coined for this class of compounds as most of them were '*hydrates of carbon*' or could be represented by the general formula, $C_x(H_2O)_y$.

The carbohydrates are nowadays broadly defined as **polyhydroxy aldehydes** or **ketones** and **their derivatives** or as substances that yield one of these compounds on hydrolysis.

Carbohydrates are usually classified in 3 groups :

A. Monosaccharides or **Monosaccharoses** (*mono*^{G} = one; *sakcharon*^{G} = sugar). The monosaccharides, often called *simple sugars*, are compounds which possess a free aldehyde (—CHO) or ketone (= CO) group and 2 or more hydroxyl (—OH) groups. They are, in fact, the simplest sugars and cannot be hydrolyzed into smaller units. Their **general formula** is $C_n(H_2O)_n$ or $C_nH_{2n}O_n$.

B. Oligosaccharides or Oligosaccharoses (*oligo*^{G} = few). These are *compound sugars* that yield 2 to 10 molecules of the same or different monosaccharides on hydrolysis. Accordingly, an oligosaccharide yielding 2 molecules of monosaccharide on hydrolysis is designated as a dissaccharide and the one yielding 3 molecules of monosaccharide as a trisaccharide and so on. The **general formula** of disaccharides is $C_n(H_2O)_{n-1}$ and that of trisaccharides is $C_n(H_2O)_{n-2}$ and so on. A few examples are :

Disaccharides – Sucrose, Lactose, Maltose, Cellobiose, Trehalose, Gentiobiose, Melibiose

Trisaccharides – Rhamninose, Gentianose, Raffinose (= Melitose), Rabinose, Melezitose

Tetrasaccharides – Stachyose, Scorodose

C. **Polysaccharides** or **Polysaccharoses** (*poly*^{G} = many). These are also *compound sugars* and yield more than 10 molecules of monosaccharides on hydrolysis. These may be further classified depending on whether the monosaccharide molecules produced as a result of the hydrolysis of polysaccharides are of the same type (homopolysaccharides) or of different types (heteropolysaccharides). Their **general formula** is $(C_6H_{10}O_5)_n$.

Some common examples are :

Homopolysaccharides – Starch, Glycogen, Inulin, Cellulose, Pectin, Chitin

Heteropolysaccharides – "Specific soluble sugar" of pneumococcus type III, Hyaluronic acid, Chondrotin

Monosaccharides– Structural Aspects

Stereoisomerism is an important character of monosaccharides. Stereoisomers are the compounds that have the same structural formulae but differ in their spatial configuration.

A carbon is said to be asymmetric when it is attached to four different atoms or groups. The number of asymmetric carbon atoms (n) determines the possible isomers of a given compound which is equal to 2^n. Glucose contains 4 asymmetric carbons and thus has 16 isomers.

Glyceraldehyde–the reference carbohydrate

Glyceraldehyde (triose) is the simplest mono-saccharide with one asymmetric carbon atom. It exists as two stereoisomers and has been chosen as the reference carbohydrate to represent the structure of all other carbohydrates.

D and L isomers

The D and L isomers are mirror images of each other. The spatial orientation of –H and –OH groups on the carbon atom (C_5 for glucose) that is adjacent to the terminal primary alcohol carbon determines whether the sugar is D- or L-isomer. If the –OH group is on the right side, the sugar is of D-series, and if on the left side, it belongs to L-series. The structures of D- and L-glucose based on the reference monosaccharide, D- and L-glyceraldehyde (glycerose) are depicted in Fig.

It may be noted that the naturally occurring monosaccharides in the mammalian tissues are mostly of D-configuration. The enzyme machinery of cells is specific to metabolise D-series of monosaccharides.

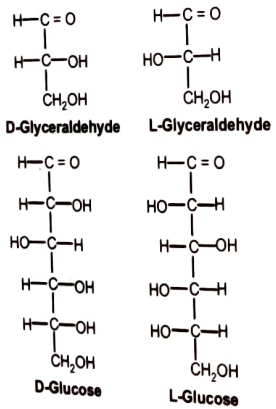


Fig. : D – and L – forms of glucose compared with D– and L- glyceraldehydes.

Optical activity of sugars

Optical activity is a characteristic feature of compounds with asymmetric carbon atom. When a beam of polarized light is passed through a solution of an optical isomer, it will be rotated either to the right or left. The term **dextrorotatory (+)** and **levorotatory (–)** are used to compounds that respectively rotate the plane of polarized light to the right or to the left.

Epimers

If two monosaccharides differ from each other in their configuration around a single specific carbon (other than anomeric) atom, they are referred to as epimers to each other Fig. (Structures of epimers) For instance, **glucose and galactose are epimers** with regard to carbon 4 (C_4-epimers). That is, they differ in the arrangement of –OH group at C_4. Glucose and mannose are epimers with regard to carbon 2 (C_2-epimers).

Structure of Glucose

For a better understanding of glucose structure, let us consider the formation of hemiacetals and hemiketals, respectively produced when an aldehyde or a ketone reacts with alcohol.

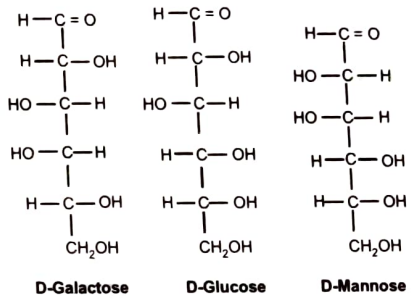
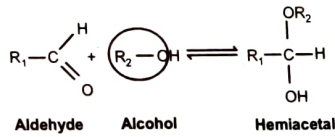


Fig. : Structures of epimers (glucose and galactose are C_4-epimers while glucose and mannose are C_2-epimers).



The hydroxyl group of monosaccharides can react with its own aldehyde or keto functional group to form hemiacetal and hemiketal. Thus, the aldehyde group of glucose at C_1 reacts with alcohol group at C_5 to form two types of cyclic hemiacetals namely α and β, as depicted in fig.

Anomers–mutarotation

The α and β cyclic forms of D-glucose are known as anomers. They differ from each other in the configuration only around C_1 known as anomeric carbon (hemiacetal carbon). In case of α anomer, the –OH group held by anomeric carbon is on the opposite side of the group – CH_2OH of sugar ring. The reverse is true for β-anomer. The anomers differ in certain physical and chemical properties.

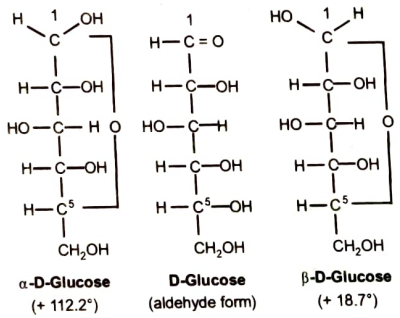


Fig. : Mutarotation of glucose representing α and β anomers (Fischer projections).

Glycosides

Glycosides are formed when the hemiacetal or hemiketal hydroxyl group (of anomeric carbon) of a carbohydrate reacts with a hydroxyl group of another carbohydrate or a non-carbohydrate (e.g. methyl alcohol, phenol, glycerol). The bond so formed is known as **glycosidic bond** and the non-carbohydrate moiety (when present) is referred to as **aglycone**.

Derivatives of Monosaccharides

There are several derivatives of monosaccharides, some of which are physiologically important

1. **Amino sugars :** When one or more hydroxyl groups of the monosaccharides are replaced by amino groups, the products formed are amino sugars e.g. D-glucosamine, D-galactosamine. They are present as constituents of heteropolysaccharides.

   N-Acetylneuraminic acid (NANA) is a derivative of N-acetylmannose and pyruvic acid. It is an important constituent of glycoproteins and glycolipids. The term **sialic acid** is used to include NANA and its other derivatives.

2. **Deoxysugars :** These are the sugars that contain one oxygen less than that present in the parent molecule. The groups $-CHOH$ and $-CH_2OH$ become $-CH_2$ and $-CH_3$ due to the absence of oxygen. D-2-Deoxyribose is the most important deoxysugar since it is a structural constituent of DNA (in contrast to D-ribose in RNA).

3. **L-Ascorbic acid (vitamin C) :** This is a water-soluble vitamin, the structure of which closely resembles that of a monosaccharide.

Disaccharides

Among the oligosaccharides, disaccharides are the most common. As is evident from the name, a disaccharide consists of two monosaccharide units (similar or dissimilar) held together by a glycosidic bond. They are crystalline, water-soluble and sweet to taste. The disaccharides are of two types

1. **Reducing** disaccharides with free aldehyde or keto group e.g. maltose, lactose.
2. **Non-reducing** disaccharides with no free aldehyde or keto group e.g. sucrose, trehalose.

Maltose

Maltose or malt sugar ($C_{12}H_{22}O_{11}$), as such, is not found in free form in the body. It is produced during the course of digestion of starch by the enzyme amylase (pancreatic).

Maltose is composed of two α-D-glucose units held together by α (1 → 4) glycosidic bond.

Sucrose

Sucrose (cane sugar) is the sugar of commerce, mostly produced by sugar cane and sugar beets. Sucrose is made up of α-D-glucose and β-D-fructose. The two monosaccharides are held together by a glycosidic bond ($\alpha_1 \rightarrow \beta_2$), between C_1 of α-glucose and C_2 of β-fructose. The reducing groups of glucose and fructose are involved in glycosidic bond, hence sucrose is a non-reducing sugar, and it cannot form osazones.

Lactose

Lactose is more commonly known as milk sugar since it is the disaccharide found in milk. Lactose is composed of β-D-galactose and β-D-glucose held together by β (1 → 4) glycosidic bond.

Lactose of milk is the most important carbohydrate in the nutrition of young mammals. It is hydrolysed by the intestinal enzyme lactase to glucose and galactose.

Inversion of sucrose

Sucrose, as such is dextrorotatory (+ 66.5°). But, when hydrolysed, sucrose becomes levorotatory (– 28.2°). The process of change in optical rotation from dextrorotatory (+) to levorotatory (–) is referred to as inversion. The hydrolysed mixture of sucrose, containing glucose and fructose, is known as **invert sugar.**

Polysaccharides

Polysaccharides (or simply glycans) consist of repeat units of monosaccharides or their derivatives, held together by glycosidic bonds. They are primarily concerned with two important functions-structural and storage of energy.

Polysaccharides are of two types

1. **Homopolysaccharides** which on hydrolysis yield only a single type of monosaccharide. They are named based on the nature of the monosaccharide unit. Thus, **glucans** are polymers of glucose whereas **fructosans** are polymers of fructose.
2. **Heteropolysaccharides** on hydrolysis yield a mixture of a few monosaccharides or their derivatives.

Homopolysaccharides

Starch

Starch is the carbohydrate reserve of plants which is the most important dietary source for higher animals, including man. High content of starch is found in cereals, roots, tubers, vegetables etc. Starch is a homopolymer composed of D-glucose units held by α-glycosidic bonds. It is known as **glucosan or glucan.**

Starch consists of two polysaccharide components-water soluble **amylose** (15-20%) and a water insoluble **amylopectin** (80-85%). Chemically, amylose is a long unbranched chain with 200—1,000 D-glucose units held by α (1 → 4) glycosidic linkages. Amylopectin, on the other hand, is a branched chain with α (1 → 6) glycosidic bonds at the branching points and α (1 → 4) linkages everywhere else. Amylopectin molecule containing a few thousand glucose units looks like a branched tree (20—30 glucose units per branch).

Inulin

Inulin is a polymer of fructose i.e., **fructosan.** It occurs in dahlia bulbs, garlic, onion etc. It is a low molecular weight (around 5,000) polysaccharide easily soluble in water. Inulin is not utilized by the body. It is used for assessing kidney function through measurement of **glomerular filtration rate (GFR).**

Glycogen

Glycogen is the carbohydrate reserve in animals, hence often referred to as **animal starch.** It is present in high concentration in liver, followed by muscle, brain etc. Glycogen is also found in plants that do not possess chlorophyll (e.g. yeast, fungi).

The structure of glycogen is similar to that of amylopectin with more number of branches. Glucose is the repeating unit in glycogen joined together by α (1 → 4) glycosidic bonds, and α (1 → 6) glycosidic bonds at branching points.

Cellulose

Cellulose occurs exclusively in plants and it is the most abundant organic substance in plant kingdom. It is a predominant constituent of plant cell wall. Cellulose is totally absent in animal body.

Cellulose is composed of β-D-glucose units linked by β **(1 → 4) glycosidic bonds.** Cellulose cannot be digested by mammals—including man—due to lack of the enzyme that cleaves β-glycosidic bonds (α amylase breaks α bonds only). Certain ruminants and herbivorous animals contain microorganisms in the gut which produce enzymes that can cleave β-glycosidic bonds. Hydrolysis of cellulose yields a disaccharide **cellobiose,** followed by **β-D-glucose.**

Cellulose, though not digested, has great importance in human nutrition. It is a major constituent of **fiber,** the non-digestable carbohydrate. The functions of dietary fiber include decreasing the absorption of glucose and cholesterol from the intestine, besides increasing the bulk of feces.

Heteropolysaccharides

When the polysaccharides are composed of different types of sugars or their derivatives, they are referred to as heteropolysaccharides or **heteroglycans.**

Mucopolysaccharides

These are heteroglycans made up of repeating units of sugar derivatives, namely amino sugars and uronic acids. Mucopolysaccharides are more commonly known as **glycosaminoglycans (GAG).** Acetylated amino groups, besides sulfate and carboxyl groups are generally present in GAG structure.

Some of the mucopolysaccharides are found in combination with proteins to form **mucoproteins** or **mucoids** or **proteoglycans.** Mucoproteins may contain up to 95% carbohydrate and 5% protein.

Mucopolysaccharides are essential components of tissue structure. The extracellular spaces of tissue (particularly connective tissue-cartilage, skin, blood vessels, tendons) consist of collagen and elastin fibers embedded in a matrix or ground substance. The ground substance is predominantly composed of GAG.

The important mucopolysaccharides include hyaluronic acid, chondroitin 4-sulfate, heparin, dermatan sulfate and keratan sulfate.

Hyaluronic acid

Hyaluronic acid is an important GAG found in the ground substance of synovial fluid of joints and vitreous humor of eyes. It is also present as a ground substance in connective tissues and forms a gel around the ovum. Hyaluronic acid serves as a lubricant and shock absorbant in joints.

Hyaluronic acid is composed of alternate units of D-glucuronic acid and N-acetyl D-glucosamine. These two molecules form disaccharide units held together by β (1 → 3) glycosidic bond. Hyaluronic acid contains about 250—25,000 disaccharide units (held by β1 → 4 bonds) with a molecular weight up to 4 million.

Chondroitin sulfates

Chondroitin 4-sulfate (Greek : chondros-cartiiage' is a major constituent of various mammalian tissues (bone, cartilage, tendons, heart, valves, skin, cornea etc.). Structurally, it is comparable with hyaluronic acid. Chondroitin 4-sulfate consists of repeating disaccharide units composed of D-glucuronic acid and N-acetyl D-galactosamine 4-sulfate.

Heparin

Heparin is an anticoagulant (prevents blood clotting) that occurs in blood, lung, liver, kidney, spleen etc. Heparin helps in the release of the enzyme lipoprotein lipase which causes clearing the turbidity of lipemic plasma.

Heparin is composed of alternating units of N-sulfo D-glucosamine 6-sulfate and glucuronate 2-sulfate.

Dermatan sulfate

The name dermatan sulfate is derived from the fact that this compound mostly occurs in the skin. It is structurally related to chondroitin 4-sulfate.

Glycoproteins

Several proteins are covalently bound to carbohydrates which are referred to as glycoproteins. The carbohydrate content of glycoprotein varies from 1 % to 90% by weight. Sometimes the term **mucoprotein** is used for glycoprotein with carbohydrate concentration more than 4%. Glycoproteins are very widely distributed in the cells and perform variety of functions. These include their role as enzymes, hormones, transport proteins, structural proteins and receptors.

Blood group substances

The blood group antigens (of erythrocyte membrane) contain carbohydrates as glycoproteins or glycolipids. N-Acetylgalactosamine, galactose, fucose, sialic acid etc. are found in the blood group substances. The carbohydrate content also plays a determinant role in blood grouping.

Lipids

General Properties

Lipids are substances of Biological origin that are soluble in organic solvents such as chloroform and ethanol and are only sparingly soluble, if at all, in water. They are heterogeneous group of compounds related to the fatty acids either actual or potential and chemically they are esters of fatty acids and some alcohols. Hence they are easily separated from other biological materials by extraction into organic solvents and may be further fractionated by such techniques as, adsorption chromatography, thin layer chromatography, and reversed phase chromatography.

Functions

The lipids having only four major biological functions

(*i*) In all cells, the major structural elements of the membranes are composed of lipids.

(*ii*) Certain lipids (triacylglyceral) serves as efficient reserves for the storage of energy.

(*iii*) Many of the vitamins and hormones found in animals are lipids or derivatives of lipids and

(*iv*) The bile acids help to solubilize the other lipid classes during digestion.

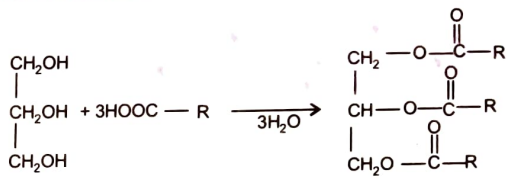
Basic Structural Components

The major structural components of the lipid are

(*i*) Fatty acid and

(*ii*) Glycerol

The lipids are esters of these two molecules.



Fatty Acids

Fatty acids are carboxylic acids with long chain hydrocarbon side groups. They are rarely free in nature, rather occur in esterified form as the major component of the various lipids.

Classification on the basis of their bonding pattern :

These are of two types

(*i*) Saturated fatty acids : Having no double bond and

(*ii*) Unsaturated fatty acids : Having atleast one, or more double bonds.

Note :

1. Most fatty acids have an even number of carbon atoms, because they are usually biosynthesized by concatenation of C_2 units.
2. Over half of the fatty acids residues of plants and animals are unsaturated.
3. In poly unsaturated fatty acids, the double bonds tend to occur at every 3rd carbon atom towards the methyl terminus of the molecule and these double bonds are almost never conjugated.
4. Triple bonds are rarely occurs in fatty acids or any other compounds of biological origine.
5. Saturated fatty acids are highly flexible molecules that can assume a wide range of conformations.
6. The melting points of saturated fatty acids increase with molecular mass.
7. Fatty acids double bonds almost always have the *cis* conformation. This puts a rigid 30° bend in the hydrocarbon chain of unsaturated fatty acids that interferes with their efficient packing to fill space.
8. The consequent reduced van-der-walls interactions cause fatty acids melting points to decrease with their degree of unsaturation.
9. Lipid fluidity increases with the degree of unsaturation of fatty acids.
10. The fatty acids <14 and > 24 carbon atoms are uncommon.
11. The most common monunsaturated fatty acids is oleic acid.
12. The coconut oil have minimum fatty acid content.
13. Mustard oil is one of the most unsaturated.

Classification on the basis of their synthesis in animal body :

Fatty acids are further classified into two groups

(i) Essential fatty acids :

These are some polyunsaturated fatty acids which cannot be synthesized in the animal body and must be supplied with food to avoid their deficiency.

They are

(a) Linoleic acid

(b) Linolenic acid

(c) Arachidonic acid

The maximum number of double bonds present in essential fatty acid is four (i.e. Arachidonic acid).

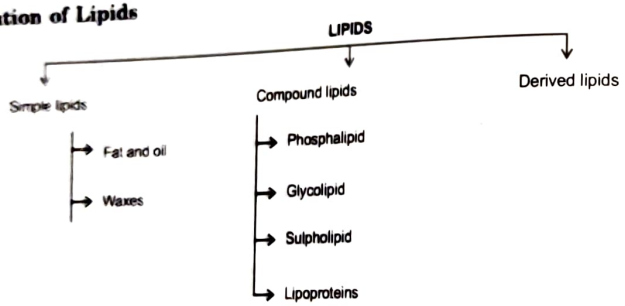
Essential fatty acids serves as precursors for synthesis of Prostaglandins.

(ii) Non-Essential Fatty acids :

These have a wide group of saturated and unsaturated fatty acids synthesized in the animal body.

The fatty acids are esterified with alcohols to produce lipids. The alcohol in this case may be glyceral (mainly) or higher alcohols (such as in waxes).

Classification of Lipids



Simple Lipids

They are esters of fatty acids with various alcohols. They are grouped into two, viz fat and oil and waxes.

Fats and Oils

These are esters of fatty acids with glycerols generally called as *triacylglycerols*.

Triacylglycerols : The fats and oils that occur in plants and animals consist largely of mixtures of **triacylglycerols** (also referred to as *triglycerides or neutral fats*). These *nonpolar, water-insoluble substances are fatty acid triesters of glycerol.*

Triacylglycerols function as energy reservoirs in animals and are therefore their most abundant class of lipids even though they are not components of biological membranes.

Triacylglycerols differ according to the identity and placement of their three fatty acid residues. The so-called simple triacylglycerols contain one type of fatty acid residue and are named accordingly. For example, triacylglycerds or tristearin contains three stearic acid residues, whereas trioleoylglycerol or triolein has three oleic acid residues. The more common mixed triacylglycerols contain two or three different types of fatty acid residues and are named according to their placement on the glycerol moiety.

Fats and oils (which differ only in that fats are solid and oils are liquid at room temperature) are complex mixtures of simple and mixed triacylglycerols whose fatty acid compositions vary with the organism that has produced them. Plant oils are usually richer in unsaturated fatty acid residues than are animal fats, as the lower melting points of oils imply.

Triacylglycerols function as energy reserves : *Fats are a highly efficient form in which to store metabolic energy.* This is because fats are less oxidized than are carbohydrates or proteins and hence yield significantly more energy on oxidation. Further more, fats being nonpolar substances, are stored in anhydrous form, whereas glycogen, for example , binds about twice its weight of water under physiological conditions. Fats therefore provide about six times the metabolic energy of an equal weight of hydrated glycogen.

In animals, **adipocytes** (fat cells) are specialized for the synthesis and storage of triacylglycerols. Whereas other types of cells have only a few small droplets of fat dispersed in their cytosol, adipocytes may be almost entirely filled with fat globules. **Adipose tissue** is most abundant in a subcutaneous layer and in the abdominal cavity. The fat content of normal humans (21% for men, 26% for women) enables them to survive starvation for 2 to 3 months. In contrast, the body's glycogen supply, which functions as a short-term energy store, can provide for the body's metabolic needs for less than a day. The subcutaneous fat layer also provides thermal insulation, which is particularly important for warm-blooded aquatic animals, such as whales, seals, geese and penguins, which are routinely exposed to low temperatures.

Waxes

Waxes are formed by combination of a long chain fatty acid with a long chain alcohol. They have high melting points and are not digested by animals. A common example is Beeswax, a combination of myricyl alcohol ($C_3OH_{61}OH$) and palmitic acid. In the human body the commonest waxes are esters of cholesterol. Vitamin A and Vitamin D esters are palmitic or steric acid esters. The waxes form water insoluble coating on hair and strin in animals and stem, leaves and fruits of plants for protection and pollination.

Compound Lipids

These are esters of fatty acids containing groups in addition to an alcohol and a fatty acid.

Classification : These are of following types :

(I) Phospholipid

(II) Glycolipids

(III) Sulpholipids and Sulfatides

(IV) Lipoproteins

(I) Phospholipids

Alcohol + Fatty acid + Phosphoric acid + Nitrogen – containing bases and other substituents.

The phospholipids are present in large amount in nerve tissue, brain, liver, kidney, pancreas and heart. These are the major part of the biological membranes.

On the basis of Alcohol these are of 3 types :

(i) Glycerophospholipid or Phosphoglycerides :

These are the major phospholipids found in the membranes and contains two fatty acid molecule or 'tails' esterified to the first and second hydroxyl group of Glyceral. The IIIrd hydroxyl group of glyceral forms an ester linkage with phosphoric acid. In addition the phosphoglyceride contains a second alcohol, which is also esterified with phosphoric acid. This is referred to as 'head' alcohol group. The various phosphoglycerides differs in their head alcohol group and all of them contains two non-polar tails. Usually the unsaturated fatty acid esterified to the middle hydroxyl group of glycerol, and saturated with first hydorxyl group. The glycerophospholipids have an assymetric carbon atom at their position α and have L- configuration, e.g.

(a) Leathines (Phosphatidyl cholines): Soyabean and the yeasts are the main plant sources and the glandular and nervous tissues in animal kingdom. These are important for the transport and utilization of other lipids in liver. In their absence accumulation of lipid occurs in liver causes fatty liver (30% lipid in place of 3-4%) ultimately leads to cause fibrotic changes '*Cirrhosis*'. It is also precursors of neurotransmeters.

On complete hydrolysis, lecithin yields, choline, phosphoric acid, glyceral, and two moles of fatty acid. But partial hydrolysis by lecithinase (found in snake venom) causes removal of only one fatty acid to yields Lysolecithins causes rapid rupture of R.B.Cs.

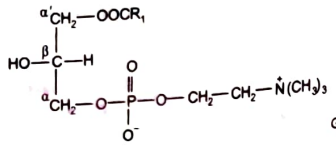


Fig. Structure of α-lecithin

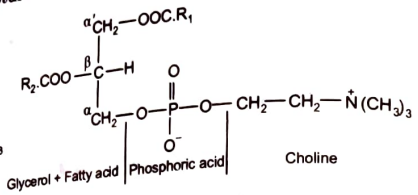


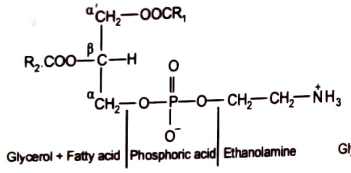
Fig. Structure of α-lysolecithin

(b) Cephalin : The sources is same as lecithins. The structure is also similar but the choline is replaced by *Ethanolamine or serine*. The major difference between cepholin and lecithins are:

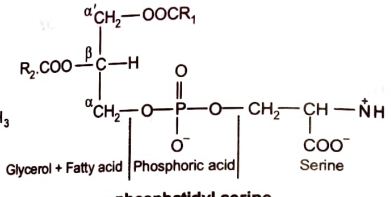
The cephalins are more acidic.

The cephalins are less soluble in alcohol.

Venom containing lecithinase also split off fatty acid from cephalin leaving hemolytic *lysocepholins*.



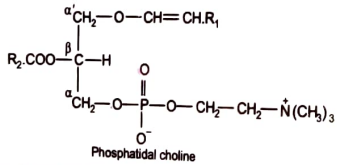
α-phosphatidyl ethanolamine



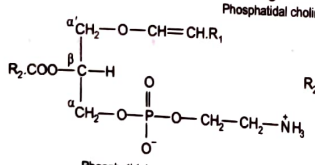
α-phosphatidyl serine

Fig. Two cephalins

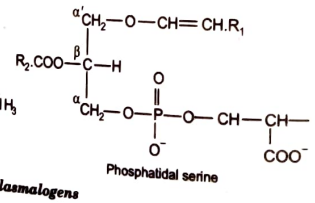
(c) Plasmalogens : These are the glycerophospholipids in which C-1 substituent to the glycerol moity is bonded to it via an α-β unsaturated ether linkage in the cis-configuration rather than an ester linkage. They constitute 210% of the total phosphalipid of the brain and muscle.



Phosphatidal choline



Phosphatidal ethanolamine

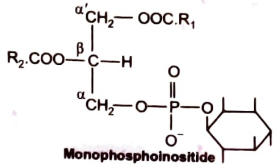


Phosphatidal serine

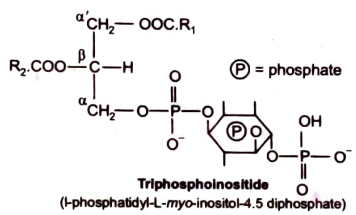
Fig. Three plasmalogens

(ii) Phosphoinositides :

These are mainly transport lipids and considerably found in brain tissue and soyabean oil. On hydrolysis, the phosphoinositides yields one mole of glyceral, two moles of fatty acids, one mole of inosital and 1,2 or 3 moles of phosphoric acid. These may also called glycolipids in as much they contains carbohydrate residues.



Monophosphoinositide



Triphosphoinositide
(I-phosphatidyl-L-*myo*-inositol-4.5 diphosphate)

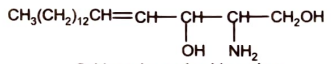
Fig. Two phosphoinositides

- The phosphatidyl inosited acts as second messenger in Ca^{2+} dependent harmone secretion.
- These are more acidic than the other phosphalipids.

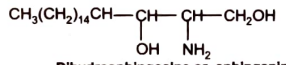
(iii) Phosphosphingosides (Sphingomyelins): These are found in the myelin sheath of nerves and lack in plants and other microrganisms.

In Niemann-Pick disease caused by an autosomal recessive condition, there is accumulation of the sphingomyelins in the brains take place.

They lack of glyceral, and there present a nitrogeneous base, i.e. Sphingosine (18c-amino alcohol) or Dihydrosphingosine besides choline in place of glyceral. These are electrically charged and contains phosphocholines as their polar head group.



Sphingosine or 4-sphingenine
(1,3-dihydroxy-2-amino-*trans*-octadec-4-ene)



Dihydrosphingosine or -sphinganine

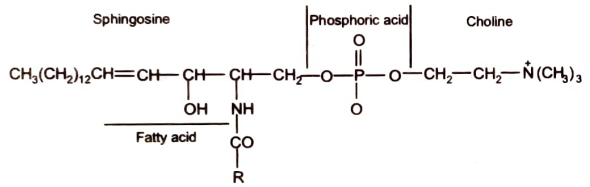


Fig. Structure of sphingomyelin

- In sphingosine molecule, $-NH_2$ group binds to fatty acid by an amide linkage to produce ceramide.
- The clinical features of Niemann-Pick disease are :
  - (*a*) Lack of enzyme sphingomyelinase.
  - (*b*) Enlargement of liver and spleen.
  - (*c*) Mental retardation
  - (*d*) Anaemia and leucocytosis

Gaucher disease : It is due to defficiency of the enzyme *glucosyl ceramide hydrolase* (Glucocere brosidase) which cleaves the ceramide from ceramide Triheraside causing deposition of the later in mainly lungs and bone marrow causing hepatosplenomegaly. This disease effects lungs and cause destruction of bones.

II Glycolipids (Cerebrosides or Glycosphingosides)

(i) Cerebrosides : These are the simplest **sphingoglycolipids** (alternatively **glycosphingo-lipids**), are ceramides with head groups that consist of a single sugar residue. **Galacto**-cerebrosides, which are most prevalent in the neuronal cell membranes of the brain have a β-D-galactose head group.

Glucocerebrosides, which instead have a β-D-glucose residue, occur in the membranes of other tissues, *Cerebrosides, in contrast to phospholipids, lack phosphate groups and hence are most frequently nonionic compounds.* The galactose residues of some galactocerebrosides, however, are sulfated at their C3 positions to form ionic compounds known as **sulfatides**. More complex sphingoglycolipids have unbranched oligosaccharide head groups up to four sugar residues.

(*ii*) **Gangliosides** form the most complex group of sphingoglycolipids. They are ceramide oligosaccharides that include among their sugar groups at least one sialic acid residue(N-acetylneuraminic acid and its derivatives). Gangliosides are primarily components of cell surface membranes and constitute a significant fraction (6%) of brain lipids. Other tissues also contain gangliosides but in lesser amounts.

Gangliosides have considerable physiological and medical significance. Their complex carbohydrate head groups, which extend beyond the surfaces of cell membranes, act as specific receptors for certain pituitary glycoprotein hormones that regulate a number of important physiological functions. Gangliosides are also receptors for bacterial protein toxins such as cholera toxin . There is considerable evidence that gangliosides are specific determinants of cell-cell recognition, so they probably have an important role in the growth and differentiation of tissues as well as in carcinogenesis. Disorders of ganglioside breakdown are responsible for several hereditary sphingolipid storage diseases, such as Tay-Sachs disease, which are characterized by an invariably fatal neurological deterioration.

III Sulfolipids and Sulfatides

Sulfolipids : A glycolipid that contains sulfur is widely distributed in plants. It is localized in the chloroplast but is also found in the chromatophores of photosynthetic bacteria. As the sulfur in this compounds present as a sulfonic group in a hexose, this may be included under a class of compounds called sulfolipids. Its structure is given in the figure below.

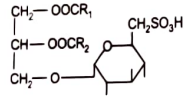


Fig. Structure of a sulfolipid

(6-sulfo-6-dexoy-α-glucosyl monoglyceride))

Sulfatides : A sulfate ester analogue of phrenosin, abundant in white matter of brain is another sulfur-containing glycolipid. In it, the sulfate is present in ester linkage at C3 of the galactose portion of the molecule. Members of this group of cerebroside sulfuric esters have been designated as sulfatides.

The classification of compound lipids as presented here is now becoming obsolete. It is, however, more logical to classify them according to the main alcohol component.

According to this system, the compound lipids fall under two categories :

(*i*)in which the main alcohol component is glycolipid (glycerophosphatides), and

(*ii*)in which sphingosine is the main alcohol component (sphingolipids).

The further classification of these two categories depends on the nature of the other alcohol components.

IV Lipoproteins

Proteins forming complexes with lipids (cephalin, lecithin, cholesterol) are called lipoproteins soluble in water but insoluble in organic solvents.

e.g., lipovitellin and lipovitellenin egg.

The lipoproteins are in reality the temporary intermediates in the process of transfer of lipids from the site of absorption to the site of utilization. The classification of lipoproteins is frequently based on an operational definition, i.e. migration of the fraction in a density gradient separation.

On this basis, the lipoproteins have been classified into following 4 categories Table.

(*a*) *Very high density lipoproteins (VHDLs).* These have densities greater than 1.21.

(*b*) *High density lipoproteins (HDLs).* These possess densities range of 1.063 to 1.21.

(*c*) *Low density lipoproteins (LDLs).* Their density range between 1.05 and 1.063.

(*d*) *Very low density lipoproteins (VLDLs).* Their density range is from 0.93 to 1.05.

Table, also lists the compositions of the fractions floated at the respective densities, alongwith proposed functions.

Table : Classification of lipoproteins based on density

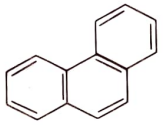
	Name	*Density range*	*Composition by weight(in blood plasma)*	*Half-life*	*Function*
1.	Very high density lipoprotein	> 1.21	62% protein 28% lipid	> 5 days	Phospholipid transport (contains 8-15% of serum phospholipids)
2.	High density lipoprotein	1.063-1.21	50% protein 50% lipid	3-5 days	Cholesterol transport Lipid transport
3.	Low density lipoprotein	1.05-1.063	22% protein 12% triglyceride 20% phospholipids 46% cholestrol	3 -5 days	Lipid transport (congenital absence accompanied by intestinal malabsorption and distorted shape of erythrocyte)
4.	Very low density lipoprotein	0.93-1.05	Largely protein triglyceride and phospholipid; composition variable	3-4 hours	Triglyceride transport (related to chylomicron units

Derived lipids

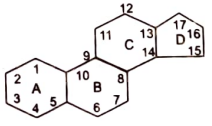
Derived lipid family includes hydrolysis products of simple and compound lipids also various other compounds such as steroids, terpenes, fatty acids, alcohols, fatty aldehydes ketones etc.

Steroids

The steroids are one of the most studied class of biological compounds and are often found in association with fat. Since there is no fatty acid in steroids, hence they are non saponifiable. All steroids may be considered as derivatives of a fused and fully saturated ring system called *cyclopentano-perhdrophenanthrene* or *sterane*.



Phenanthrene



Sterane

Some common examples are given below—

(*i*) Cholesterol

Cholesterol is exclusively found in animals and is the most abundant animal sterol. It is widely distributed in all cells and is a major component of cell membranes and lipoproteins. Cholesterol (Greek: chole–bile) was first isolated from bile. Cholesterol literally means 'solid alcohol from bile.

(*ii*) Coprostanol (= coprosterol)

It occurs in feces and is produced in the intestine as a result of bacterial action on the double bond of cholesterol. The A/B junction is *-cis* in coprostanol in contrast to *-trans* in cholesterol.

(iii) Cholestanol

It occurs as a minor constituent of the sterols of blood and other tissues. Hence the A/ B junction is in *trans* form.

(iv) Ergosterol

It is present in ergot (hence its nomenclature), yeast and the mould *Neurospora*. Its parent hydrocarbon is ergostane, $C_{28}H_{50}$. Ergosterol has a molecular formula, $C_{28}H_{43}OH$ with one OH group at C_3 and 3 double bonds at C_5, C_7 and C_{22}. It is also optically active, $[\alpha]_D$-135°.

Rupture of the ring B by UV radiation produces vitamin D_2 which is chemically known as ergocalciferol. A similar compound cholecalciferol (or vitamin D_3) is, however, obtained from 7-dehydrocholesterol on irradiation with UV light.

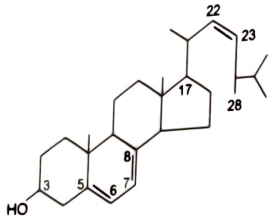


Fig. Structure of ergosterol

(v) Lanosterol (= cryptosterol)

It is a major constituent of wool fat and is also present a minor quantities in liver and yeast. Lanosterol is a C 30 compound with twin methyl groups at C 4 and a third angular methyl group on C 14. There are 2 double bonds at C_8 and C_{24}. It is an intermediate in the biosynthesis of cholesterol.

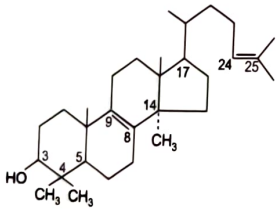


Fig. Structure of lanosterol

A number of sterols have also been obtained from various plant and animal sources, e.g. **stigmasterol** (from soyabean and wheat germ oils), **spinasterol** (spinach and cabbage), **sitosterol** (many higher plants), **ostreasterol** (oysters) and **chondrillasterol** (marine sponges).

(vi) Bile Acids

The bile acids are the important end products of cholesterol metabolism in higher plants. About 20 natural bile acids have been characterized. All these are derived from a C 24 parent steroid, cholanic acid and resemble coprostanol in having rings A and B in cis form. The most abundant bile acids in human bile are cholic acid (25–60% of the total bile acids), chenodexycholic acid (30–50%) and deoxycholic acid (5–25%). Various bile acids differ from each other in the number and position of OH groups which are all in a configuration. In the bile acids, the number of OH group(s) may be 1, 2 or 3 and the position of OH group(s) may be any of the following : 3, 6, 7, 11, 12 and 23. The side chain is usually made up of 5 carbon atoms and bears the carboxyl group.

Table : Some important natural bile acids.

	Name	Formula	Melting point, 0°C	Position of OH group(s)	Source
1.	Monohydroxycholanic acid	$C_{23}H_{38}(OH).COOH$			
	Lithocholic acid		186	3	Man, Ox
2.	Dihydroxycholanic acids	$C_{23}H_{37}(OH)_2.COOH$			
	Deoxycholic acid		172	3, 12	Man, Ox
	Chenodeoxycholic acid		140	3, 7	Man, Ox, Hen
	α-hydrodeoxycholic acid		197	3,6	Pig
3.	Trihydroxycholanic acid	$CH_{23}H_{36}(OH)_3.COOH$			
	Cholanic acid		195	3,7,12	Man, Ox

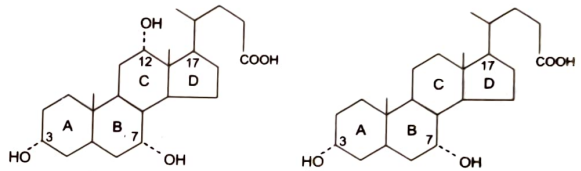


Fig. Structure of two principal bile acids

These acids are coupled in amide linkage to the amino acids glycine ($NH_2—CH_2—COOH$) and taurine ($NH_2—CH_2—CH_2—SO_3H$) to form glycocholic and taurocholic acids respectively. The reaction is characteristic of COOH group and involves the side chain only. The salts of these conjugated acids have high surface activity and hence promote the intestinal absorption of lipids like cholesterol. They are water-soluble and are powerful detergents.

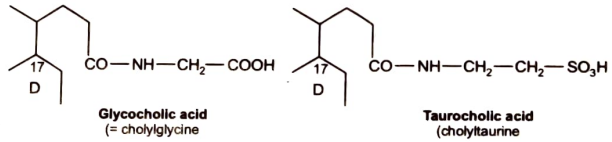


Fig. Side chain of two bile-acid amides

(vii) Terpenes

Among the nonsaponifiable lipids found in plants are many hydrocarbons known as terpenes (*from turpentine*). In general, these hydrocarbons and their oxygenated derivatives have lesser than 40 carbon atoms. The simplest terpenes are called *monoterpenes* and conform to the formula $C_{10}H_{16}$ (equivalent to 2 isoprene units), those with the formula $C_{15}H_{24}$ are called as *sesquiterpenes*, with $C_{20}H_{32}$ as *diterpenes* and with $C_{30}H_{48}$ as *triterpenes*. Terpenes with 40 carbon atoms (or *teraterpenes*) include compounds called carotenoids.

(2-methyl-1, 3-butadiene)

Fig. Structure of isoprene

(viii) Lycopene

Lycopene ($C_{40}H_{56}$) is the main pigment of tomato, paprika and many other fruits. It is a highly unsaturated, unbranched, long chain hydrocarbon (a polyene) and is composed of two identical units ($C_{20}H_{28}$), joined by a double bond between C_{15} and $H_{15'}$. Each of these units may, in turn, be considered have been derived from 4 isoprene units (C_5H_8). A molecule of lycopene in all contains as many as 13 double bonds, of which 11 are conjugated.

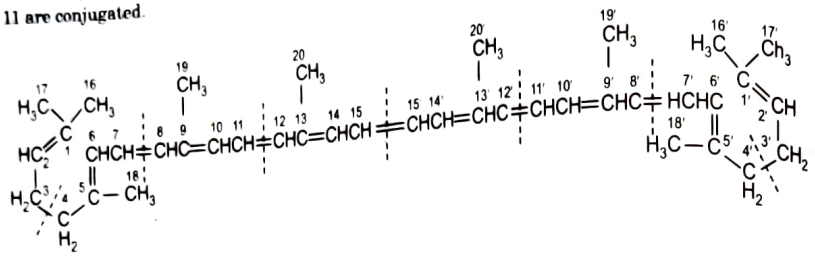


Fig. Structure of lycopene

(ix) Carotene

Another group of naturally occurring carotenoids, with the same molecular formula as that of lycopene, is **carotene**. Carotene was first isolated by Wackenroder (1831) from carrots (this was the origin of the name *carotin*, which was later on modified to *carotene*).

The three types of carotenes are :

(*i*) α-carotene–violet crystals ; m.p. 187°C ; optically active (dextrorotatory).

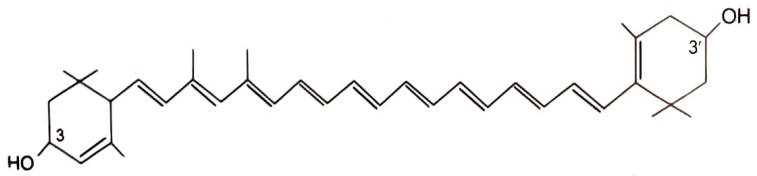
(*ii*) β-carotone–red crystals ; m.p. 183°C ; optically inactive.

(*iii*) γ-carotene–dark red crystal; m.p. 152-154°C; optically inactive.

(The prefixes α, β and γ in the carotenes were assigned not on the basis of structural properties but rather as a result of observations of their elution patterns on a $CaCO_3$ chromatographic column, eluted with petroleum ether).

(x) Xanthophylls

Xanthophylls are characterized by the presence of hydroxyl groups in the ionone rings of carotenes, in the para position to the long chain. The xanthophyll of the leaf (*lutein*, $C_{40}H_{56}O_{20}$) is derived from *α-carotene* while that of corn (*zeaxanthine*) from *β-carotene*. The xanthophyll of crustaceans (*astaxanthine*) is, however, more rich in oxygen. Astaxanthine is responsible for the appetizing redness of boiled lobsters.



(3, 3'-dithdroxy-α-carotene)

Fig. Structure of lutein

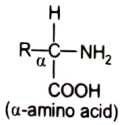
Eicosanoids

Eicosanoids is a family of very potent biological signalling molecules that act as short-range messengers affecting tissues near the cells that produce them. All 3 subclasses of eicosanoids (prostaglandins, thromboxanes and leukotrienes) are unstable and insoluble in water. These signalling molecules generally do not move far from the tissue that produced them and act primarily on cells very near their point of release. Eicosanoids are derivatives of the C-20 polyunsaturated fatty acid arachidonate.

Amino Acids

Amino acids are the monomers of proteins. Alpha-amino acids have an amino and a carboxyl group attached to the same (α) carbon atom.

Although there are over 200 different amino acids found in nature, only about one-tenth (20) of these occur in proteins. Proteins of plant, animal and microbial life contain the same 20 amino acids.

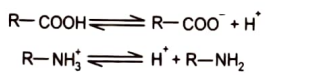


(α-amino acid)

General properties of amino acids

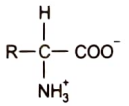
1. Protonic equilibria of amino acids

(*a*) Amino acids have at least two ionizable groups i.e. -COOH and $-NH_3$. The former dissociates more easily than the latter. In solution, two forms of these groups, one-charged and one neutral, exist in protonic equilibrium with each other.

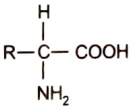


R-COOH and- $R-NH_3^+$ represent the protonated or acid partners in these equilibria. R–COO^- and $R–NH_2$ are the conjugated bases (i.e. proton acceptors) of the corresponding acids. Although both R–COOH and $R-NH_3^+$ are weak acids, R–COOH is a several thousand times stronger acid than is $R-NH_3^+$.

(*b*) At pH 7.4, carboxyl groups exist almost entirely as the conjugated base i.e. R–COO^-. Most amino groups exist in the form, $R-NH_3^+$. In blood and most tissues, amino acid structures are drawn as follows :



The following structure cannot exist at any pH but is frequently used as a convenience when the chemistry of amino acids is discussed.

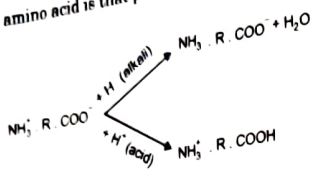


(*c*) P_k of an acid is simply the negative log of the dissociation constant

$$P_k = -\log K$$

P_k values for α-amino groups of free amino acids is about 9.8.

(d) The isoelectric pH (P^I) of an amino acid is that pH at which it has no net charge and hence does not move in an electric field.



Addition of acid or alkali depresses one type of ionisation so that the amino acid behaves as a base or an acid. The ion at the isoelectric point which carries + and - charges internally neutralised is called **"Zwitterion"**. The three types of ions are represented in Fig.

In an acid solution, the amino acid acts as a base yielding cations. When current is allowed to pass through the solution, the amino acid migrates to the cathode or positive pool. In an alkaline solution, it behaves as an acid forming anions. In the electric field, the amino acid migrates to the anode or negative pool. On account of these opposite reactions depending on the acidity or alkalinity of the solution, the amino acids are called ampholytes.

Since $Pk_1(RCOOH) = 2.35$ and $Pk_2\left(RNH_3^+\right) = 9.69$,

then isoelectric pH (P^I) of alanine is

$$P^I = \frac{Pk_1 + Pk_2}{2} = \frac{2.35 + 9.69}{2} = 6.02$$

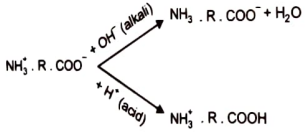
$$P^I \text{ of Aspartic acid} = \frac{Pk_1 + Pk_2}{2} = \frac{2.09 + 3.86}{2} = 2.98$$

Thus P^I of lysine and arginine is 9.7 and 10.8 respectively.

The ability to perform calculations of this type is of significant value in clinical laboratory to assess the mobility of known compounds in electric fields and to select appropriate buffers for separation of one from another.

Zwitterion

1. The ion at isoelectric point which carries + and – charges is called zwitterion.

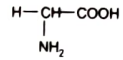
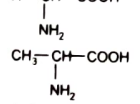
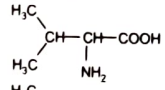
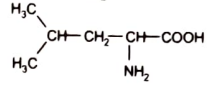
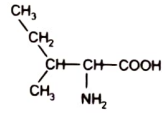
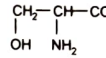
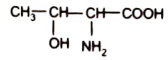
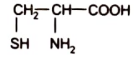
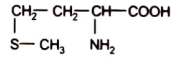
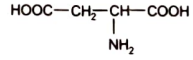
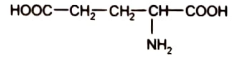
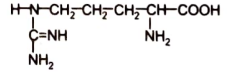
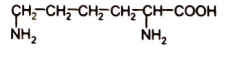
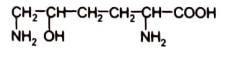
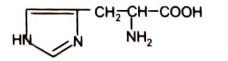


2. In acid solution zwitterion combines with H^+ ions to form base yielding cation, when the current is passed, the amino acid migrates towards cathode.
3. In alkali solution, zwitterion, combines with OH^- to form acid yielding anions and migrates towards anode during the change of current.
4. Zwitterion is the ampholyte, i.e. it is both a proton donor and proton acceptor.
5. Generally, the acidic and basic strength of the zwitterion are different and a solution of pure amino acid in water is not neutral.
6. Proteins like the amino acids, contain 6 acidic and basic groups exist in solution as zwitterion.

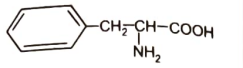
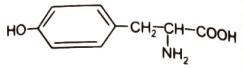
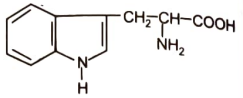
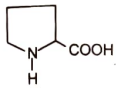
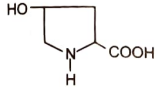
2. Structures of amino acids

For many purposes, it is conveninet to subdivide the amino acids in proteins into 7 classes as in the following table. In addition to their common names, systematic chemical names are also included in this table.

TABLE : L-α-amino acids found in proteins

Group	Trivial name	Symbol	Systematic name	Structural formula
I.	**With aliphatic side chains**			
	Glycine	Gly (G)	Amino acetic acid	
	Alanine	Ala (A)	2-Amino propanoic acid	
	Valine	Val (V)	2-Amino-3-methyl butanoic acid	
	Leucine	Leu (L)	2-Amino-4-methyl pentanoic acid	
	Isoleucine	*lie* (I)	2-Amino-3-methyl pentanoic acid	
II.	**With side chains containing hydroxyl (OH) groups**			
	Serine	Ser (S)	2-Amino-3-hydroxy propanoic acid	
	Threonine	Thr(T)	2-Amino-3-hydroxy -butanoic actd	
III.	**With side chains containing sulfur atoms**			
	Cysteine	Cys (C)	2-Amino-3-mercapto propanoic acid	
	Methionine	Met (M)	2-Amino-4-(methylthio) butanoic acid	
IV.	**With side chains containing acidic group**			
	Aspartic acid	Asp (D)	Amino succinic acid	
	Glutamic acid	Glu (E)	2-Amino glutaric acid	
V.	**With side chains containing basic groups**			
	Arginine	Arg (R)	2-Amino-5-guanido valeric acid	
	Lysine	Lys (K)	2, 6-Diamino hexanoic acid	
	Hydroxy lysine	Hyl.	2, 6-Diamino-5-hydroxy-hexanoic acid	
	Histidine	His (H)	2-Amino-1 H-imidazole-4-propanoic acid	

Continue

Group	Trivial name	Symbol	Systematic name	Structural formula
VI.	**Containing aromatic rings**			
	Phenyl alanine	PHe (F)	2-Amino-3-phenyle-propanoic acid	
	tyrosine	Tyr (Y)	2-Amino-3-(4-hydroxy phenyl) propanoic acid	
	Tryptophan	Trp (W)	2-Amino-3-(3-indolyl) propanoic acid	
VII.	**Imino acids**			
	Proline	Pro (P)	2-Pyrrolidine carboxylic-acid	
	Hydroxy proline Hyp.		4-hydroxy-2-pyrrolidine-carboxylic acid	

3. Optical isomers of amino acids

Except glycine, each amino acid has at least one asymmetric carbon atom and hence is optically active. Although D-amino acids occur in cells and even in polypeptides, they are not present in proteins.

Various other amino acids--Homocysteine, Homoserine, Ornithine, Citrulline, Arginosuccinic acid, Dopa, 3-monoiodotyrosine, 3, 5-Diiodotyrosine, 3,5,3'-triiodotyrosine, Thyroxine, β–Alanine, Taurine etc.—in free or combined states fulfil important functions in metabolic processes other than as constituents of proteins. Many additional amino acids occur in plants or in antibiotics. Over 20 D-amino acids occur naturally.

Physical Properties of Amino Acids

1. Amino acids are soluble in polar solvents such as water and ethanol but they are insoluble in nonpolar solvents such as benzene or ether.
2. Their melting point is above 200°C.
3. The aromatic amino acids tryptophan, tyrosine, histidine and phenylalanine absorb ultraviolet light.

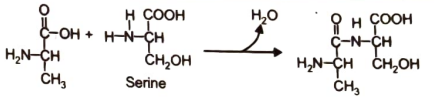
Chemical Reactions of Amino Acids

1. Ninhydrin test : Ninhydrin is a powerful oxidizing agent which causes oxidative decarboxylation of α-amino acids yielding CO_2, NH_3 and an aldehyde. The reduced ninhydrin then reacts with the liberated ammonia forming a blue complex. Proline and hydroxyproline produce a yellow rather than a purple colour with ninhydrin.
2. A variety of colour reactions specific for particular functional groups in amino acids are known which are useful in both qualitative and quantitative identification of particular amino acids. These are given below.
3. Formation of peptide bonds : Peptide bond formation involves removal of one mole of water between the α-amino group of one amino acid and the α-carboxyl group of a second amino acid.

Peptides

Definition : A peptide consists of two or more amino acids linked by a peptide bond formed between the carboxyl group of one amino acid and the amino group of another with the removal of one mole of water during a peptide bond.

Name of the test	Reagents	Colour	Amino acid detected
Nitroprusside	Sodium nitroprusside in dil. NH_4OH	Red	Cysteine
Hopkins Cole	Glyoxylic acid in 36N. H_2SO_4	Purple	Tryptophan
Millon	$HgNO_3$ in HNO_3, heat	Red	Tyrosine
Xanthoproteic	Boiling conc. HNO_3	Yellow	Tyrosine, Tryptophan, Phenylanine
Sakagauchi	α-Naphthol and Sodium hypochlorite	Red	Arginine
Pauly	Diazotized sulfanillic acid in alkaline solution	Red	Histidine



Alanyl-Serine (Ala-Ser) a dipeptide

Polypeptides are formed by long peptide chains containing large numbers of peptide bonds. Polymers of up to 100 amino acids are termed polypeptides and those with more than 100 are generally termed proteins.

Primary structure of peptides

The sequence of amino acids in a polypeptide is referred to as its primary structure. Chemical methods are required to determine the primary structure. Physical technics such as X-ray crystallography are employed for higher order of protein structure.

The N-terminal (amino terminal) amino acid is always shown at the left and the C-terminal (carboxyl terminal) amino acid at the right of the polypeptide chain.

Fig. Primary Structure of Peptide

Even small changes in the primary structure of proteins may produce remarkable physiologic effects. Substitution of a single amino acid for another in a sequence of 100 or more amino acids may abolish biologic activity with serious consequences (e.g., Sickle cell disease). The biochemical basis for many inherited metabolic diseases are known owing to tbe introduction of new chemical and physical methods to determine protein structure.

Synthesis of Peptides

Peptides may be synthesized by a reaction between an activated carboxyl group such as an acid chloride of one amino acid and the amino group of another, e.g., cysteine acid chloride and lysine.

After the peptide bond is formed, the blocking group is removed, leaving the desired peptide.

Physiologically Active Peptides

1. In some cases, they are the basis of products of protein.
2. In others, they may be hormones, antibiotics, precursors of bacterial cell walls, or even potent poisons.
3. The widely distributed tripeptide glutathione is required for the action of several enzymes including insulin. Glutathione reductase functions either in insulin degradation or in the formation of disulfide bonds.
4. Other naturally occurring important peptides are Bradykinin and Kallidin which are smooth muscle hypotensive agents are liberated from specific plasma proteins by treatment with snake venom.

Glutathione

1. It is a tripeptide consisting of glutamic acid, cysteine and glycine.
2. It is converted to disulphide form by easy dehdrogenation and involved in oxidation-reduction reaction.

Oxytocin and Vasopressin

1. These are cyclic peptide hormones consisting of 8 amino acids found in pituitary gland.
2. Oxytocin functions on uterine muscle and aids in the ejection of milk, whereas vasopressin acts in the reabsorption of water from the distal and collecting tubules of the kidney.

Carnosine

1. It is a dipeptide consisting of β–alanine and histidine.
2. It is a dipeptide of voluntary muscle.

Bradykinin

1. It is a nonapeptide consisting of 9 amino acids and the most potent pain-producing substance.
2. It mediates the production of Prostaglandin E_2 from arterial walls.
3. Kallidin I and Kallidin II consisting of 10 amino acids have relaxant effects on smooth muscles.
4. The decapeptide (10 amino acids) P_4 is present in dorsal horns of spinal cord and intestine and is involved in voluntary movement and pain relief.

Angiotensin

1. The enzyme Renin, being released from, kidney acts on plasma globulin fraction to liberate a peptide Angiotensin I consisting of ten amino acids—having slight effect on blood pressure.

Proteins

Proteins may be defined as the high molecular weight mixed polymers of a-amino acids joined together with peptide linkage (–CO–NH–).

Proteins are the chief constituents of all living matter. They contain carbon, hydrogen, nitrogen and sulphur and some contain phosphorus also.

Biological Importance or Biochemical Functions

1. Proteins are the essence of life processes.
2. They are the fundamental constituents of all protoplasm and are involved in the structure of the living cell and in its function.
3. Enzymes are made up of proteins.
4. Many of the hormones are proteins.
5. The cement substances and the reticulum which bind or hold the cells as tissues or organs are made up partly of proteins.
6. They execute their activities in the transport of oxygen and carbon dioxide by hemoglobin and special enzymes in the red cells.
7. They function in the homostatic control of the volume of the circulating blood and that of the interstitial fluids through the plasma proteins.
8. They are involved in blood clotting through thrombin, fibrinogen and other protein factors.
9. They act as the defence against infections by means of protein antibodies.
10. They perform hereditary transmission by nucleoproteins of the cell nucleus.

Protein hydrolysing enzymes

1. Pepsin : In Gastric Juice.
2. Trypsin, Chymotrypsin and Carboxypeptidases : In Pancreatic Juice.
3. Amino-peptidases, Dipeptidases and Polypeptidases : In intestinal juice.

Classification of Proteins

1. Simple proteins

Class of Proteins	Characteristics	Examples
(*i*) Albumins	Soluble in water, coagulfable by heat and precipitated at high salt concentrations.	Serum albumin, egg albumin, lactalbumin (Milk), leucosin (wheat), legumelin (soyabeans).
(*ii*) Globulins	Insoluble in water, soluble in dilute salt solutions and precipitated by half saturated salt solutions.	Serum globulin, vitellin (egg yolk), tuberin (potato), myosinogen (muscle), legumin (peas).
(*iii*) Glutelins	Insoluble in water but soluble in dilute acids and alkalis. Mostly found in plants.	Glutenin (wheat), oryzenin (rice).
(*iv*) Prolamines	Insoluble in water and absolute alcohol but soluble in 70 to 80 per cent alcohol.	Gliadin (wheat), zein (maize).
(*v*) Protamines	Basic proteins of low molecular weight. Soluble in water, dilute acids and alkalis. Not coagulable by heat.	Salmine (salmon sperm).
(*vi*) Histones	Soluble in water and insoluble in very dilute ammonium hydroxide.	Globin of hemoglobin and thymus histones.
(*vii*) Scleroproteins	Insoluble in water, dilute acids and alkalis.	Keratin (hair, horn, nail, hoof and feathers), collagen (bone, skin), elastin (ligament).

2. Conjugated Proteins

Class of Proteins	***Characteristics***	***Examples***
(*i*) Nucleoproteins	Composed of simple basic proteins (protamines or histones) with nucleic acids, found in nuclei. Soluble in water.	Nucleoprotamines and nucleohistones.
(*ii*) Lipoproteins	Combination of proteins with lipids, such as fatty acids, cholesterol and phospholipids etc.	Lipoproteins of egg-yolk, milk and cell membranes, lipoproteins of blood.
(*iii*) Glycoproteins	Combination of proteins with carbohydrate (mucopolysaccharides).	Mucin (saliva), ovomucoid (egg white), osseomucoid (bone), tendomucoid (tendon).
(*iv*) Phosphoproteins	Contain phosphorus radical as a prosthetic group.	Caseinogen (milk), ovovitellin (egg yolk).
(*v*) Metalloproteins	Contain metal ions as their prosthetic groups. The metal ions generally are Fe, Co, Mg, Mn, Zn, Cu etc.	Siderophilin (Fe), ceruloplasmin (Cu).
(*vi*) Chromoproteins	Contain porphyrin (with a metal ion) as their prosthetic groups.	Haemoglobin, myoglobin, catalase, peroxidase, cytochromes.
(*vii*) Flavoproteins	Contain riboflavin as their prosthetic groups.	Flavoproteins of liver and kidney

3. Derived Proteins

Class of Proteins	Characteristics	Examples
A. *Primary derivatives*		
(*i*) Proteins	Derived in the early stage of protein hydrolysis by dilute acids, enzymes or alkalis.	Fibrin from fibrinogen.
(*ii*) Metaproteins	Derived in the later stage of protein hydrolysis by slightly stronger acids and alkalis.	Acid and alkali metaproteins.
(*iii*) Coagulated proteins	They are denatured proteins formed by the action of heat. X-rays, ultraviolet rays etc.	Cooked proteins, coagulated albumins.
B. *Secondary derivatives (Due to cleavage of peptide bonds)*		
(*i*) Proteoses	Formed by the action of pepsin or trypsin. Precipitated by saturated Solution of ammonium sulphate, inoaglubale by heat.	Albumose from albumin, globulose from globulin.
(*ii*) Peptones	Further stage of cleavage than the proteoses. Soluble in water, incoagulable by heat and not precipitated by saturated ammonium sulphate solutions.	
(*iii*) Peptides	Compounds containing two or more amino acids. They may be di-, tri-, and porypeptides. ==	Glycyl-alanine, leucyl-glutamic acid.

Bonds Relating to Protein Structure

Protein structures are confirmed by

1. Strong bonds : Peptide acid disulphide.
2. Weak bonds : Hydrogen and hydrophobic.

1. Strong bonds

A. Peptide bonds

The primary structure of proteins is derived from peptide bonds which can be established by the following evidences :

(*i*) Proteases produce polypeptides by hydrolyzing proteins. They can also hydrolyze peptide bonds.

(*ii*) The protein, insulin, is synthesized by linking amino-acids by peptide bonds.

(*iii*) Polypeptides, like proteins, react with biuret reagent which is suggestive of 2 or more peptide bonds.

B. Disulphide bond

1. The disulphide bond may interconnect two parallel chains through cysteine within each polypeptide shown in the Figure.

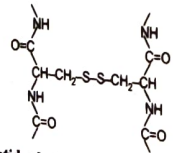


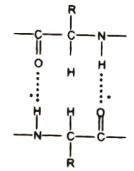
Fig. : Two peptide chains united by a disulphide linkage

2. The -bond is not broken under the usual conditions of denaturation.
3. Performic acid is used to oxidize insulin to separate the protein molecule into its constituent polypeptide chains without affecting the other part of the molecule.

2. Weak bonds

A. Hydrogen bonds

1. The hydrogen bond appears from the sharing of hydrogen atoms between the nitrogen and the carbonyl oxygen of different peptide bonds as shown below by star mark.



2. Each hydrogen bond is quite weak.
3. During denaturation of a protein, the hydrogen and hydrophobic bonds are broken.

B. Hydrophobic bonds

1. The nonpolar side chains of neutral amino acids are closely associated with one another in proteins.
2. No true bond exists.
3. Nonetheless, they play an important role in maintaining protein structure.

Structure of Proteins

The structure of protein is considered by several levels of organization on the basis of modern views. The levels of organization are :

1. Primary level of organization.
2. Secondary level of organization.
3. Tertiary level of organization.
4. Quarternary level of organization.

1. Primary structure

Peptide bond is formed by the amino acids linked by carboxyl group of one amino acid with the α-amino group of another amino acid as shown :

$$H_2N–CH_2–CO–NH–CH_2–COOH$$

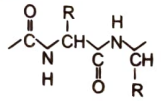
Glycyl-Glycine

Thus, they form peptide bonds by several amino acids as

–Ala–Gly–gly–His–leu–

–Ala–Gly–His–Gly– leu–

The primary structure ultimately becomes as



2. Secondary structure

Globular proteins indicate a coiled structure in which peptide bonds are folded in a regular manner. The foldings are the results of linking of the carboxyl- and amino groups of the peptide chains by means of hydrogen bonds and disulfide bonds. Such foldings are referred to as the secondary structure of the protein. Present evidence suggests that in many proteins, the hydrogen bonding produces a regular coiled arrangement, called α-helix.

The β-Pleated sheet

1. Pauling and corey also proposed a second ordered structure, the β-plated sheet β becuase it was their second structure, the α-helix being the first). In the α-helix the polypeptide chain is condensed, in the β-pleated sheet it is almost fully extended.

2. In a β-pleated sheet when the adjacent polypeptide chains run in opposite directions (N to C terminus), the structure is termed an antiparallel β-pleated sheet. But when the chains run in the same direction, it is termed parallel.
3. Regions of β-pleated structure are present in many proteins, and both parallel and antiparallel forms occur.
4. The α-helix is stabilized by hydrogen bonding between peptide bonds 4 residues apart in a primary structural sense, stabilization of the β-pleated sheet results from formation of hydrogen bonds between peptides far removed from one another in a primary structural sense.

3. Tertiary structure

The globular protein if completely is composed of a series of single helix, these molecules will have elongated structures with a larger axial ratio (length: breadth). Many proteins are spherical. The special structure in three dimensions is maintained by covalent or other bonds. Many globular proteins lack in disulphide bonds, yet they are stable in solution. The other bonds like hydrogen bonds, salt bonds and hydrophobic or non-polar bonds are also involved in the stability of the molecule. This has been described as the tertiary structure of protein.

4. Quarternary structure

1. When a protein consists of two or more peptide chains held together by non-covalent interactions or by covalent cross-link, is referred to as quarternary structure.
2. In quarternary structure there are several monomeric units and the primary, secondary, tertiary structures may combine. The polymerization of these monomers or subunits is described as quarternary structure.
3. Glutamic dehydrogenase is tetramer and aldolase molecule is trimer.

Structures of protein related to biological functions of protein

1. Enzymes are made up of peptide by peptide linkage (–CONH–) by different amino acid sequences. This follows the primary structure of protein.
2. Many of the hormones also follow the same method forming peptides which include the primary structure of protein. If any alteration takes place in the proper sequence, there will be the abnormalities in the function.
3. The globular protein (e.g., many enzymes) have folded, coiled polypeptide chains and axial ratios of less than 10 and generally not greater than 3 to 4. Fibrous proteins have axial ratio greater than 10.
4. Proteins that bind nucleotides show a nucleotide binding domain of tertiary structure.
5. Most fibrous proteins fulfil structural roles in skin, connective tissue, or fibres such as hair, silk, wool. The typical sequence of amino acid of fibrous protein follow specific secondary tertiary structures.
6. Silk fibroin contains small quantities of bulky amino acids; these bulky residues interrupt the β-sheet regions and show the flexibility essential for a successful colour.
7. Tropocollagen, the important unit of collagen, consists of 3 polypeptide chains.

   Each polypeptide forms a left handed helix (not an α helix) having 3 residues per turn. 3 left handed helical polypeptide then entwine to form a "right hand" triple or super, helix i.e., stabilized by H bond formed between individual polypeptide chain. Mature collagen fibres show the stable and extended helicle conformation.

Principles of determination of molecular weight of protein

The proteins have a large molecular weight and they are large macromolecules.

Molecular weight is determined by physical methods such as:

1. Analytic ultracentrifugation
2. Sucrose density gradient centrifugation
3. Gel filtration
4. Polyacrylamide Gel Electrophoresis

Denaturation of Proteins

Denaturation may be defined as the disruption of the secondary, tertiary and quarternary structure of the native protein resulting in the alterations of the physical, chemical and biological characteristics of the protein by a variety of agents.

Denaturing agents

1. Physical agents : Heat, surface action, ultraviolet light, ultrasound, high pressure etc.

2. Chemical agents : Acids, alkalis, heavy metal salts, urea, ethanol, guanidine detergents etc.

Urea and guanidine probably interfere with the hydrogen bonds between peptide linkages. Acids and alkalis probably attack directly the hydrogen bonds in the secondary and tertiary structure of proteins.

Physical alterations

Many proteins, especially of the globular type, can be crystallized in the native state. But denatured proteins cannot be crystallized.

Chemical alterations

The denatured protein is greatly decreased in solubility at its isoelectric point. The chemical groups are exposed to chemical reactions and more readily detected as a result of the unfolding process in denaturation. Among these are sulphydryl group of cysteine, the disulphide group of cystine and the phenolic group of tyrosine.

Biological alterations

The digestability of certain denatured proteins by proteolytic enzymes is increased. Enzymatic or hormonal activity is usually destroyed by denaturation. The antigenic or antibody functions of proteins are frequently altered.

If the denaturation is severe, the protein molecules become insoluble and precipitation results as well as the changes in the properties of the proteins are permanent and *"irreversible"*. In case of mild denaturation, there is *"reversible denaturation"* leading to the slight changes in the properties of the protein which can be restored to the native state after suitable treatment.

Important Tests of Proteins

Colour reactions

1. *Biuret test :* To 2 ml of test solution add an equal volume of 10% NaOH and one drop of 10% $CuSO_4$ solution. A violet colour formation indicates the presence of peptide linkage.
2. *Ninhydrin test :* To 1 ml Ninhydrin solution add 1 ml protein solution and heat. Formation of violet colour indicates the presence of α-amino acids.

Ramachandran Plot

The Ramachandran plot in fig. illustrates the positions of the various regular secondary structures, as well as some additional conceivable ones. Note that all helices with a given integral number of residues per turn lie on one of the set of numbered lines drawn across the map. Particularly important are line corresponding to the flat ribbon ($n = 2$) and the points corresponding to closed rings ($n = 5$). As these are passed, the handedness of the helix changes. This can easily be understood from the following thought experiment. Imagine unwinding a right-hand helix until it approaches a flat ribbon.

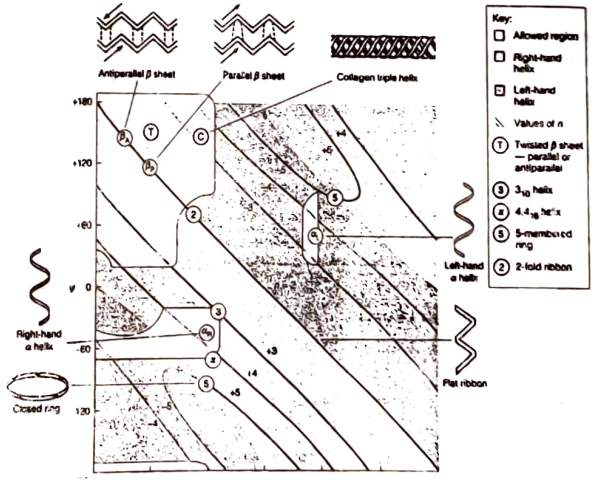


Fig. A Ramachandran plot.

A map of this type can be used to describe the backbone conformation of any polypeptide residue as well as some simple secondary structures of proteins. The coordinates are the bound angles ϕ and ψ. The white areas correspond to allowed conformations, which vary according to the bulkiness of the side chains; this plot assumes that all residues are L-alanines. The colored lines running across the graph correspond to various values of n (residues per turn); the line bisecting the graph corresponds to $n = 2$. The helix is right-handed when n is positive (tan regions), left-handed when n is negative (blue regions). The circles with symbols correspond to important secondary structures, as shown in margins.

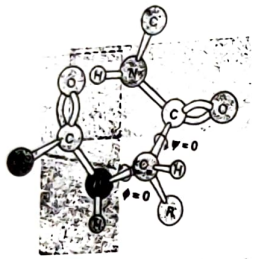


Fig. A sterically nonallowed conformation.

The conformation $\phi = 0$, $\psi = 0$ is not allowed in any polypeptide chain because of the steric clash between the carbonyl oxygen and amino proton.

The value of n will decrease until, at $n = 2$, the flat ribbon structure is reached. If winding is continued in the same direction, a left-hand helix begins to form. Similarly, in a progression to broader and broader helices, the pitch continually decreases. A polypeptide chain can form a five-membered ring with pitch = 0. This lies at the border between right and left helices.

One of the most useful features of Ramachandran maps is that they allow us to describe very simply which structures are sterically possible and which are not. Every point on the surface of the map corresponds to a pair of ϕ, ψ angles and thus to a conceivable secondary structure. But for many pairs of ϕ, ψ values some atoms in the chain would approach closer than their van der Waals radii would allow. Example is shown in fig. Such conformations are sterically excluded. Ramachandran and other researchers have examined the entire map surface, using models and computers to determine which conformations are actually allowed. The allowed ϕ, ψ pairs lie in the white areas in fig. Clearly, only a small fraction of the conceivable conformations is actually possible. All of the regular structures we have discussed fall into or very close to these regions.

Although Fig. shows the left-hand α helix lying on the edge of an allowed region, it is, in fact, not nearly as favored as the right-hand form. This difference is a consequence of the fact that all amino acids in proteins are of the L-form. With L-amino acids, steric interference between the side chains and the backbone of the helix is less with a right-hand helix than with a left-hand helix.

Nucleic Acid

One of the hallmarks of living organism is their ability to reproduce. The information that makes each individual life form unique must be preserved and passed on the progeney. All life on earth uses nucleic acid for storage of genetic information. With the exception of some viruses, the biomolecule utilized for storage information is Deoxyribonucleic acid (DNA). This molecule is remarkably well suited for its task because of its chemical stability and its ability to encode vast amount of information using a simple 3- letter code.

The structural components of nucleic acid, i.e., Nucleotides are not only have functions as genetic information molecules but they play other functions like metabolism of fat, protein, lipid etc. and also as messengers (CAMP) and as energy currency (ATP, ADP etc.).

Structure of Nucleic Acid

The nucleic acid are polymers of nucleotides.

Nucleotides

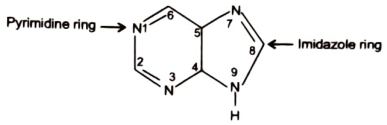
Nucleotides are composed of a nitrogeneous base, a pentose sugar and phosphoric acid molecule.

Nitrogeneous Base

There are two types of Nitrogeneous bases are found in nucleic acids.

(i) Purines Derivatives :

These are all derived from their parent compound purine. Purine are two ring nitrogen compound. Each purine molecule is dicyclic, formed with nine atoms. There is one pyrimidine ring of six atoms and an imidazole ring of five atoms. The atoms are numbered anticlockwise in pyrimidine ring but clockwise in imidazole ring. It has a melting point of 216°C.



Purine

The prevalent purine derivatives found in nucleic acids are adenine and guanine.

Adenine
(6-aminopurine)

Guanine
(2-amino-6-oxypurine)

Fig. Major purine derivaties.

Adenine : These are ($C_5H_5N_5$), found on both RNA and DNA, is a white crystalline purine base , with MW=135.15 Daltons and a M.P. 360-365°C.

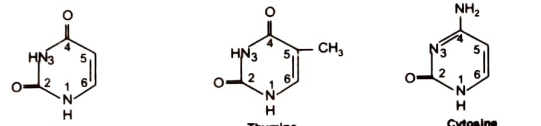
Guanine ($C_5H_5ON_5$) : These also found in both RNA and DNA, is a colourless, insoluble crystalline substance, with MW = 151.15 Daltons. It was first isolated from guano (bird manure), hence so named.

Other purine derivatives are also found in plants, e.g., caffeine (1, 3, 7 trimethyl - xan- thine) is present in coffee and tea and theobromine (3,7-dimethylxanthine) is found in tea and cocoa. Both caffeine and theobromine have important pharmacological properties.

***(ii)* Pyrimidine Derivatives.** These are all derived from parent heterocyclic compound Pyrimidine, which contains a six membered ring with two nitrogen atoms and three double bonds. It has a melting point of 22°C and Boiling point of 123°C. The atoms are numbered clockwise.

Common pyrimidine derivatives found in nucleic acids :

Uracil ($C_4H_4O_2N_2$) : These are found in RNA molecules only, is a white, crystalline pyrimidine base with MW = 112.10 Daltons and =a M.P. 338°C. Only rarely does uracil occur in DNA.



Uracil
(2,4-dioxypyrimidine
or pyrimidinedione)

Thymine
(5-methyl-2,4-dioxypyrimidine
or 5-methyluracil)

Cytosine
(2-oxy-4-aminopyrimidine
or aminopyrimidone)

Fig. Major pyrimidine derivatives

Thymine ($C_5H_6O_2N_2$) : These are found in DNA molecules only, has MW = 126.13 Daltons. It was first isolated from thymus, hence so named. Only rarely does thymine occur in RNA.

Cytosine ($C_4H_5ON_3$) : These are found in both RNA and DNA, is a white crystalline substance, with MW = 111.12 Daltons and a M.P. 320.325°C.

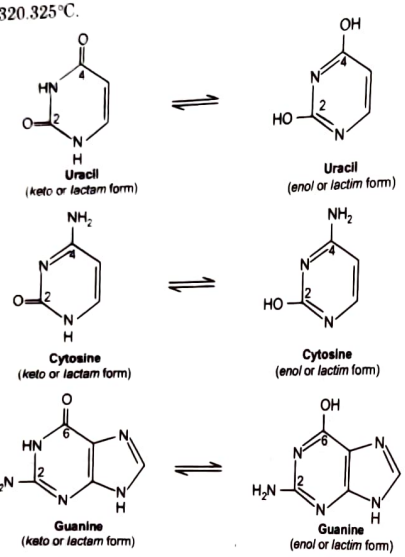


Fig. Tautomeric forms of uracil, cytosine and guanine

Nucleosides and Nucleotides

Nucleosides

The nucleosides are compounds in which nitrogenous bases (purines and pyrimidines) are conjugated to the pentose sugars (ribose or deoxyribose) by a β- glycosidic linkage. This is the configuration in the polymric nucleic acids. The β-glycosidic linkage involves the C-l' of sugar and the hydrogen atom of N-9 (in case of purines) or N-l (in case of pyrimidines), thus eliminating a molecule of water. Therefore, the purine nucleosides are N-9 glycosides and the pyrimidine nucleosides are N-l glycosides. Like the O-glycosides, the nucleosides are stable in alkali. Purine nucleosides are readily hydrolyzed by acid whereas pyrimidine nucleosides are hydrolyzed only after prolonged treatment with concentrated acid.

Table : Nucleosides

Base	Sugar	Nucleoside	Trivial name*	Abbreviation
Ribonucleosides				
Adenine	Ribose	Adenine ribonucleoside	Adenosine	AR
Guanine	Ribose	Guanine ribonucleoside	Guanosine	GR
Cytosine	Ribose	Cytosine ribonucleoside	Cytidine	CR
Thymine	Ribose	Thymine ribonucleoside	Thymidine	TR
Uracil	Ribose	Uracil ribonucleoside	Uridine	UR
Deoxyribo-nucleosides				
Adenine	Deoxyribose	Adenine deoxyribonucleoside	Deoxyadenosine	AdR
Guanine	Deoxyribose	Guanine deoxyribonucleoside	Deoxyguanosine	GdR
Cytosine	Deoxyribose	Cytosine deoxyribonucleoside	Deoxycytidine	CdR
Thymine	Deoxyribose	Thymine deoxyribonucleoside	Deoxythymidine**	TdR
Uracil	Deoxyribose	Uracil deoxyribonucleoside	Deoxyuridine	UdR

Nucleotides

A nucleotide is formed of one molecule of deoxyribose, one molecule of phosphoric acid and one of the four nitrogenous bases. Thus a nucleotide is a **nucleoside monophosphate** (dNMP). Since, there are four nitrogenous bases, there are four types of nucleotides namely:

1. Deoxyadenylic acid or dAMP = Adenine + Deoxyribose + Phosphoric acid
2. Deoxyguanylic acid or dGMP = Guanine + Deoxyribose + Phosphoric acid
3. Deoxycytidylic acid or dCMP = Cytosine + Deoxyribose + Phosphoric acid
4. Deoxythymidylic acid or dTMP = Thymine + Deoxyribose + Phosphoric acid

Nucleotides are Covalently Linked to form Polynucleotide Chain

DNA is a marcromolecule formed by the linking of several thousands of nucleotides. These are called monomers or building blocks of DNA. In a nucleotide, the phosphate (phosphoric acid) molecule is attached to fifth carbon atom (C – 5′) of the deoxyribose molecule. The phosphate molecule of a nucleotide is joined to the third carbon atom of the deoxyribose of the next nucleotide by covalent bond. Such 5′ – 3′ linkages between adjacent nucleotides are called phosphodiester bonds. Thus, the adjacent nucleotides are connected together forming the sugar-phosphate-sugar-phosphate chain in which sugar and phosphate molecules are arranged in alternate fashion. The nitrogenous bases are attached to the first carbon atom of the deoxyribose and are directed at right angle to the long axis of the polynucleotide chain. They are stacked one above the other like the steps in a ladder.

Classification and Function of Nucleic Acid

There are two types of nucleic acids are found, i.e., DNA. and RNA differes at their pentose sugar residues, at their way of organization and their nitrogeneous base composition in some extent along with their function.

Structure and Properties of DNA

Composition and Structure : (Watson and Crick Model).

Nucleic acids are the polynucleotides. There are 2 types of nucleic acids viz DNA and RNA.

DNA is the chemical basis of heredity. Avery, Mac leod and McCarty in 1944 first demonstrated in a series of experiments that DNA contained the genetic information. These authours referred DNA as **"transforming factor".**

Chemistry of DNA

The four types of monomeric units of DNA are

(*i*) adenylate

(*ii*) guanylate

(*iii*) cytidylate and

(*iv*) thymidylate

- The monomeric units form a single strand of DNA which are held in polymeric form by 3', 5'-phosphodiester bridges.
- The informational content of DNA is available in the sequence in which these monomers are discovered.
- The polymer possesses a polarity ; one end has a 5' -hydroxyl or phosphate terminus while other has a 3'-phosphate or hydroxyl moiety.
- Since the genetic information is available in monomeric units within the polymers, there is sentence of a mechanism reproducing of replicating specific information.
- In DNA molecules the concentration of Adenosine (A) nucleotides equals that of thymidine nucleotides (A = T). The concentration of Guanosine (G) nucleotides equals that of cytidine (C) nucleotides (G = C). This accelerates Watson, Crick and Wilkins in 1950 to propose a model of double stranded DNA molecule.
- The 2 strands of this double-stranded molecule are held together by hydrogen bonds between the purine and pyrimidine bases.
- The 2 strands of the double helical molecule are anti-parallel, i.e. one strand runs in the 5' to 3' direction and the other in the 3' to 5' direction. Since the information resides in the sequence of nucleotides on one strand, the opposite strand is considered as "antisense", i.e. the complement of the "sense" strand.
- Three (3) hydrogen bonds hold the guanosine nucleotide to the cytosine nucleotide and A-T pair is held together by 2 hydrogen bonds. Therefore, they are represented as G ≡ C and A = T. G – C bond is stronger by 50%.

- The B form has a pitch of 3.4 nm per turn and within the single turn 10 base pairs exist.
- The double-stranded structure in solution can be melted by increasing temperature of decreasing salt concentration.
- The double-stranded DNA molecule shows the properties of a fiber and it is a viscous material in solution and loses its viscosity upon denaturation.
- In the major and minor grooves winding along the molecule parallel to the phosphodiester backbones, specific proteins interact with DNA molecules.
- In some organisms such as bacteria, bacteriophage and many DNA-containing animal viruses, the two ends of the DNA molecules are jointed to create a closed circle with no terminus. This does not destroy the polarity of the two molecules but it eliminates all 3' and 5' free hydroxyl and phosphoryl groups.
- Heat, acid and alkali denature DNA.

Characteristics of different forms of DNA

Characteristics	A-DNA	B-DNA	C-DNA	Z-DNA
Conditions	75% relative humidity; Na^+, K^+, Cs^+ ions	92% relative humidity; Low ion strength	60% relative humidity; Li^+ ions	Very high salt concentration
Shape	Broadest	Intermediate	Narrow	Narrowest
Helix sense	Right-handed	Right-handed	Right-handed	Left-handed
Helix diameter	25.5Å	23.7Å	19.0Å	18.4Å
Rise per base pair ('h')	2.3Å	3.4Å	3.32Å	3.8Å
Base pairs per turn of helix ('n')	11	10.4	9.33	12 (= 6 dimers)
Helix pitch (h × n)	25.30 Å	35.36Å	30.97Å	45.60Å

DNAs with Unusual Structures

A number of other sequence-dependent structural variations in DNA have been detected that may serve locally important functions in DNA metabolism. These are :

(i) Bent DNA

Some sequences cause bends in the DNA helix. Bends are produced whenever 4 or more adenine residues appear sequentially in one of the two strands. Six adenines in a row produce a bend of about 18°. Bending may be important in the binding of some proteins to DNA.

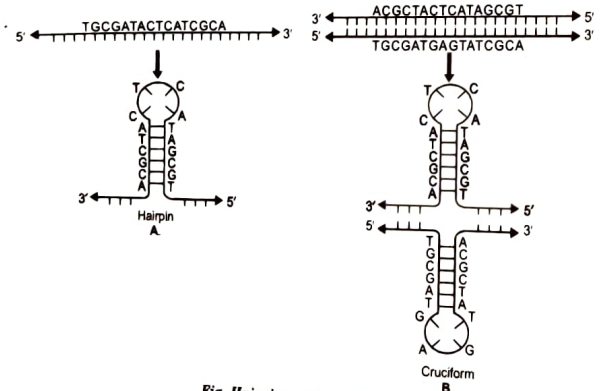


Fig. Hairpins and cruciforms

Bases (enclosed in rectangle) are the asymmetric sequences that can pair alternatively with a complementary sequence in the same or opposite strand.

(a) Hairpin : When only a single strand of palindromic DNA (or RNA) is involved, a hairpin is formed

(b) Cruciform : When both the strands of a double helical DNA are involved, a cruciform is formed.

(ii) Palindromic DNA

The term palindromic DNA is applied to regions of DNA in which there are inverted repetitions of base sequence with two fold symmetry occuring over two strands. Such sequences are self-complementary within each of the strands and therefore have the potential to form hairpin (when only a single strand is involved) or cruciform (when both strands of a duplex DNA are involved) structures, involving intrastrand base pairing.

When the inverted sequence occurs within each individual strand of the DNA, the sequence is called a **mirror repeat**. Mirror repeats do not have complementary sequences is within the same strand and cannot form hairpin or cruciform structures. Sequences of these types are found in virtually every large DNA molecule and can involve a few or up to thousands of base pairs.

Fig. Palindromic DNAs and mirror repeats

(iii) H-DNA

H-DNA is usually found in polypyrimidine or polypurine segments that contain within themselves a mirror repeat. One simple example is a long stretch of alternating T and C residues. A striking feature of H-DNA is the pairing and interwinding of 3 strands of DNA to form a triple helix. Triple-helical DNA is produced spontaneously only within long sequences containing only pyrimidmes (or purines) in one strand. Two of the three strands in the H-DNA triple helix contain pyrimidines and the third contains purines.

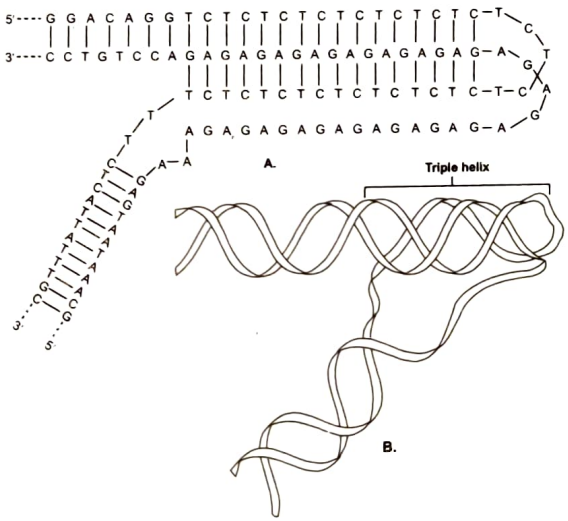


Fig. H-DNA

(iv) Single-stranded DNA

The DNA molecules are not always a double-helical structure. Robert Sinsheimer, in 1959, discovered that the DNA in ϕX174, a small virus that infects Escherichia coli, is single-stranded. His finding was based on the following evidences.

(a) The base ratios (A/T and G/C) of ϕ XI74 DNA do not approach unity.

(b) The ϕ XI74 DNA behaves as a randomly coiled polymer whereas the DNA double helix behaves as a rigid rod.

(c) The amino groups of the bases of ϕ X174 DNA react readily with formaldehyde. In contrast, the bases in the double-helical DNA do not react with this reagent.

The discovery of this single-stranded DNA raised doubts as to whether the semiconservative scheme of replication as proposed by Watson and Crick, is universally true. However, it was soon found that ϕX174 DNA is single-stranded for only a part of its life span and the infected E.*coli* cells contain a double helical form of ϕ X174 DNA. This double helical DNA is called *replicative form*,, since it acts as a template for the synthesis of DNA of the progeny virus. Viruses that contain single-stranded RNA also replicate *via* a double-stranded replicative form.

Table : Differences between double-stranded and single-stranded DNAs

	dsDNA		*ssDNA*
1.	It is linear or filamentous types.	1.	It is somewhat star-shaped (stellate).
2.	DNA double helix behaves as a rigid rod	2.	ϕ X174 DNA behaves as a randomly coiled polymer.
3.	A/T ratio approaches unity.	3.	A/T ratio is 0.77.
4.	G/C ratio also approaches unity.	4.	G/C ratio is 1.3.
5.	It is resistance to the action of formaldehyde.	5.	The amino groups of the bases of ϕ X 174 DNA react readily with formaldehyde.
6.	Ultra-violet absorption increases from 0 to 80°C	6.	Ultra-violet absorption increases from 20° to 90°C.

Properties of DNA

(i) Physical Properties :

(a) Molecular Weight

The DNA. molecules are among the largest known. It is difficult to isolate DNA without fragmentation. The entire DNA of bacteria such as *E. coli* has a molecular weight of 2.6×10^9 and viral DNA range is from about 1 to 350×10^6. The residue weight of a single nucleotide is ≈ 660. Thus there are about 1500 nucleotide per million molecular weight of DNA.

(b) Length

The length by DNA of various organisms are given in following table

Organism Viruses	Number of b.p.	Length in μm.
Polyoma or SV 40	s.1	1.7
λ phage	48.6	17
T 2 phage	166	56
Vaccinia	190	65
Bacteria		
Mycoplasma	760	260
E. coli	4000	1360
Eukaryots		
Yeast	13,500	4,600
Drossophila	1,65,000	56,000
Human	2,900,000	9,90,000

(c) Melting Point DNA

Solutions of carefully isolated, native DNA are highly viscous at pH 7.0 and at 25°C. When such a condition is subjected to extremes of temperature above 80°C, the DNA melts. The temperature at which half of the DNA melts is known as T_m (Melting temperature).

The value of T_m depends upon G ≡ C content of DNA Higher the G ≡ C content, higher will the T_m. The value of T_m also depends upon ionic strength of the solution, lower the ionic strength lower the T_m.

(ii) Chemical Properties

(a) Denaturation and Renaturation of DNA

Renaturation of a DNA molecule is a rapid one-step process as long as a double-helical segment of a dozen or more residues still unites the two strands. When the temperature or pH is, returned to the range in which most organisms live, the unwound segments of the two strands spontaneously rewind or **anneal** to yield the intact duplex. However, if the two strands are completely separated, renaturation occurs in two steps.

In the first, relatively slow steps, the two strands "find" each other by random collisions and form a short, segment of complementary double helix.The second steps, is much faster: the remaining unpaired bases successively come into register as base pairs, and the two strands "zipper" themselves together to form the double helix.

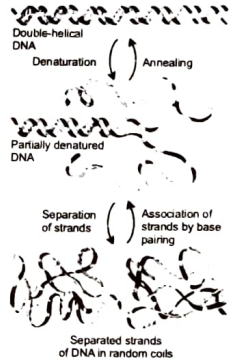


Fig. Reversible denaturation and annealing (renaturation) of DNA.

The close interaction between stacked bases in a nucleic acid has the effect of decreasing its absorption of UV light relative to that of a solution with the same concentration of free nucleotides, and the absorption is decreased further when two complementary nucleic acids are paired. This is called the *hypochromic effect*. Denaturation of a double-stranded nucleic acid produces the opposite result, an increase in absorption called the *hyperchromic effect*. The transition from double-stranded DNA to the single-stranded, denatured form can thus be detected by monitoring the absorption of UV light.

Viral or bacterial DNA molecules in soluction denature when they are heated slowly. Each species of DNA has a characteristic denaturation temperature or melting point (t_m) : the higher its content of G ≡ C base pairs, the higher the melting point of the DNA. This is because G ≡ C base pairs, with three hydrogen bonds, require more heat energy to dissociate than A = T base pairs.

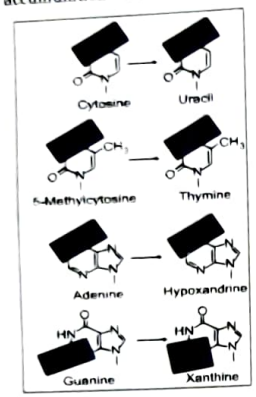
(b) Hybrid Formation :

The ability of two complementary DNA strands to pair with one another can be used to detect similar DNA sequences in two different species or within the genome of a single species. If duplex DNAs isolated from human cells and from mouse cells are completely denatured by heating then mixed and kept at 65°C for many hours, much of the DNA will anneal. Most of the mouse DNA strands anneal with complementary mouse DNA strands to form mouse duplex DNA, similarly, most human DNA strands anneal with complementary human DNA strands.

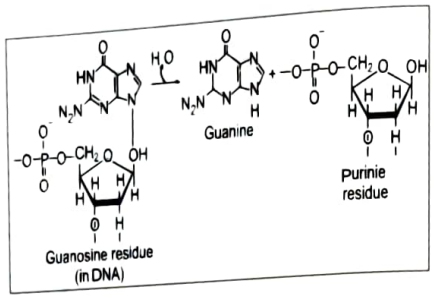
However, some strands of the mouse DNA will associate with human DNA strands to yield **hybrid duplexes,** in which segments of a mouse DNA strand form base-paired regions with segments of a human DNA strand.

(c) Non enzymetic Transformation :

Purines and pyrimidines, along with the nucleotides of which they are a part, undergo a number of spontaneous alterations in their covalent structure. The rate of these reactions is generally very slow, but they are physiologically significant because of the cell's very low tolerance for alterations in its genetic information. Alterations in DNA structure that produce permanent changes in the genetic information encoded therein are called **mutations**, and much evidence suggests an intimate link between the accumulation of mutations and the processes of aging and carcinogenesis.



(a) Deamination



(b) Depurination

(a) Deamination reactions. Only the base is shown. (b) Depurination, in which a purine is lost by hydrolysis of the N-β-glycosyl bond. The deoxyribose remaining after depurination is readily converted from the β-furanose to the aldehyde form

Fig. Some well-characterized nonenzymatic reactions of nucleotides.

The slow cytosine deamination reaction seems innocuous (having no adverse effect) enough, but is almost certainly the reason why DNA contains thymine rather than uracil. The product of cytosine deamination (uracil) is readily recognized as foreign in DNA and is removed by a repair system. If DNA normally contained uracil, recognition of uracils resulting from cytosine deamination would be more difficult, and unrepaired uracils would lead to permanent sequence changes as they were paired with adenines during replication. Cytosine deamination would gradually lead to a decrease in G ≡ C base pairs and an increase in A = U base pairs in the DNA of all cells. Over the millennia, cytosine deamination could eliminate G ≡ C base pairs and the genetic code that depends on them. Establishing thymine as one of the four bases in DNA may well have been one of the crucial turning points in evolution, making the long-term storage of genetic information possible.

Another important reaction in deoxyribonucleotides is the hydrolysis of the *N*-β-glycosyl bond between the base and the pentose (Fig. *b*). This occurs at a higher rate for purines than for pyrimidines. As many as one in 10^5 purines (10,000 pre mammalian cell are) lost from DNA every 24 hours under typical cellular conditions. Depurination of ribonucleotides and RNA is much slower and generally is not considered physiologically significant. In the test tube, loss of purines can be accelerated by dilute acid. Incubation of DNA at pH 3 causes selective removal of the purine bases, resulting in a derivative called apurinic acid.

(d) Methylation

Certain nucleotide based in DNA molecules are enzymatically methylated. Adenine and cytosine are methylated more often than guanine and thymine. Methylation is generally confined to certain sequences or regions of a DNA molecule. In some cases the function of methylation is well understood; in others the function remains unclear. All known DNA methylases use S-adenosylmethionine as a methyl group donor. *E. coli* has two prominent methylation systems.

One serves as part of a defense mechanism that helps the cell to distinguish its DNA from foreign DNA by marking its own DNA with methyl groups and destroying (foreign) DNA without the methyl groups (this is known as a restriction-modification system). The other system methyl adenosine residues within the sequence (5')GATC(3') to N^6-methyladenosine. This is mediated by the Dam (*DNA adenine methylation*) methylase, a component of a system that repairs mismatched base pairs formed occasionally during DNA replication.

***(iii)* Biological Properties**

The amino acid sequence of every protein in a cell, and the nucleotide sequence of every RNA is specified by nucleotide sequence in the cell's DNA. A segment of a DNA molecule that contins the information required for the synthesis of a functional biological product, whether protein or RNA is refferred to as gene. A cell typically has many thousands of *Genes*, and DNA molecule not surprisingly, tend to be very large. The storage and transmission of biological information are the only known biological function of DNA. There are two classical experiments to proove that DNA is a genetic material.

(a) Griffith experiment

In 1928, **Frederick Griffith** encountered a phenomenon now known as genetic transformation. Colonies of virulent strain (pathogenic) of pneumonia causing bacterium, *Diplococcus pneumoniae* grown on nutrient agar, have a **smooth (S)** glistering appearance owing to the presence of a type specific, polysaccharide (a polymer of glucose and glucuronic acid) capsule. The avirulent (non-pathogenic) strains, on the other hand, lack this capsule and they produce dull, **rough (R)** colonies.

In the course of his work, Griffith injected laboratory mice with live R-II pneumococci; the mice suffered no illness because R-II pneumococci was avirulent. But, when the mice were injected with virulent S-III pneumococci, the mice suffered from pneumonia and died. However, when he injected the heat killed S-III bacteria in mice, they did not suffer from pneumonia. But, when the mice were injected with the mixture of living avirulent R-II and heat killed S-III virulent, the unexpected symptoms of pneumonia appeared and high mortality resulted in them. By postmortems the dead mice, it was found that their heart blood had both R-II and S-III pneumococci. From these results, Griffith concluded that the presence of the heat-killed S-III bacteria must have caused a **transformation** of the living R-II bacteria, so as to restore to them the capacity for capsule formation they had earlier lost by gene mutation. This was called **"Griffith effect"** or more popularly **"bacterial transformation".**

Griffith could not understand the cause of bacterial transformation and that is first of all identified by **Oswald Avery, Colin MacLeod** and **Maclyn McCarty** (1944). They partially purified the **transforming principle** from the cell extract (*i.e.*, cell free extract of S-III bacteria) and demonstrated that it was DNA.

(b) Hershey and chase experiment

In their experiment, two batches of isotopically labelled bateriophage T_2 particles were prepared. One was labelled with p^{32} in the phosphate group of DNA, the other with S^{35} in the sulphur containing amino acids of the protein coat (capsid).

The two batches of the labelled phages were then allowed to infect separate suspensions of bacteria. Each suspension of phage-infected cell was agitated in a blender to shear the viral capside from the bacteria. The bacteria and empty viral coats (called ghosts) were then seperated by centrifugation. The cells infected with the ^{32}p-labelled phage were found to contain ^{32}p, indicating that the labelled viral DNA had entered the cell; the viral Ghosts contain no radioactivity.

The cell infected with the ^{35}s-labelled phage were found to have no-radioactivity after blender treatement, but the viral Ghosts, contained ^{35}s. Progeny virus particles were produced in both batches of bacteria sometime after viral coats were removed, indicating that the genetic message for their multiplication had been introduced by viral DNA not by viral protein.

Structure and Properties of RNA

Ribonucleic acid, like DNA is a long, unbranched and usually single stranded macromolecule consisting of nucleotides joined by 3' → 5' phosphodiester bonds. The number of ribonucleotids in RNA ranges from as few as 75 to many thousands.

There are 3 major classes of RNA

(i) Ribosomal RNA :

The term Ribosomal RNA has been proposed by Kurland in 1960. Ribosomal RNA is found in ribosomes of cells and is also called insoluble RNA. It is remarkable to note that rRNA from all sources has G-C content more than 50%. The rRNA molecule appears as a single unbranched polynucleotide strand (primary structure). At low ionic strength the molecule shows a compact rod with random coiling. But at high ionic strength, the molecule reveals the presence of compact helical region with complimentry base pairing and looped outer region. The main function of rRNA is to attract and provide large surface for spreading of m-RNA over ribosomes during translocation process of protein synthesis.

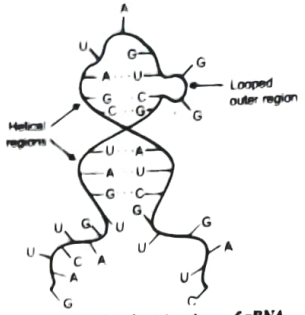


Fig. Molecular structure of rRNA

(ii) Messenger RNA (mRNA) :

- The messenger RNA is single-stranded and complementary to the sense strand of DNA.
- The 5' terminus of messenger RNA is "capped" by a 7- methylguanosine triphosphate which is linked to an 2'- O methyl ribonucleoside at its 5'-hydroxyl through the 3 phosphate. The protein-synthesizing machinery begins translating the mRNA into proteins at the 5' or capped terminus. The 3'-hydroxyl terminus has attached a polymer of adenylate residues 20-250 nucleotides in length.
- It is the most heterogeneous in size and stability.
- It passes from nucleus to cytoplasm conveying information in a gene to the protein-synthesizing machinery where each serves as a template on which a specific sequence of amino acids is polymerized to form a specific protein molecule, the ultimate gene product.
- It has a large molecular weight of 30,000 to 50,000 to have the coded information corresponding to long polypeptide chains.
- In mammalian nuclei, the immediate products of gene transcription are another class of RNA molecule which are quite large and heterogeneous in size. These heterogeneous nuclear RNA (HnRNA) may exceed 10^7 daltons whereas the mRNA molecules are generally smaller than 2×10^6 daltons. These HnRNA molecules are processed to generate the mRNA molecules which then enter the cytoplasm to serve as templates for protein synthesis.

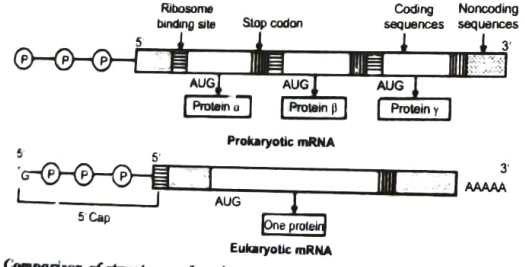


Fig. Comparison of structures of prokaryotic and eukaryotic messenger RNA molecules

(iii) Transfer RNA (tRNA) or soluble RNA (sRNA) :

- The transfer RNA molecules amount to 10 to 20 per cent of the total cellular RNA molecules.
- They consist of about 75 nucleotides and have molecular weight of 25,000.
- There are at least 20 tRNA molecules in every cell, one corresponds to each of the 20 amino acids required for protein synthesis.
- They serve as adaptors for the translation of the information in the sequence of nucleotides of the mRNA into specific amino acids.
- The primay structure allows extensive folding and intrastrand complementarity to generate a significant secondary structure which appears like a cloverleaf as in the figure below.

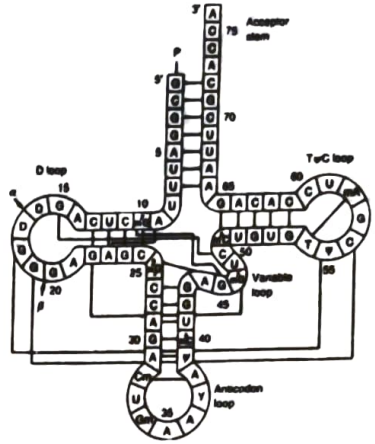


Fig. Cloverleaf structure of tRNA.

- All tRNA molecules have a common CCA sequence at the 3' termini. The carboxyl groups of amino acids are attached to the 3'-hydroxyl group of the adenosyl moiety through an ester bond.
- The *anticodon* loop at the end of a base paired stem recognizes the triplet nucleotide or codon of the template mRNA.
- In nearly all tRNA molecules, there is a loop containing the nucleotides of ribothymine and pseudouridine for binding amino acyl tRNA by ribosome and another loop containing the minor base dihydrouracil recognition of tRNA.
- They are quite stable in prokaryotes and less stable in eukaryotes.

(iv) Tertiary structure of t-RNA :

The tertiary structure of RNA molecules results from base stacking and hydrogen bonding between different parts of the molecule. tRNA provides a number of examples. In solution, tRNA is folded into a compact "L-shaped" conformation, stabilized by Watson-Crick base-pairing and base interactions involving more than two nueleotides. This is true for other RNAs. Figure shows a well-defined base triplet from bacterial ribosomal RNA. Bases can donate hydrogen atoms to bond with the phosphodiester backbone. The 2'-OH of the ribose is also an important donor and acceptor for hydrogen bond formation. All these interactions contribute to the folded shape of an RNA molecule.

Other tertiary features were deduced from the three dimensional structure of RNAs in solution or in crystal form. The term "tetraloop" refers to specific sequences (e.g.,UUCG or GAGA) that are often found at the closed end of RNA hairpins. These tetraloops impart extra stability to RNA hairpins by forming internal hydrogen bonds between based and sugars. Tetraloops can hydrogen bond with regions of the double helix or with internal loops in other parts of an RNA, thereby forming a complex tertiary structure.

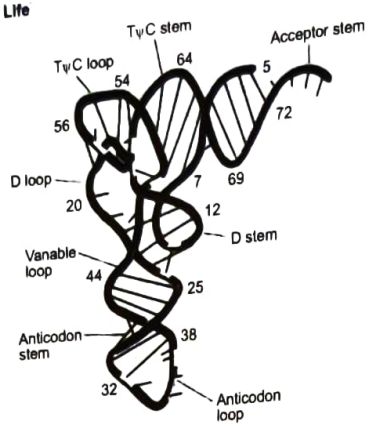


Fig. Tertiary structure on tRNA.

Tertiary folding of the cloverleaf structure of tRNA. Hydrogen bonds are indicated by cross rungs.

Vitamins

Vitamins may be regarded as organic compounds required in the diet in small amounts to perform specific biological functions for normal maintenance of optimum growth and health of the organism.

Classification of vitamins

There are around 16 vitamins, essential for humans. They are classified as fat soluble (A, D, E and K) and water soluble (C and B-group) vitamins. The B-complex vitamins may be sub-divided into energy-releasing (B_1, B_2, B_6, biotin etc.) and hematopoietic (folic acid and B_{12}).

Fat soluble vitamins–general

The four vitamins, namely vitamin A, D, E, and K are known as fat or lipid soluble. Their availability in the diet, absorption and transport are associated with fat. They are soluble in fats and oils and also the fat solvents (alcohol, acetone etc.). Fat soluble vitamins can be stored in liver and adipose tissue.

Water soluble vitamins–general

B-complex vitamins and vitamin C are water soluble vitamins. The water soluble vitamins are a heterogenous group of compounds since they differ chemically from each other. The only common character shared by them is their solubility in water. Most of these vitamins are readily excreted in urine and they are not toxic to the body.

Properties and Functions of Various Vitamins

Vitamin A

The fat soluble vitamin A, as such is present only in foods of animal origin. However, its provitamins carotenes are found in plants.

Chemistry

In the recent years, the term vitamin A is collectively used to represent many structurally related and biologically active molecules.

1. **Retinol (vitamin A alcohol) :** It is a primary alcohol containing *β-ionone ring*. The side chain has two isoprenoid units.
2. **Retinal (vitamin A aldehyde) :** This is an aldehyde form obtained by the oxidation of retinol.
3. **Retinoic acid (vitamin A acid) :** This is produced by the oxidation of retinal.

4. **(β-Carotene (provitamin A) :** This is found in plant foods. It is cleaved in the intestine to produce two moles of retinal. In humans, this conversion is inefficient, hence β-carotene possesses about one-sixth vitamin A activity compared to that of retinol.

Biochemical Functions

Vitamin A is necessary for a variety of functions such as vision, proper growth and differentiation, reproduction and maintenance of epithelial cells.

Vitamin A and vision : The biochemical function of vitamin A in the process of vision was first elucidated by George Wald (Nobel Prize 1968). The events occur in a cyclic process known as **Rhodopsin cycle** or **Wald's visual cycle.**

Other biochemical functions of vitamin A

1. Retinol and retinoic acid function almost like steroid hormones. They regulate the protein synthesis and thus are involved in the cell growth and differentiation.
2. Vitamin A is essential to maintain healthy epithelial tissue.
3. Carotenoids (most important β-carotene) function as antioxidants and reduce the risk of cancers initiated by free radicals and strong oxidants. (β-Carotene is found to be beneficial to prevent heart attacks. This is also attributed to the antioxidant property.

Recommended dietary allowance (RDA)

The RDA of vitamin A for adults is around 1,000 retinol equivalents (5,000 IU) for man and 800 retinol equivalents (4,000 IU) for woman. One International Unit (IU) equals to 0.3 μg of retinol. The requirement increases in growing children, pregnant women and lactating mothers.

Dietary sources

Animal sources contain (preformed) vitamin A. The best sources are liver, kidney, egg yolk, milk, cheese, butter. Fish (cod or shark) liver oils are very rich in vitamin A.

Vegetable sources contain the provitamin A-carotenes. Yellow and dark green vegetables and fruits are good sources of carotenes e.g. carrots, spinach, amaranthus, pumpkins, mango, papaya etc.

Vitamin A deficiency

Deficiency manifestations of the eyes : Night blindness (nyctalopia) is one of the earliest symptoms of vitamin A deficiency. The individuals have difficulty to see in dim light.

Severe deficiency of vitamin A leads to ***xerophthalmia***. This is characterized by dryness in conjunctiva and cornea and keratinization of epithelial cells. In certain areas of conjunctiva, white triangular plaques known as Bitot's spots are seen.

If xerophthalmia persisits for a long time, corneal ulceration and degeneration occur. This results in the destruction of cornea, a condition referred to as ***keratomalacia***, causing total blindness. Vitamin A deficiency blindness is most common in children of the developing countries.

Vitamin D

Vitamin D is a fat soluble vitamin. It resembles sterols in structure and functions like a hormone.

Chemistry

Cholecalciferol (vitamin D_3) is found in animals. Ergocalciferol (vitamin D_2) is formed from ergosterol and is present in plants. Both the sterols are similar in structure except that ergocalciferol has an additional methyl group and a double bond. ***Ergocalciferol*** and ***cholecalciferol*** are sources for vitamin D activity and are referred to as ***provitamins***.

During the course of cholesterol biosynthesis 7-dehydrocholesterol is formed as an intermediate. On exposure to sunlight, 7-dehydro- cholesterol is converted to cholecalciferol in the skin (dermis and epidermis).

Metabolism and Biochemical Functions

Vitamins D_2 and D_3 as such are not biologically active. They are metabolized identically in the body and converted to active forms of vitamin D.

Biochemical function

Calcitriol (1, 25 DHCC) is the biologically active form of vitamin D. It ***regulates the plasma levels of calcium and phosphate***. Calcitriol acts at 3 different levels (intestine, kidney and bone) to maintain plasma calcium.

1. **Action of calcitriol on the intestine :** Calcitriol increases the intestinal absorption of calcium and phosphate. In the intestinal cells, calcitriol binds with a cytosolic receptor to form a calcitriol-receptor complex. This complex then approaches the nucleus and interacts with a specific DNA leading to the synthesis of a specific ***calcium binding protein***. This protein increases the calcium uptake by the intestine.
2. **Action of calcitriol on the bone :** In the osteoblasts of bone, calcitriol stimulates calcium uptake for deposition as calcium phosphate. Thus calcitriol is essential for bone formation.
3. **Action of calcitriol on the kidney :** Calcitriol is also involved in minimizing the excretion of calcium and phosphate through the kidney, by decreasing their excretion and enhancing reabsorption.

Recommended dietary allowance (RDA)

The daily requirement of vitamin D is 400 International Units or 10 μg of cholecalciferol. In countries with good sunlight (like India), the RDA for vitamin D is 200 IU (or 5 μg cholecalciferol).

Dietary sources

Good sources of vitamin D include fatty fish, fish liver oils, egg yolk etc. Milk is not a good source of vitamin D.

Deficiency symtoms

Deficiency of vitamin D leads to demineralization of bone. The result is ***rickets in children and osteomalacia in adults.***

Rickets in children is characterized by bone deformities due to incomplete mineralization, resulting in soft and pliable bones and delay in teeth formation. In rickets, the plasma level of calcitriol is decreased and ***alkaline phosphatase activity is elevated.***

In case of osteomalacia (adult rickets) demineralization of the bones occurs (bones become softer), increasing their susceptibility to fractures.

Renal rickets (renal osteodystrophy)

This is seen in patients with chronic renal failure. Renal rickets is mainly due to decreased synthesis of calcitriol in kidney. It can be treated by administration of calcitriol.

Vitamin E

Vitamin E (tocopherol) is a naturally occurring antioxidant. It is essential for normal reproduction in many animals, hence known as ***anti-sterility vitamin***.

Chemistry

Vitamin E is the name given to a group of tocopherols, possessing vitamin E activity. About eight tocopherols have been identified–α, β, γ, δ etc. Among these, ***α-tocopherol*** is the most active. The tocopherols are derivatives of ***6-hydroxy chromane (tocol) ring*** with isoprenoid (3 units) side chain. The antioxidant property is due to the chromane ring.

Biochemical functions

Most of the functions of vitamin E are related to its ***antioxidant*** property.

The biochemical functions of vitamin E, related either directly or indirectly to its antioxidant property, are given hereunder

1. Vitamin E is essential for the membrane structure and integrity of the cell
2. It prevents the peroxidation of polyunsaturated fatty acids in various tissues and membranes i t protects RBC from hemolysis by oxidizing agents (e.g., H_2O_2).
3. It is closely associated with reproductive functions and prevents sterility. Vitamin E preserves and maintains germinal epithelium of gonads for proper reproductive function.
4. Vitamin E protects liver from being damaged by toxic compounds such as carbon tetrachloride.
5. It works in association with vitamins A, C and β-carotene, to delay the onset of cataract.

6. Recent studies have shown that high intake of vitamin E (about 200-300 mg/day) protects against the development of heart diseases. It is believed that vitamin E prevents the oxidation of LDL. The oxidized LDL have been implicated to promote heart diseases.

Recommended dietary allowance (RDA)

A daily consumption of about 10 mg (15 IU) of α-tocopherol for man and 8 mg (12 IU) for woman is recommended. One mg of α-tocopherol is equal to 1.5 IU. Vitamin E supplemented diet is advised for pregnant and lactating women.

Dietary sources

Many vegetable oils are rich sources of vitamin E. Wheat germ oil, cotton seed oil, peanut oil, corn oil and sunflower oil are the good sources of this vitamin. It is also present in meat, milk, butter and eggs.

Deficiency symtoms

The symptoms of vitamin E deficiency vary from one animal species to another. In many animals, the deficiency is associated with sterility, degenerative changes in muscle, megaloblastic anaemia and changes in central nervous system.

Toxicity of vitamin E

Among the fat soluble vitamins (A, D, E, K), vitamin E is the least toxic. No toxic effect has been reported even after ingestion of 300 mg/day for 23 years.

Vitamin K

Vitamin K is the only fat soluble vitamin with a specific coenzyme function. It is required for the production of blood clotting factors, essential for coagulation.

Chemistry

Vitamin K exists in different forms. Vitamin K_1, (phylloquinone) is present in plants. Vitamin K_2, (menaquinone) is produced by the intestinal bacteria and also found in animals. Vitamin K_3 (menadione) is a synthetic form.

All the three vitamins (K_1, K_2, K_3) are ***naphthoquinone derivatives***. Isoprenoid side chain is present in vitamins K_1 and K_2. All the three are stable to heat. Their activity is, however, lost by oxidizing agents, irradiation, strong acids and alkalies.

Biochemical functions

The functions of vitamin K are concerned with blood clotting process. It brings about the ***post-translational*** (after protein biosynthesis in the cell) ***modification of certain blood clotting factors.*** The clotting factors **II** (prothrombin), **VII, IX** and **X** are synthesized as inactive precursors (zymogens) in the liver. Vitamin K acts as a coenzyme for the ***carboxylation of glutamic acid*** residues present in the proteins and this reaction is catalysed by a carboxylase (microsomal).

Recommended dietary allowance (RDA)

Strictly speaking, there is no RDA for vitamin K, since it can be adequately synthesized in the gut. It is however, recommended that half of the body requirement is provided in the diet, while the other half is met from the bacterial synthesis. Accordingly, the suggested RDA for an adult is 70-140 μg/day.

Dietary sources

Cabbage, cauliflower, tomatoes, alfa alfa, spinach and other green vegetables are good sources. It is also present in egg yolk, meat, liver, cheese and dairy products.

Deficiency symptoms

The deficiency of vitamin K is uncommon, since it is present in the diet in sufficient quantity and/or is adequately synthesized by the intestinal bacteria.

Deficiency of vitamin K leads to the lack of active prothrombin in the circulation. The result is that blood coagulation is adversely affected. The individual bleeds profusely even for minor injuries. ***The blood clotting time is increased.***

Antagonists of vitamin K

The compounds—namely heparin, bishydroxy-coumarin—act as anticoagulants and are antagonists to vitamin K. The salicylates and dicumarol are also antagonists to vitamin K. Dicumarol is structurally related to vitamin K and acts as a competitive inhibitor in the synthesis of active prothrombin.

Ascorbic Acid (Vitamin C)

Vitamin C is a water soluble versatile vitamin. It plays an important role in human health and disease.

Biosynthesis and metabolism

Many animals can synthesize ascorbic acid from glucose via uronic acid pathway. However, ***man***, other primates, guinea pigs and bats cannot synthesize ascorbic acid due to the ***deficiency of*** a single enzyme namely ***L-gulono-lactone oxidase***.

Biochemical functions

Most of the functions of vitamin C are related to its property to undergo reversible oxidation–reduction *i.e.*, interconversion of ascorbic acid and dehydroascorbic acid.

1. Collagen formation : Vitamin C plays the role of a coenzyme in hydroxylation of proline and lysine while protocollagen is converted to collagen (i.e. post-translational modification). The hydroxylation reaction is catalysed by ***lysyl hydroxylase*** (for lysine) and ***prolyl hydroxylase*** (for proline). This reaction is dependent on vitamin C.
2. Bone formation
3. Iron and hemoglobin metabolism
4. Tryptophan metabolism
5. Tyrosine metabolism
6. Folic acid metabolism
7. Synthesis of corticosteroid hormones
8. Sparing action of other vitamins
9. Immunlogical function
10. Preventive action on chronic diseases

Recommended dietary allowance (RDA)

About 60-70 mg vitamin C intake per day will meet the adult requirement. Additional intakes (20-40% increase) are recommended for women during pregnancy and lactation.

Dietary sources

Citrus fruits, gooseberry (amla), guava, green vegetables (cabbage, spinach), tomatoes, potatoes (particularly skin) are rich in ascorbic acid. High content of vitamin C is found in adrenal gland and gonads. Milk is a poor source of ascorbic acid.

Deficiency symptoms

The deficiency of ascorbic acid results in ***scurvy***. This disease is characterized by spongy and sore gums, loose teeth, anemia, swollen joints decreased immunocompetence, delayed wound healing, haemorrhage, osteoporosis etc.

Thiamine (Vitamin B_1)

Thiamine (anti-beri beri or antineuritic vitamin) is water soluble. It has a specific coenzyme, ***thiamine pyrophosphate (TPP)*** which is mostly associated with carbohydrate metabolism.

Biochemical functions

The coenzyme, thiamine pyrophosphate or cocarboxylase is intimately connected with the energy releasing reactions in the carbohydrate metabolism.

1. The enzyme ***pyruvate dehydrogenase*** catalyses (oxidative decarboxylation) the irreversible conversion of pyruvate to acetyl CoA. This reaction is dependent on TPP, besides the other coenzymes.
2. **α-Ketoglutarate dehydrogenase** is an enzyme of the citric acid cycle. This enzyme is comparable with pyruvate dehydrogenase and requires TPP.
3. **Transketolase** is dependent on TPP. This is an enzyme of the hexose monophosphate shunt.

4. **The branched chain α-keto acid dehydrogenase (decarboxylase)** catalyses the oxidative decarboxylation of branched chain amino acids (valine, leucine and isoleucine) to the respective keto acids. This enzyme also requires TPP.
5. TPP plays an important role in the **transmission of nerve impulse**. It is believed that TPP is required for acetylcholine synthesis and the ion translocation of neural tissue.

Recommended dietary allowance (RDA)

The daily requirement of thiamine depends on the intake of carbohydrate. A dietary supply of 1-1.5 mg/day is recommended for adults (about 0.5 mg/1,000 Cals of energy).

Dietary sources

Cereals, pulses, oil seeds, nuts and yeast are good sources. Polishing of rice removes about 80% of thiamine. Vitamin B_1 is also present in animal foods like pork, liver, heart, kidney, milk etc.

Deficiency symptoms

The deficiency of vitamin B_1, results in a condition called ***beri-beri*** [Sinhalese : I cannot (said twice)]. Beri-beri is mostly seen in populations consuming exclusively polished rice as staple food. The early symptoms of thiamine deficiency are loss of appetite (anorexia), weakness, constipation, nausea, mental depression, peripheral neuropathy, irritability etc. Numbness in the legs complaints of 'pins and needles sensations' are reported.

Carbohydrate metabolism is impaired. Accumulation of ***pyruvate*** occurs in the tissues which is harmful. Pyruvate concentration in plasma is elevated and it is also excreted in urine.

Thiamine antagonists

Pyrithiamine and ***oxythiamine*** are the two important antimetabolites of thiamine.

Riboflavin (Vitamin B_2)

Riboflavin through its coenzymes takes part in a variety of cellular oxidation–reduction reactions.

Coenzymes of riboflavin

Flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD) are the two coenzyme forms of riboflavin. The ribitol (5 carbon) is linked to a phosphate in FMN. FAD is formed from FMN by the transfer of an AMP moiety from ATP.

Biochemical functions

The flavin coenzymes (mostly FAD and to a lesser extent FMN) participate in many ***redox reactions*** responsible for energy production. The functional unit of both the coenzymes is isoalloxazine ring which serves as an acceptor of two hydrogen atoms (with electrons). FMN or FAD undergo identical reversible reactions accepting two hydrogen atoms forming $FMNH_2$ or $FADH_2$.

Enzymes that use flavin coenzymes (FMN or FAD) are called ***flavoproteins***. Many flavoproteins contain metal atoms (iron, molybdenum etc.) which are known as ***metalloflavoproteins***.

The coenzymes, FAD and FMN are associated with several enzymes involved in carbohydrate, lipid, protein and purine metabolisms, besides the electron transport chain.

Recommended dietary allowance (RDA)

The daily requirement of riboflavin for an adult is 1.2-1.7 mg. Higher intakes (by 0.2-0.5 mg/day) are advised for pregnant and lactating women.

Dietary sources

Milk and milk products, meat, eggs, liver, kidney are rich sources. Cereals, fruits, vegetables and fish are moderate sources.

Deficiency symptoms

Riboflavin deficiency symptoms include ***cheilosis*** (fissures at the corner of the mouth), ***glossitis*** (tongue smooth and purplish) and ***dermatitis***. Assay of the enzyme ***glutathione reductase*** in erythrocytes will be useful in assessing the riboflavin deficiency.

Antimetabolite : Galactoflavin is an antimeta bolite of riboflavin.

Niacin

Niacin or nicotinic acid is also known as pellagra preventive (P.P.) factor of Goldberg. The coenzymes of niacin (NAD^+ and $NADP^+$) can be synthesized by the essential amino acid, tryptophan.

Recommended dietary allowance (RDA)

The daily requirement of niacin for an adult is 15-20 mg and for children, around 10-15 mg. Very often, the term niacin equivalents (NE) is used while expressing its RDA. One NE = 1 mg niacin or 60 mg of tryptophan.

Dietary sources

The rich natural sources of niacin include liver, yeast, whole grains, cereals, pulses like beans and peanuts. Milk, fish, eggs and vegetables are moderate sources. The essential amino acid tryptophan can serve as a precursor for the synthesis of nicotinamide coenzymes. On an average, 1 g of a good quality protein containing about 60 mg of tryptophan is equivalent to 1 mg of niacin ***(conversion ratio 60 : 1)*** for the synthesis of nicotinamide coenzymes.

Deficiency symptoms

Niacin deficiency results in a condition called pellagra (Italian: rough skin). The disease ***pellagra*** involves skin, gastrointestinal tract and central nervous system. The symptoms of pellagra are commonly referred to as ***three Ds***. The disease also progresses in that order ***dermatitis, diarrhea, dementia,*** and if not treated may rarely lead to death (4th D).

Pyridoxine (Vitamin B_6)

Vitamin B_6 is used to collectively represent the three compounds namely ***pyridoxine, pyridoxal and pyridoxamine***.

Recommended dietary allowance (RDA)

The requirement of pyridoxine for an adult is 2-2.2 mg/day. During pregnancy, lactation and old age, an intake of 2.5 mg/day is recommended.

Dietary sources

Animal sources such as egg yolk, fish, milk, meat are rich in B_6. Wheat, corn, cabbage, roots and tubers are good vegetable sources.

Deficiency Symptoms

Pyridoxine deficiency is associated with ***neurological symptoms*** such as depression, irritability, nervousness and mental confusion. Convulsions and peripheral neuropathy are observed in severe deficiency. These symptoms are related to the decreased synthesis of biogenic amines (serotonin, GABA, norepinephrine and epinephrine). Demyelination of neurons is also observed in B_6 deficiency.

Pyridoxine antagonists

Isoniazid, deoxypyridoxine and methoxy pyridoxine are the antagonists of vitamin B_6.

Biotin

Biotin (formerly known as anti-egg white injury factor, vitamin B_7 or vitamin H) is a sulfur containing B-complex vitamin. It directly participates as a coenzyme in the carboxylation reactions.

Williams and his associates (1940) showed that the egg white injury in rats and chicks was due to the presence of an anti-vitamin in egg white. The ***egg-white injury factor*** was identified as a glycoprotein–***avidin*** and ***biotin*** was called as ***anti-egg white injury factor.***

Recommended dietary allowance (RDA)

A daily intake of about 100-300 µg is recommended for adults. In fact, biotin is normally synthesized by the intestinal bacteria. However, to what extent the synthesized biotin contributes to the body requirements is not clearly known.

Dietary sources

Biotin is widely distributed in both animal and plant foods. The rich sources are liver, kidney, egg yolk, milk, tomatoes, grains etc.

Deficiency symptoms

The symptoms of biotin deficiency include anemia, loss of appetite, nausea, dermatitis, glossitis etc.

Biotin ***deficiency is uncommon***, since it is well distributed in foods and also supplied by the intestinal bacteria.

Antagonists

Desthiobiotin, biotin sulphonic acid are biotin antagonists.

Pantothenic Acid

Pantothenic acid (Greek : pantos-everywhere), formerly known as chick anti-dematitis factor (or filtrate factor) is widely distributed in nature. It's metabolic role as ***coenzyme A*** is also widespread.

Recommended dietary allowance (RDA)

The requirement of pantothenic acid for humans is not clearly known. A daily intake of about 5-10 mg is advised for adults.

Dietary sources

Pantothenic acid is one of the most widely distributed vitamins found in plants and animals. The rich sources are egg, liver, meat, yeast, milk etc.

Deficiency symptoms

Pantothenic acid deficiency manifestations have not been reported in humans. Dr. Gopalan, a world renowned nutritionist from India, linked the ***burning feet syndrome*** (pain and numbness in the toes, sleeplessness, fatigue etc.) with pantothenic acid deficiency.

Folic Acid

Folic acid or folacin (Latin : folium-leaf) is abundantly found in green leafy vegetables. It is important for one carbon metabolism and is required for the synthesis of certain amino acids, purines and the pyrimidine—thymine.

Recommended dietary allowance (RDA)

The daily requirement of folic acid is around 100 µg. In the women, higher intakes are recommended during pregnancy (300 µg/day) and lactation (150 µg/day).

Dietary sources

Folic acid is widely distributed in nature. The rich sources are green leafy vegetables, whole grains, cereals, liver, kidney, yeast and eggs. Milk is rather a poor source of folic acid.

Deficiency symptoms

Folic acid deficiency is probably the most common vitamin deficiency, observed primarily in the pregnant women, in both developed (including USA) and developing countries (including India).

The ***macrocytic anemia*** (abnormally large RBC) associated with megaloblastic changes in bone marrow is a characteristic feature of folate deficiency.

Formiminoglutamate (FIGLU), formed in histidine metabolism ***accumulates and is excreted in urine***. Histidine load test utilizing the excretion of FIGLU in urine is used to assess folic acid deficiency.

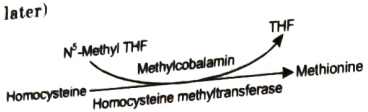
Folic acid antagonists

Aminopterin and amethopterin (also called as methotrexate) are structural analogues of folic acid. They competitively ***inhibit dihydrofolate reductase*** and block the formation of THF. The biosynthesis of purines, thymine nucleotides and hence DNA is impaired. This results in the blockage of cell proliferation. ***Aminopterin and methotrexate are successfully used in the treatment of many cancers, including leukemia.***

Cynocobalamin (Vitamin B_{12})

Vitamin B_{12} is also known as ***anti-pernicious anemia vitamin***. It is a unique vitamin, synthesized by only microorganisms and not by animals and plants. It was the last vitamin to be discovered.

1. Synthesis of methionine from homocysteine : Vitamin B_{12} as methylcobalamin is used in this reaction. This is an important reaction involving N^5-methyl tetrahydrofolate from which tetrahydrofolate is liberated. This metabolic step signifies the interrelation between vitamin B_{12} and folic acid (for details, see later)



2. Isomerization of methymalonyl CoA to succinyl CoA : The degradation of odd chain fatty acids, certain amino acids (valine, isoleucine etc.) and pyrimidines (thymine and uracil) produce directly or through the mediation of propionyl CoA, an improtant compound methylmalonyl CoA.

Recommended dietary allowance (RDA)

A daily intake of about 3 μg of vitamin B_{12} is adequate to meet the adult requirements. For children, 0.5-1.5 μg/day is recommended. During pregnancy and lactation, the requirement is 4 μg/day.

Dietary sources

Foods of animal origin are the only sources for vitamin B_{12}. The rich sources are liver, kidney, milk, curd, eggs, fish, pork and chicken. Curd is a better source than milk, due to the synthesis of B_{12} by *Lactobacillus*.

Vitamin B_{12} is synthesized only by microorganisms (anaerobic bacteria). Plants cannot synthesize, hence B_{12} is never found in plant foods. Animals obtain B_{12} either by eating foods, derived from other animals or from the intestinal bacterial synthesis.

Deficiency symptoms

The most important disease associated with vitamin B_{12} deficiency is ***pernicious anemia***. It is characterized by low hemoglobin levels, decreased number of erythrocytes and neurological manifestations.

The excretion of ***methylmalonic acid*** (elevated) in urine and estimation of serum B_{12} level are used to assess B_{12} deficiency.

Treatment

Vitamin B_{12}, is administered in therapeutic doses (100-1000μg) intramuscularly. Folic acid administration can also reverse hematological abnormalities observed in B_{12} deficiency. However, the neurological symptoms persist. Therefore, a combined supplementation of B_{12} and folate is employed to treat the patients with megaloblastic anemias.

Vitamin Like Compounds

Besides the vitamins described above, there are m-any other compounds present in foods as accessory factors. Although not essential in the diet, they perform many important functions in the body.

Choline

Best and Huntsman (1934) found that choline deficiency in rats produced fatty liver and established its nutritional importance. However, in a strict sense, choline is not a vitamin, since it is adequately synthesized in the body (from serine). It is also available from many dietary sources (e.g. milk, eggs, liver, cereals etc.).

Biochemical functions

1. Choline, as a component of **phospholipids** (lecithins) is involved in membrane structure and lipid transport.

2. Choline prevents the accumulation of fat in liver (as **lipotropic factor**). It promotes the synthesis of phospholipids and lipoproteins and the disposal of **triacylglycerols** from liver.
3. Due to the presence of three methyl groups (one carbon fragments), choline is actively involved in **one carbon metabolism**.
4. Choline is a precursor for the synthesis of acetlcholine which is required for transmission of nerve impulse.

Inositol

Inositol is ***hexahydroxy-cyclohexane***. It is also known as myo-inositol or meso-inositol.

Biochemical functions

1. Inositol is required for the synthesis of ***phosphatidylinositol*** (lipositol) which is a constituent of cell membrane.
2. It acts as lipotropic factor (along with choline) and prevents the accumulation of fat in liver.
3. For some hormones, inositol acts as a ***second messenger*** at the membrane level for the release of Ca^{2+} ions.
4. Inositol concentration in the heart muscle in high, the significance of which however, is not known.
5. Phytin is hexaphosphate of inositol found in plants. It prevents the absorption of iron and calcium from the intestine.

Lipoic Acid

Lipoic acid (thioctic acid) is a sulfur containing fatty acid (6, 8-dithiooctanoic acid). It exists in an oxidized and reduced form. Lipoic acid is fat as well as water soluble.

Biochemical functions

Lipoic acid is involved in the decarboxylation reactions along with other vitamins (thiamine, niacin, riboflavin and pantothenic acid). The conversion of pyruvate to acetyl CoA and α-ketoglutarate to succinyl CoA require lipoic acid.

Para Aminobenzoic Acid

Para aminobenzoic acid (PABA) is a structural constituent of folic acid. PABA may be regarded as a ***vitamin in another vitamin*** (folic acid)

The specific functions of PABA in humans, except that it is a component of folic acid, have not been identified.

PABA is synthesized by the bacteria and is essential for their growth. The sulfa drug sulfanilamide (p-amino benzene sulfanilamide) is a structural analogue of PABA. Sulfanilamide competes with PABA and acts as a bacteriostatic agent. Ingestion of large doses of PABA will compete with the action of drugs and therefore should be avoided during sulfanilamide therapy (trade name—sulfonamides).

Bioflavonoids

Szent-Gyorgi and his associates (1936) observed that flavonoids, isolated from lemon peel (known as citrin) were responsible for maintenance of normal capillary permeability. The term ***vitamin P*** (P for permeability) was used to this group of substances. However, they are commonly known as bioflavonoids. Bioflavonoids act as antioxidants and protect ascorbic acid from being destroyed.

Antivitamins

Antivitamins are antagonistic to (Oppose and block) the action of vitamins. They usually have structural similarities with vitamins. Administration of antivitamins causes vitamin deficiencies. The common antivitamins are already discussed as antagonists for each vitamin.

EXERCISE - I

MCQ TYPE QUESTIONS

1. Which one of the following is the sweetest sugar?
   (a) Fructose (b) Glucose
   (c) Galactose (d) Sucrose
2. In which form does the food transported in plants
   (a) Sucrose (b) Fructose
   (c) Glucose (d) Lactose
3. Which of the following fats is least harmful for heart?
   (a) Saturated fat (b) Cholesterol
   (c) Polyunsaturated fat (d) Oils
4. DNA is not present in one of the following
   (a) Mitochondria (b) Chloroplast
   (c) Bacteriophage (d) Tobacco mosaic virus
5. DNA strands are antiparallel because of the presence of
   (a) H-bonds
   (b) Peptide bonds
   (c) Disulphide bonds
   (d) Phosphate-diester bonds
6. Which is wrong about nucleic acids?
   (a) DNA is single stranded in some viruses
   (b) RNA is double stranded occasionally
   (c) Length of one helix is 45 Å in B-DNA
   (d) One turn of Z-DNA has 12 bases
7. The nuclease enzyme, which begins its attack from free end of a polynucleotide, is
   (a) Polymerase (b) Endonuclease
   (c) Exonuclease (d) Kinase
8. The enzyme enterokinase helps in the conversion of
   (a) Caseinogen into casein
   (b) Trypsinogen into trypsin
   (c) Pepsinogen into pepsin
   (d) Proteins into polypeptides
9. Cellulose, the most important constituent of plant cell wall is made up of
   (a) Branched chain of glucose molecule linked by β-1,4 glycosidic bond in straight chain and α-1,6 glycosidic bond at the site of branching
   (b) Unbranched chain of glucose molecules linked by β -1,4 glycosidic bond
   (c) Branched chain of glucose molecule linked by α-1,6 glycosidic bond at the site of branching
   (d) Unbranched chain of glucose molecules linked by α-1,6 glycosidic bond
10. Some of the enzymes, which are associated in converting fats into carbohydrates, are present in
    (a) Microsomes (b) Glyoxysomes
    (c) Liposomes (d) Golgi bodies
11. Which of the following has carbohydrate as prosthetic group?
    (a) Glycoprotein (b) Chromoprotein
    (c) Lipoprotein (d) Nucleoprotein
12. The major role on minor elements inside living organisms is to act as
    (a) Co-factors of enzymes
    (b) Building blocks of important amino acids
    (c) Constituent of hormones
    (d) Binder of cell structure
13. Carbohydrates are commonly found as starch in plants storage organs. Which of the following five properties of starch make it useful as a storage material?
    (1) Easily translocated
    (2) Chemically non-reactive
    (3) Easily digested by animals
    (4) osmotically inactive
    (5) Synthesized during photosynthesis
    The useful properties are
    (a) (1), (3) and (5) (b) (1) and (5)
    (c) (2) and (3) (d) (2) and (4)
14. Steroid hormones are almost similar in structure to
    (a) Triglyceride
    (b) Tyrosine
    (c) Coenzyme-A
    (d) Cholesterol
15. Glycosidic link is broken in digestion of
    (a) Starch (b) Protein
    (c) Lipid (d) All of these
16. The high energy bonds of ATP are between
    (a) C-O (b) O-P
    (c) C-N (d) C-C
17. The specificity of any protein and its physical and enzymatic properties depends upon
    (a) Absence of amino acids
    (b) Linear sequence of amino acids
    (c) Amino acid without any sequence
    (d) Number of amino acids
18. High energy compounds
    (a) Are produced, when ATP loses two of its phosphate groups
    (b) Are compounds, which link exergonic and endergonic process
    (c) Are produced in respiration only
    (d) Refer the oxidation of sugar, which releases large energy during respiration
19. The environmental factor that influences the elongation of polypeptide chain is
    (a) humidity (b) oxygen
    (c) light (d) temperature
20. Arachidonic Acid is
    (a) non essential fatty acid
    (b) polyunsaturated fatty acid
    (c) both A and B
    (d) none of the above
21. The numbers of molecule of fatty acid present in one molecule of fat is
    (a) one (b) two
    (c) three (d) four
22. Which of the following amino acids is not optically active?
    (a) glycine (b) valine
    (c) leucine (d) isoleucine
23. Which of the following can synthesize all of the amino acids ?
    (a) Animals (b) Plants
    (c) both A and B (d) None of these
24. Calmodulin is a
    (a) chlorophyll binding protein
    (b) cadmium binding protein
    (c) calcium binding protein
    (d) carotene binding protein
25. Which of the following vitamin is required for activation or act as coenzyme of dehydrogenases?
    (a) vitamin A (b) vitamin B
    (c) vitamin C (d) vitamin K
26. Which one of the following is an example of a conjugated protein?
    (a) Casein (b) Albumin
    (c) Globulin (d) Globin
27. Which of the following minerals needed for the activity of enzymes, required for the synthesis of oligosaccharides and glycoproteins?
    (a) Mn (b) Mg
    (c) Cu (d) Mo

28. The major fibrous proteins that provide external protection to vertebrates are
    (a) collagen (b) elastin
    (c) keratins (d) none of these
29. Nucleic acids have multiple negative charges due to
    (a) sugars
    (b) phosporyl groups
    (c) associated protein
    (d) purine and pyrimidines
30. Which one of the following is an example of glycoprotein?
    (a) haemoglobin (b) lecithin
    (c) mucin (d) casein
31. The spatial arrangement produced by the twisting and folding of the peptide chains in a protein is referred to as its
    (a) primary structure (b) secondary structure
    (c) tertiary structure (b) quaternary structure
32. Cholestrol is the precursor of
    (a) progesterone
    (b) testosterone
    (c) estradiol and cortisol
    (d) all of the above
33. Several plants that depend upon animals for seed dispersal accumulate sugar(s) in their ripe fruits. The sugar most widely used for the purpose is
    (a) sucrose (b) lactose
    (c) fructose (d) trehalose
34. The stored form of sugar in animals is a
    (a) homopolysaccharide
    (b) heteroplysaccharide
    (c) oligosaccharide
    (d) disaccharide
35. Vitamin containing a cobalt cyanide linkage is
    (a) A (b) B1
    (c) B6 (d) B12
36. Vitamin B_{12} is available to ruminants by
    (a) Plants
    (b) Micro-organism in caecum
    (c) Animals
    (d) All of the above
37. Recently discovered vitamin having anti-cancer properties is
    (a) Vitamin B5 (b) Vitamin B15
    (c) Vitamin B17 (d) Vitamin Q

38. Prolonged deficiency of nicotinic acid in human diet may lead to
(a) Beriberi (b) Pellagra
(c) Scurvy (d) Rickets

39. Burning feet syndrome is due to deficiency of
(a) Vitamin D (b) Vitamin A
(c) Vitamin B3 (d) Vitamin B5

40. The disease due to inflammation of vermiform appendix of digestive system is known as
(a) Amoebic dysentery (b) Intestinal cancer
(c) Appendicitis (d) None of the above

41. Mental retardation in children suffering from galactosemia can be avoided by
(a) Giving them more milk
(b) Giving them milk free diet
(c) Giving them fortified with vitamins
(d) Giving them more proteinous diet

42. Anti-infection vitamin is
(a) B12 (b) A
(c) D (d) K

43. Vitamin essential for formation of collagen is
(a) A (b) E
(c) B12 (d) C

44. Which vitamin is destroyed by ultraviolet rays
(a) A (b) D
(c) E (d) K

45. E.coli in human colon behave as
(a) Parasite (b) Commensal
(c) Saprophyte (d) Mutualism

46. Vitamin which helps in carbohydrate metabolism is
(a) Riboflavin (b) Thiamine
(c) Folic acid (d) Pantothenic acid

47. Cow's milk is slightly yellowish in colour due to the presence of
(a) Carotene
(b) Riboflavin
(c) Xanthophyll
(d) Xanthophyll and Carotene

48. Which combination is incorrect?
(a) Niacin- Pellagra (b) Thiamine - Beriberi
(c) Vitamin K- Sterility (d) Vitamin D - Rickets

49. What does the doctor advice to the patients suffering from high blood cholesterol?
(a) Red Mutton with fat layer
(b) Vegetable and margarine
(c) Vegetable oil such as ground nut oil
(d) Pure deshi ghee and butter

50. Deficiency of copper causes
(a) Pellagra
(b) Anaemia and damage to CNS
(c) Influenza
(d) Xerophthalmia

51. Which of the following vitamins can be synthesized by the bacteria inside the gut?
(a) B_1 (b) A
(c) D (d) K

52. One of the factors required for the maturation of erythrocytes is
(a) Vitamin D (b) Vitamin A
(c) Vitamin B_{12} (d) Vitamin C

53. Zinc is associated with
(a) Glycosylation (b) Immunization
(c) Sulfation (d) Transcription

54. Which one of the following pairs is not correctly matched
(a) Recombinant DNA.....DNA formed by joining of segments.....of DNA from different sources
(b) Purine.....Nitrogenous bases Cytosine, thymine and uracil
(c) ATP.....The principal energy carrying compound in the cell
(d) r-RNA.....RNA molecules found in ribosomes

55. Hydrolysis of nucleic acid yields
(a) Only sugar
(b) Phosphoric acid only
(c) Nitrogenous base only
(d) All the above

56. DNA differs from RNA
(a) In the nature of sugar alone
(b) In the nature of purines alone
(c) In the nature of sugar and pyrimidines
(d) None of the above

57. Strands of DNA are bonded by
(a) Hydrogen (b) Carbon
(c) Oxygen (d) Nitrogen

58. DNA is not present in one of the following
(a) Mitochondria (b) Chloroplast
(c) Bacteriophage (d) Tobacco mosaic virus

59. DNA strands are antiparallel because of the presence of
(a) H-bonds
(b) Peptide bonds
(c) Disulphide bonds
(d) Phosphate-diester bonds

60. In DNA molecule, which of the following base pair is present
(a) Cytosine and adenine
(b) adenine and thymine
(c) adenine and guanine
(d) Cytosine and thymine

61. Which one of the following is widely distributed in a cell?
(a) DNA (b) RNA
(c) Chloroplast (d) Chromoplast

62. The Shine-Dalgarno sequence exists only in
(a) Prokaryotes (b) Eukaryotes
(c) Ribosomes (d) Mitochondria

63. The important requirement for all carrier in drug delivery system is that the biomolecules should retain their conformation state during
(a) Adsorption
(b) Desorption
(c) Both (a) & (b)
(d) No need to retain conformation

64. Structure analysis can be done by
(a) Inline probing
(b) Hydroxyl radical probing
(c) Chemical probing
(d) All of these

65. Dural polarization interferometry is capable of measuring
(a) Biomolecules conformational changes
(b) Construction of 3D structure
(c) Biomolecules charge
(d) Biomolecules light absorption capacity

66. In a conformation, the large scale geometric structure of the molecule is understood to be
(a) Conserved (b) Fluctuate
(c) Oscillate (d) Rotate

67. Charge pumping technique is investigated to identify
(a) Biomolecular charge polarity
(b) Biomolecular conformation
(c) Biomolecule viscosity factor
(d) Biomolecule surface tension

68. In eukaryotic organisms chaperones are known
(a) Heat shock proteins
(b) Housekeeping proteins
(c) Stress proteins
(d) Cardiovascular proteins

69. Guanine nucleotide binding proteins are
(a) Glycoproteins
(b) Glucoproteins
(c) Oxidoreductive proteins
(d) Nucleoproteins

70. In the lipid-free state, the N-terminal domain of human apolipoprotein E3 (apoE3) adopts an
(a) Stem loop structure
(b) Four helical bundle
(c) Coiled structure
(d) Ribbon shaped structure

71. The aminoacid that is restricted in ramchandran plot
(a) Leucine (b) Lysine
(c) Glycine (d) Serine

72. In ramchandran plot limited number of possible combinations of ψ and φ is 0
(a) Histidine (b) Phenylalanine
(c) Proline (d) Alanine

73. Allowed regions is also called as
(a) Low energy region
(b) High energy region
(c) Brown allowed region
(d) Generously allowed region

74. Disallowed regions generally involve steric-hindrance between
(a) Side chain C-beta methylene group and main chain atoms
(b) Side chain C-alpha methylene group and main chain atoms
(c) Side chain C-beta methylene group and side chain atoms
(d) Side chain C-alpha methylene group and side chain atoms

75. The arrangement of hydrogen bonds in helix axis is
(a) Diagonal (b) Perpendicular
(c) Antiparallel (d) Parallel

76. A helical wheel is
(a) 2D (b) 3D
(c) 1D (d) No dimension

77. "The beta - pleated sheet structure is an alternative secondary structure" It was predicted by
(a) Pauling and corey (b) Ramachandran
(c) Walter flemming (d) Van beneden

78. The energy of a Keesom interaction depends on
(a) Inverse of sixth power of distance
(b) Directly proportional to distance
(c) Inversely proportional to four power of distance
(d) Directly proportional to four power distance

79. Hydrogen bond is a type of
(a) Dipole - induced dipole
(b) Dipole - dipole interaction
(c) London dispersion forces
(d) Ion - dipole

80. Term used to describe the interaction of a coordinately unsaturated transition metal with a C-H bond
(a) Agnostic bond (b) Aurophilicity
(c) Metal aromaticity (d) Intercalation

81. Coulomb's Law is formulated as
(a) F1 = q1 * q2/r2
(b) F1 = q1 + q2/r2
(c) F1 = r2/q1 * q2
(d) F1 = r2 (q1 + q2)

82. The only metal that is not solid at room temperature
(a) Iodine (b) Mercury
(c) Gold (d) Silver

83. London dispersion forces occurs due to
(a) Presence of mobile electrons
(b) Random motion of electrons
(c) Presence of lattice structure
(d) Delocalization of electrons

NUMERICAL TYPE QUESTIONS

1. Chair conformation of six membered rings have a dihedral angle of
2. A protein with n residues has
3. The counting which starts from left hand corner is from
4. Due to delocalization of the carbonyl pi electrons & the nitrogen lone pair the OMEGA angle tends to be
5. A right handed helical structure with average torsion angles
6. The atomic distance between the N and O measures

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. Disaccharide molecules that contain $\beta(1 \rightarrow 4)$ glycosidic linkage are
(a) sucrose and maltose
(b) sucrose and isomaltose
(c) maltose and isomaltose
(d) lactose and cellobiose

2015

2. Which one of the following amino acids has the highest probability to be found on the surface of a typical globular protein in aqueous environment?
(a) Ala (b) Val
(c) Arg (d) Ile

3. Levinthal's paradox is related to
(a) protein secretion
(b) protein degradation
(c) protein folding
(d) protein trafficking

4. How many different protein sequences of 100 residues can be generated using 20 standard amino acids ?
(a) 100^{20} (b) 100×20
(c) 20^{100} (d) $100! \times 20!$

5. Match the reagents in Group I with their preferred cleavage sites in Group II.

Group I
P. Cyanogen bromide
Q. *o*-Iodosobenzoate
R. Hydroxylamine
S. 2-Nitro-5-thiocyanobenzoate

Group II
1. Carboxyl side of methionine
2. Amino side of methionine
3. Carboxyl side of tryptophan
4. Amino side of cysteine
5. Asparagine-glycine bonds

(a) P-1, Q-3, R-5, S-4
(b) P-2, Q-3, R-1, S-4
(c) P-1, Q-2, R-5, S-4
(d) P-4, Q-2, R-5, S-3

2014

6. The 4-amino or 4-keto group of pyrimidine bases is located in the
(a) major groove of the double stranded DNA
(b) minor groove of the double stranded DNA
(c) minor groove of the B form DNA but not the A form DNA
(d) major groove of the B form DNA but not the A form DNA

7. Amino acid residue which is most likely to be found in the interior of water-soluble globular proteins is
(a) Threonine (b) Aspartic acid
(c) Valine (d) Histidine

8. If a plant is shifted to cold temperature, which of the following changes would take place in its membrane?
(a) Ratio of unsaturated to saturated fatty acids would increase
(b) Ratio of unsaturated to saturated fatty acids would decrease
(c) Absolute amount of both fatty acids would increase keeping the ratio same
(d) Absolute amount of both fatty acids would remain unchanged

9. Which of the following statements with respect to the orientation of the nitrogenous bases to the pentose sugars, and the puckering of the sugar, is correct?
(a) Anti, and 2'-endo in A form DNA
(b) Anti, and 2'-endo in B form DNA
(c) Syn, and 3'-endo in A form DNA
(d) Syn, and 3' -endo in B form DNA

2013

10. Which one of the following aminoacids in proteins does **NOT** undergo phosphorylation?
(a) Ser (b) Thr
(c) Pro (d) Tyr

2012

11. The molarity of water in a water : ethanol mixture (15 : 85, v/v) is approximately
(a) 0.85 (b) 5.55
(c) 8.5 (d) 55.5

2011

12. The protein in eukaryotes which is subjected to degradation undergoes
(a) phosphorylation (b) carboxylation
(c) ubiquitation (d) methylation

2010

13. Match the chemicals in Group I with the possible type/class in Group II

Group I	Group II
P. Picloram	1. Vitamin
Q. Zeatin	2. Auxin
R. Thiamine	3. Amino Acid
S. Glutamine	4. Cytokinin

(a) P-2, Q-4, R-1, S-3
(b) P-4, Q-1, R-2, S-3
(c) P-3, Q-1, R-2, S-4
(d) P-4, Q-2, R-1, S-3

Common Data for Question (14 - 15) :

The width of the lipid bilayer membrane is 30 Å. It is permeated by a protein which is a right handed α-helix.

14. The number of h-helical turns permeating the membrane is
(a) 5.6 turns
(b) 3.5 turns
(c) 6.5 turns
(d) 5.0 turns

15. The number of amino acid residues present in the protein is
(a) 15
(b) 18
(c) 17
(d) 20

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (a)	2. (a)	3. (c)	4. (d)	5. (a)	6. (c)	7. (c)	8. (b)	9. (b)	10. (b)
11. (a)	12. (a)	13. (d)	14. (d)	15. (a)	16. (b)	17. (b)	18. (b)	19. (d)	20. (b)
21. (c)	22. (a)	23. (b)	24. (c)	25. (b)	26. (a)	27. (a)	28. (c)	29. (b)	30. (c)
31. (d)	32. (d)	33. (c)	34. (a)	35. (d)	36. (b)	37. (c)	38. (b)	39. (c)	40. (c)
41. (b)	42. (a)	43. (d)	44. (c)	45. (d)	46. (b)	47. (b)	48. (c)	49. (c)	50. (b)
51. (a)	52. (c)	53. (b)	54. (b)	55. (d)	56. (c)	57. (a)	58. (d)	59. (a)	60. (b)
61. (b)	62. (a)	63. (c)	64. (d)	65. (a)	66. (a)	67. (a)	68. (a)	69. (a)	70. (b)
71. (c)	72. (c)	73. (a)	74. (a)	75. (d)	76. (a)	77. (a)	78. (a)	79. (b)	80. (a)
81. (a)	82. (b)	83. (b)							

Numerical Type Questions

1. 60° 2. 2n variables 3. –180 to +180 4. 0 or 180° 5. phi = (–57) and psi = (–47)
6. 0.28nm

EXERCISE – II

MCQ Type questions

1. (d)	2. (c)	3. (c)	4. (c)	5. (a)	6. (a)	7. (c)	8. (a)	9. (b)	10. (c)
11. (c)	12. (c)	13. (a)	14. (a)	15. (d)					

EXPLANATIONS

EXERCISE - I

MCQ TYPE QUESTIONS

1. Fructose is the most common form of sugar. It is the sweetest among naturally occurring sugars. It has sweetening index of 170 (whereas the sweetening index of glucose is 70).
2. The union makes sucrose more stable than other sugars because both its anomeric carbon atoms are protected from oxidative attack. It is because of this reason that sucrose is used for transporting carbohydrates in plants.
3. The fatty acids having more than or double bond called polyunsaturated fatty acids. Fats having such fatty acids are termed polyunsaturated fats. The latter are commended by physician for persons having cardiovascular disease as their use lowers the blood cholesterol level.
4. Because plant viruses have RNA as genetic material.
5. The two strands of DNA molecules run in opposite or antiparallel direction due to presence of hydrogen bond because two base i.e. one in each chain of DNA molecule, joined together by hydrogen bonds.
6. The DNA duplex model proposed by Watson and Crick is right handed spiral and is called B-DNA. It is one complete turn is 34Å long and has 10 base pairs.
7. Exonuclease activity means cleavage of nucleotides only at the end. Endonuclease breaks DNA strand at internal position. DNA polymerase I has 3'-5' exonuclease activity which removes any nucleotide which mispair during elongation of growing strand. A small segment of polymerase I also show 5'-3' exonuclease activity which removes DNA segment which comes as an obstruction in growing DNA strand. Polymerase catalyses the elongation of a polymeric molecule. Kinase is an enzyme that can transfer a phosphate group.
8. The enzyme enterokinase which is also known as enteropeptidase helps in conversion of trypsinogen into trypsin. Trypsin acts on proteins and breaks them for digestion. It occurs in the brush borders of duodenum.
9. Cellulose is the most abundant carbohydrate. It is composed of 1600-6000 glucose molecules joined together. These polymers form long twisting macromolecules of cellulose. The chains are unbranched and linear. The successive glucose residues are joined together by β-1-4, linkages.
10. Glyoxysomes are microbodies found in fungi and germinating oil seeds that are involved in converting fat into carbohydrates. They contain enzymes for oxidation of fatty acids and glyoxylate cycle. Glyoxylate cycle is a metabolic pathway in plants and microorganisms that is a modified form of the Kreb's cycle. It utilizes fats as a source of carbon and enables the synthesis of carbohydrates from fatty acids by avoiding the stages of the Kreb's cycle in which carbon dioxide is released.
11. Glycoproteins are proteins that contain sugars like carbohydrates as prosthetic group. In most glycoproteins, the linkage is between asparagine and N-acetyl-D-glucosamine. Some glycoproteins are immunoglobulins, membrane proteins and muscle proteins. There are formed in Golgi apparatus in the process of glycosylation and are important components of plasma membrane in which they extend throughout the lipid bilayer.
12. Minor elements are those which are required in quantity less than 1 mg/ gm of dry matter but are essential for proper growth and development of an organism. *e.g.* Cl, Mn, B, Zn, Cu, Mo. These elements work as non-protein cofactor in enzymes. They also take part in oxidation reduction reactions.
13. Starch is the major storage carbohydrate of plants. In most plant species it is accumulated in the chloroplast of leaves, whereas in storage organ it accumulates in amyloplast as reserve starch. It is the osmotically inactive form of photosynthetic product and is a hexosan polysaccharide made of large number of glucose unit so, chemically non-reactive.
14. Steroids are a type of lipid. Some steroids are hormones such as corticosteroids made in the adrenal cortex and the sex hormones such as testosterone and oestrogen. Cholesterol is the precursor of steroid hormones possessing a complex basic structure.
15. The bond formed between two mono-saccharides as a result of condensation (removal of 1 molecule of water) is called a glycosidic bond and it normally forms between carbon atoms 1 and 4 of neighboring units. It is broken during the digestion of starch.
16. High energy phosphate bonds are universally symbolized by the symbol ~ and high energy phosphate groups as ~ P. ATP can be designated as A-R-P~P~P and ADP as A-R-P~P. In other words, the ADP-ATP system is a carrier of phosphoryl ($-PO_2^{2-}$) groups from high energy phosphate compounds.
17. The specificity of any protein and its physical and enzymatic properties depend on the linear sequence of the amino acids. Different types of amino acids arranged in a specific manner produce different proteins. Change in amino acid sequences results in different types of diseases. E.g. Sickle cell anaemia.

18. The ATP-ADP system functions as a carrier of chemical energy because ADP is able to accept a phosphate group during coupled energy yielding reaction (exothermic) of catabolism and the ATP so formed is able to donate its terminal phosphate group in coupled energy-requiring reactions (endothermic).
19. After the formation of 70s-mRNA-fmet.tRNA-fmet complex, elongation of polypeptide chain is brought about by the regular addition of amino acids. In bacteria at 370C about 15 amino acids per second are added to a growing polypeptide chain, so that in 20 seconds a protein with 300 amino acids can be synthesized.
20. Arachidonic Acid is a poly unsaturated fatty acid or essential fatty acid with four cis double bonds, which are the sources of its flexibility and give it the capacity to react with molecular oxygen. It is present in the membrane of the body's cells and is a precursor in the production of eicosanoids: the prostaglandins, thromboxanes, prostacyclin and the leukotrienes.
21. Fatty acid esters of the alcohol glycerol are called are called acylgyecerols, neutral fats or glycerides. When all three hydroxyl groups of glycerol are esterified with fatty acids, the structure is called a triacylglycerol. Triacylglycerols make up the great bulk of the neutral fats in nature.
22. If a carbon atom is attached to four different groups, it is asymmetric and therefore exhibits optical isomerism. The amino acids (except glycine) possess four distinct groups. (R, H, COO^-, NH_3) held by α-carbon. Thus all the amino acids (except glycine where R = H) have optical isomers.
23. In plants, photosynthesis plays a vital role in amino acid synthesis. This is due to the carbohydrates metabolism to create all amino acids. Therefore, bacteria, yeast, mould and plants are unique because they are able to produce all needed amino acids by themselves.
24. Calmodulin (CaM) is a Ca^{2+} binding protein that is a key component of the calcium second messenger system and is involved in controlling many of the biochemical processes of cells. Calmodulin is a small, acidic protein approximately 148 amino acids long. Calmodulin consist of 4 EF- hand "motifs" or domains, each of which binds a calcium ion. It typically binds 0, 2 or 4 calcium ions, and binds and regulates different proteins in each state. Calmodulin-stimulated protein phosphatase is the major calmodulin -binding protein in the brain.
25. Vitamin B is required for activation or acts as coenzymes of dehydrogenase .Vitamin A is anti-infective, essential for synthesis of visual pigments, Vitamin C is essential for formation of collagen, cartilage, bones, teeth ,connective tissue etc. and Vitamin K acts as antihaemorrhagic factor essential for synthesis of prothrombin in liver which is required for blood clotting.
26. Casein is a conjugated protein consisting mostly of phosphorous. A conjugated protein is that functions in interaction with other chemical groups attached by covalent bonds or by weak interactions. Conjugated proteins are classified on the basis of the chemical nature of their prosthetic groups. Casein is the predominant phosphoprotein found in fresh milk. When coagulated with rennet, casein is sometimes called paracasein. The term caseinogens is used for the uncoagulated protein and casein for the coagulated protein.
27. Manganese is cofactor of many enzymes hydrolase like, decarboxylase and transferase enzymes, glycoprotein and proteoglycan synthesis, mitochondrial superoxide dismutase. Manganeese ions helps the enzymes catalyzes the synthesis of oligosaccharides and glycoproteins. Magnesium acts a cofactor for enzyme kinase. Copper is a constituent of oxidase enzyme, e.g. cytochromo oxidase. Molybdenum is a cofactor of enzyme nitrogenase nitrate reductase which catalyzes fixation of nitrogen.
28. Keratin is an extremely tough protein substance in hair, nails and horny tissues. It is insoluble in water, weak acids or alkalis and is unaffected by most proteolytic enzymes. Collagen is a fibrous insoluble protein found in connective tissue, including skin, bone, ligaments and cartilage. It represents about 30% of the total protein. Elastin is a protein substance forming the principle constituent of yellow elastic tissue, comprising about 30% of this tissue.
29. Nucleic acids store, transmit and express genetic information. Nucleic acids are composed of subunits called nucleotides. Nucleotides contain a phosphate group, a sugar and a ring of carbon and nitrogen atoms. The ring is also known as the base because it can accept hydrogen ions. Nucleotides are linked together by bonds between the phosphate group of one nucleotide and the sugar of the next one. In this fashion, nucleotides form long chains.
30. Mucin is a glycoprotein found in mucus. It is present in saliva and bile and salivary glands, in the skin, connective tissues, tendon, and cartilage. It is formed from mucigen and in water forms a slimy solution.
31. Proteins that contain 2 or more polypeptide chain associated by non-covalent forces are said to exhibit quaternary structure. That is quaternary structure of protein refers t way in which two or more polypeptide chains, each the product of a gene sequence are associated to yield a more complex protein structure. In these multimeric proteins the individual polypeptide chains are termed as protomers or subunits. Different spatial orientation o subunits confirm altered properties on the oligomer and permit multimeric protein to play unique role in intracellular regulation.

32. Cholestrol is the precursor of all steroids in the body, e.g. corticosteroids, sex hormones, bile acid and vitamin D. It is also important amphipathic lipids which allow it to play a structural role in the membrane and in the outer layer of lipoproteins. Cholestrol is synthesized in liver from acetyl CoA and is regulated partly by influx of dietary cholesterol in cholesterol rich chylomicron remanant.
33. Several plants that depend upon animal for seed dispersal, accumulate sugar(s) in their ripe fruits. The sugar most widely used for this purpose is fructose. It is fruit sugar and is sweetest among other sugars.
34. In animals, the stored form of sugar is glycogen. Glycogen is a homopolysaccharide food reserve of animals, bacteria, fungi. It is mainly stored inside liver (up to 0.1 kg) and muscles; it is popularly called as animal starch. It has about 30,000 glucose residues and a molecular weight of about 4.8 million. Glucose residues are arranged in a highly branched bush like chains. There are two types of linkages 1-4 linkages in the straight part and 1-6 linkages in the area of branching.
35. Vitamin B_{12} is a dark red compound containing cobalt.
36. Vitamin B_{12} is not found in plants. However it is considered that spirulina (algae) contains B_{12}. Vitamin B_{12} is synthesized by intestinal bacteria which in fact are main source of vitamin B_{12}. Many microorganisms (bacteria) of the stomach of ruminant mammals also synthesize large quantities of B_{12}.
37. Vitamin B17 has been recently explained to be found in water melon. It is supposed to have anti-cancer property.
38. Pellagra disease resulting from deficiency of nicotinic acid is characterized by scaly dermatitis, inflammation of mucus membrane, diarrhoea and mental illness.
39. Paralysis of muscle fatigue and burning feet syndrome is related with deficiency of pantothenic acid (vitamin B_5).
40. In man, attached to caecum is a twisted, coiled tube, measuring about 3 inches in length, is called the vermiform appendix. Inflammation of the vermiform appendix is called appendicitis.
41. Giving them milk free diet because milk is toxic to galactosemic appendicitis.
42. Anti infection vitamin is B12. It also maintains the integrity of epithelial tissue and prevents the infection.
43. Vitamin C is necessary for formation of collagen fibre, in this way vitamin C helps for healing to the wound.
44. The activity of vitamin E is destroyed by U.V. rays and oxidation.
45. E. coli lives in mutual association in human colon as it obtains food from the intestine and in turn helps to produce vitamin B_{12}.
46. It acts as the coenzyme of pyruvate carboxylase. Its active factor is thymine pyrophosphate. It causes decarboxylation of pyruvic acid.
47. Because riboflavin is an orange- yellow compound, so cow's milk appears slightly yellowish.
48. Vitamin K (phylloquinone) is required for the formation of clotting factors by liver, thereby its deficiency leads to delay in coagulation i.e. haemoglobin.
49. Vegetable oil contains low molecular weight lipids.
50. Copper takes part in haemoglobin synthesis.
51. The chemical name of B_1 is thiamine. It can be synthesized by bacteria inside the gut.
52. Vitamin B_{12} is also called cyanocobalamine. It is essential for the formation and maturation of erythrocytes.
53. Zinc takes part in immune reaction.
54. Because cytosine, thymine and uracil are single ring compound i.e. Pyrimidines while adenine and guanine are double ring i.e. purines.
55. Because nucleic acids are made up of sugar, phosphoric acid and nitrogenous base.
56. Because DNA possesses deoxyribose sugar and thymine (T) in place of uracil (U).
57. There is double hydrogen bond between adenine and thymine (A = T) and triple bond between cytosine and guanine (C ≡ G).
58. Because plant viruses have RNA as genetic material.
59. The two strands of DNA molecules run in opposite or antiparallel direction due to presence of hydrogen bond because two base i.e. one in each chain of DNA molecule, joined together by hydrogen bonds.
60. Adenine and thymine; because C always attaches with G and A attaches with T.
61. RNA is present both inside and outside the nucleus.
62. The Shine-Dalgarno sequence (or Shine-Dalgarno box), proposed by Australian scientists John Shine and Lynn Dalgarno, is a ribosomal binding site in the mRNA, generally located 8 base pairs upstream of the start codon AUG. The Shine-Dalgarno sequence exists only in prokaryotes.
63. One of the important requirement for all carrier in drug delivery system is that the biomolecules should retain their conformation state through out also rption or desorption steps because a conformational change would greatly reduce the activity.
65. Many biophysical techniques such as crystallography, NMR and FRET can be used to study macromolecular conformational change. Dual Polarization Interferometry is a bench top technique capable of measuring conformational changes in biomolecules in real time at very high resolution.
67. Charge pumping technique is investigated to identify biomolecular charge polarity using a nanogap-embedded biotransistor
    Charge pumping technique is investigated to identify biomolecular charge polarity.
68. In eukaryotic organisms chaperones are known as heat shock proteins.
70. The N-terminal domain of human apolipoprotein E3 (apoE3) adopts an elongated, globular four helix bundle conformation in the lipid-free state.
71. Glycine has a hydrogen atom, with a smaller van der Waals radius, instead of a methyl group at the a position. Hence it is least restricted and this is apparent in the Ramachandran plot for glycine for which the allowable area is considerably larger.
72. In contrast the Ramachandran plot for proline shows only a very limited number of possible combinations of phi and psi.
73. Allowed regions is also called as low energy region.
74. Disallowed regions generally involve steric hindrance between the side chain C-beta methylene group and main chain atoms. Glycine has no side chain and therefore can adopt phi and psi angles in all four quadrants of the Ramachandran plot. Hence it frequently occurs in turn regions of proteins where any other residue would be sterically hindered.
75. The H-bonds are almost parallet to the helix axis and the total dipole moment gives the helix a dipole moment that points from the N-term (+) to the C-term (–).
76. A helical wheel is a projection in 2-D along the helix axis and displays the orientation of the side chains on a 360 degree map with respect to the side of the helix.
77. In 1952, Pauling and Corey predicted the b-pleated sheet structure as an alternative secondary structure to the a-helix in proteins. b-stands are etrongated peptide segments with atomic distances from side chain n to side chain n + 2 of 0.7 nm.
78. The energy of a Keesom interaction depends on the inverse sixth power of the distance, unlike the interaction energy of two spatially fixed dipoles, which depends on the inverse third power of the distance.
79. A hydrogen bond is the attractive force between a hydrogen atom and an electronegative atom, such as nitrogen, oxygen, or fluorine. The hydrogen bond is often described as a strong electrostatic dipole-dipole interaction. However, it also has some features of covalent bonding: It is directional, stronger than a van der Waals interaction, produces interatomic distances shorter than sum of van der Waals radii.
80. Agostic interaction is a term in organometallic chemistry for the interaction of a coordinately unsaturated transition metal with a C-H bond, when the two electrons involved in the C-H bond enter the empty d-orbital of a transition metal, resulting in a two electron three center bond. [1] Many catalytic transformations, e.g. oxidative addition and reductive elimination, are proposed to proceed via intermediates featuring agostic interactions. Agostic interactions are observed throughout organometallic chemistry in alkyl, alkylidene, and polyenyl ligands.
81. Coulomb's Law states that the force of attraction between two objects is equal to the product of their charges (q1 and q2) divided by the square of the distance between them (r)
    F1 = q1 * q2/r2
82. Mercury is the only metal that is not solid at room temperature
83. London dispersion forces occurs due to random motion of electrons

NUMERICAL TYPE QUESTIONS

1. Chair conformation of six membered rings have a dihedratangle of 60° between adjacent substituents thus making it the most stable conformer.
2. The torsional angles of these bonds are commonly referred to as the phi and psi angles : a protein with n residues has 2n variable torsion angles. Thus, the conformation space of a protein has 2n dimensions, and each point in the space is a particular protein structure.
3. The counting which starts from left hand corner is from –180 to +180
4. The OMEGA angle tends to be planar (0 to 180°) due to delocalization of the carbonyl pi electrons & the nitrogen lone pair.
5. Alpha-helix: a right handed helical structure with average torsion angles phi = (–57) and psi = (–47)
6. The helix has a specific hydrogen bonding pattern, where the backbone C = O group of residue n bonds with the N-H of residue n+4. The atomic distance between the N and O measures 0.28nm.

EXERCISE – II

MCQ TYPE QUESTIONS

1. **Glycosidic bond** or **glycosidic linkage** is a type of covalent bond that joins a carbohydrate (sugar) molecule to another group, which may or may not be another carbohydrate. Lactose (β-D-galactopyranosyl (1→ 4) D-glucopyranose) and cellobiose (β-D-glucopyranosyl (1→4) D-glucopyranose) are disaccharides whose

monomeric units are connected by $\beta(1\rightarrow 4)$ glycosidic bonds.

2. Arginine, lysine aspartic, glutamic and histidine can present in aqueous environment.
3. Finding the native folded state of protein.
4. 20^{100} different protein sequences of 100 residues can be generated using 20 standard amino acids.
5.

Group I	Group II
P. Cyanogen bromide	1. Carboxyl side of methionine
Q. *o*-Iodosobenzoate	3. Carboxyl side of tryptophan
R. Hydroxylamine	5. Asparagine-glycine bonds
S. 2-Nitro-5-thiocy anobenzoate	4. Amino side of cysteine

6. The pyrimidine bases, Thymine and Cytosine in case of DNA have a 4-amino or a 4-keto group present with the amide ring. These 4-amino or 4-keto groups are present in the major groove of the DNA molecule.
7. Globular proteins have no systematic structures. There may be single chains, two or more chains which interact in the usual ways or there may be portions of the chains with helical structures, pleated structures, or completely random structures. Common globular proteins include egg albumin, hemoglobin, myoglobin, insulin, serum globulins in blood, and many enzymes. Water-soluble globular proteins have hydrophobic amino acids present in the interior of the protein structure forming hydrophobic bonds. So, valine is the amino acid residue most likely to be present on the interior.
8. When a plant body is shifted from a warm climate to a cold climate, the number of long chain unsaturated fatty acids present in the plant would increase to preserve and store body heat as well as reserved energy. So, ratio of unsaturated to saturated fatty acids would increase considerably in that plant.
9. Nitrogenous bases are always arranged in "anti" positions to their corresponding pentose sugar moieties. In B-form of DNA, sugar puckering occurs predominantly in the C2 position of the pentose sugar. The sugar pucker determines the shape of the a-helix, whether the helix will exist in the A-form or in the B-form. So, in C2-endo puckering occurs in B-form of DNA.
10. Proline is unique in that it is the only amino acid where the side chain is connected to the protein backbone twice, forming a five-membered nitrogen-containing ring. Proline is unable to occupy many of the main chain conformations easily adopted by all other amino acids. Proline can often be found in very tight turns in protein structures (i.e. where the polypeptide chain must change direction). It can also function to introduce kinks into alpha helices, since it is unable to adopt a normal helical conformation. Proline plays important roles in molecular recognition, particularly in intracellular signalling. The Proline side chain is very non-reactive and it does not undergo phosphorylation.
11. 18gm water in 100ml ethanol is 10M. Hence 15gm in 85ml is approximately 8.5.
12. Ubiquitination is an enzymatic, protein post-translational modification (PTM) process in which the carboxylic acid of the terminal glycine from the di-glycine motif in the activated ubiquitin forms an amide bond to the epsilon amine of the lysine in the modified protein. Ubiquitination, also known as ubiquitylation, regulates degradation of cellular proteins by the ubiquitin-proteasome system, controlling a protein's half-life and expression levels. This process involves the sequential action of ubiquitin-activating enzymes (E1), ubiquitin-conjugating enzymes (E2), and ubiquitin ligases (E3). The reaction catalyzed by each enzyme transfers a covalent bond with ubiquitin from one enzyme to the next and finally to a target protein.
13. Picloram is an auxin type of hormone absorbed by root. Zeatin is a member of the plant growth hormone family, cytokinins. Thiamine is a water-soluble vitamin of the B complex. Glutamine is one acids encoded by the standard genetic code CAA,CAG.
14. Given,

    Width of lipid bilayer membrane= 30 Å

    We know, each turn of right handed α-helix has a pitch of = 5.1 Å

    Thus, total turns = 30 / 5.1 = 5.6 turns.
15. We know, the right handed α-helix has 3.6 residues per turn in the helix.

    But number of helix turns = 5.6

    Therefore, No. of amino acids present in the protein = 5.6 * 3.6 = 20

    Therefore, No. of amino acid residues present in the protein is 20.

■■

2 CHAPTER Bioenergetics and Metabolism

BIOENERGETICS

Bio-energetics deals with the study of energy change in bio-chemical reactions. The reactions are broadly classified as exergonic (energy releasing) or endogonic (energy requiring). Bio-energetics is concern with the initial and final state of energy component of the reactant and not the mechanism of chemical reactions. For example although thermodynamics tells us that glucose and oxygen react with the release of huge amount of energy, it does not indicate that this mixture is indefinitely stable at room temperature in the absence of appropriate enzyme.

Thermodynamics, as it applied to bio-chemistry, is most frequently concerned with describing the conditions under which process occurs spontaneously.

Theromdynamic Properties

Enthalpy (H) :

It represents the total heat content of a system at constant pressure. It is expressed as-

$$H = E + Pv$$

$$\text{or } \Delta H = \Delta E + P\Delta v$$

$$\text{or } \Delta H = H_{final} - H_{initial}$$

$$\Delta H = \Delta E + \Delta nRT$$

where E is internal energy, P is pressure and v is volume. $P\Delta v$ is work done and is equal to ΔnRT, where Δn is change in number of moles, R is gas constant and T is absolute temperature.

When ΔH is –ve → Process is exothermic

When ΔH is +ve → Process is endothermic

If	$\Delta n = 0$,	then	$\Delta H = \Delta E$
If	Δn is +ve,	then	$\Delta H > \Delta E$
If	$\Delta n < 0$ or –ve,	then	$\Delta H < \Delta E$.

Entropy (ΔS) :

It is measure of randomness of the system. It is expressed as-

$$\Delta S = \frac{q_{reversible}}{\Delta T}$$

Where, $q_{reversible}$ = heat supplied isothermally and reversibly and ΔT = absolute temperature.

Free Energy (ΔG) :

"It is the total amount of energy of a system for doing useful work." It is expressed as-

$$\Delta G = \Delta H - T\Delta S$$

For a spontaneous process, the value of ΔG should be –ve.

Standard Free Energy (ΔG°) :

It is the free energy change at standard condition of the reactant i.e., at 25°C and pH 7 with the concentration of 1 mole / litre.

Relation between ΔG, ΔG° and $K_{equilibrium}$

ΔG of the reaction A → B depends on the concentation of the reactant and product. At constant temperature and pressure the following reaction can be derived

$$\Delta G = \Delta G^\circ + R\,T \ln \frac{[B]}{[A]}$$

$$\Delta G = \Delta G^\circ + R\,T \ln K_{equ}$$

$$K_{eq} = [B]_{equ} / [A]_{equ}$$

$$\Rightarrow \Delta G^\circ = -RT \ln K_{equ}$$

Where K_{equ} is the equilibrium constant. R is gas constant. [A] and [B] are actual concentration and $[A]_{equ}$ and $[B]_{equ}$ are the concentration at equilibrium of reactant and product.

In case of two sequential chemical reactions, A ⇌ B and B ⇌ C, each reactions has its own equlibrium constant and each has its characteristic standard free energy change. These values of ΔG° are additive.

$$AG^\circ_{total}(A \text{ to } C) = AG^\circ_1 (\text{for } A \rightarrow B) + AG^\circ_2 (\text{for } B \rightarrow C)$$

Law of Thermodynamics

First Law of Thermodynamics

"Energy is neither to be created nor to be destroyed but one form of energy can be transformed to another form."

The mathmatical form of first law is

$$\Delta E = q + w$$

where, ΔE = change in internal energy, q = heat supplied and w = work done on the sytem

Second Law of Thermodynamics

"Entropy of the universe is continuously increasing" or "Complete conversion of heat into work is impossible". That means when there is increase in randomness (disorder) of a given system, the entropy increases.

Third law of Thermodynamics

"The entropy of a perfect crystalline substance is zero at absolute zero tempreture".

High Energy Compounds

High energy compounds are those compounds which gives more than 25 kJ/mole energy after their degradation. The high energy compounds gives this huge amount of energy because their degradation is favoured by virtue of its product. These virtues are

(*i*) Products may stablized by tautomerization, e.g., pasphoenal pyruvate.

(*ii*) Products may stablized by charge separation, e.g., ATP.

(*iii*) Products may by stablized by ionization. e.g., ATP, 1-3-bisphosphoglycerate, acetyl CoA.

(*iv*) Products may be stablized by resonance stablization. e.g. ATP, 1-3-bisphosphoglycerate, phospho creative, acetyl Co-A etc.

(*v*) Greater degree of salvation of products, e.g., in ATP.

Explanation : In above these five conditions, the products are more stable than the substrate; so the reaction will be spontaneous, i.e., ΔG = –ve value and producing large amount of energy.

By the equation, $$\Delta G = \Delta G^\circ + RT \ln \frac{[\text{Product}]}{[\text{Substrate}]}$$

ATP and its Role in Bioenergetics

Properties of ATP

ATP is present in all living cells at concentrations of 10^{-3} to 10^{-2} mol L^{-1} of cell water. The ATP molecule is composed of three parts: adenine, D-ribose, and three phosphate groups in ester linkages. The analogous compounds containing one and two phosphate groups are designated AMP and ADP, respectively.

ATP has a strong tendency to hydrolyze to ADP and phosphate; this is predicted from thermodynamics:

$$ATP^{4-} + H_2O \rightarrow ADP^{3-} + HPO_4^{2-} + H^+$$

Under biochemical standard-state conditions ($[ATP^{4-}] = [ADP^{3-}] = [HPO_4^{2-}] = 1$ mol L^{-1}, $[H^+] = 10^{-7}$mol L^{-1}, and T = 310 K), the thermodynamic parameters for this reaction are $\Delta G^{\circ\prime} = -30.5$ kJ mol^{-1}, $\Delta H^{\circ\prime} = -20$ kJ mol^{-1}, and $\Delta S^{\circ\prime} = 34$ J K^{-1} mol^{-1}

ATP is often referred to as being energy-rich or containing high-energy phosphate ester bonds. This nomenclature should be thought of as implying that the overall process of ATP hydrolysis has large negative values of $\Delta G^{\circ\prime}$ and $\Delta H^{\circ\prime}$.

ATP in Bioenergetics

ATP is not a high-energy compound in comparison with many other biological compounds. The functions of ATP depend on its having a ΔG value for hydrolysis that is intermediate in value compared with ΔG values for hydrolysis of other phosphate esters. Thus, ATP and ADP can act as a donor-acceptor pair for *phosphoryl-group transfer*. In many cases the free energy of ATP hydrolysis is used to support reactions that would otherwise be thermodynamically unfavorable. This usually occurs via phosphorylation of one of the reactants in an otherwise unfavorable reaction.

Transphosphorylation

"Transfer of phosphoryl group from one nucleotide tri-phosphate to other nucleotide di-phosphate for specific biological roles."

$$2ADP \underset{Trinase\ Mg^{2+}}{\overset{Adenylate}{\rightleftharpoons}} ATP + AMP \quad \Delta G^\circ \approx 0$$

$$ATP + P_{cr} \underset{Creatin\ Trinase}{\overset{Mg^{2+}}{\rightleftharpoons}} ATP + C_r \qquad \Delta G^\circ = -12.5 \text{ kJ/mol.}$$

$$ATP + \text{Poly iP}_{(n)} (\text{in bacteria}) \underset{Mg^{2+}}{\overset{\text{Poly Phosphate binase}}{\rightleftharpoons}} ADP + \text{Poly iP}_{n+1} \qquad \Delta G^\circ = -20 \text{ kJ/mol.}$$

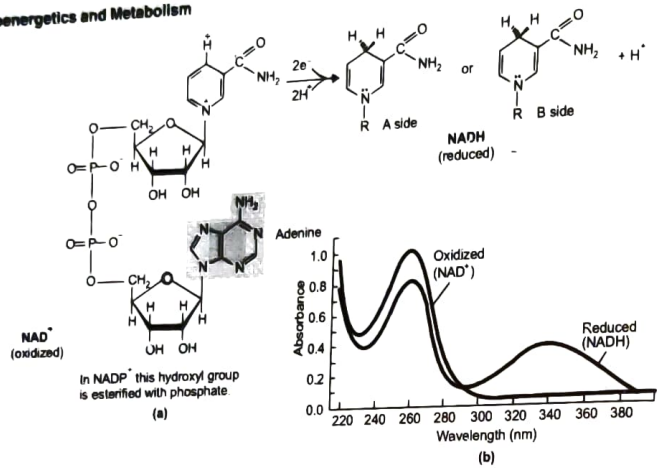
Biological Oxidation - Reduction Reactions

The transfer of a phosphoryl groups is a central feature of metabolism. Equally important, is another kind of transfer, electron transfer in oxidation reduction reactions. These reactions involves the loss of electron by one chemical species, which is thereby oxidized, and the gain of electron by another which is reduced. The flow of electrons in oxidation reduction reactions is responsible directly or indirectly for all workdone by the living organisms.

Living cells have an biological circuits of reduced compounds such as glucose as the source of electrons. As the glucose is enzymetically oxidized, the electrons released flow spontaneously through a series of electron carrier intermediates to another chemical species such as O_2. This electron flow is energonic because O_2 has high affinity for electrons than the carriers. The resulting electromotive force provides energy for ATP synthesis and synthesis of other high energy compounds for doing biological work. The biological oxidation reduction reactions often involves dehydrogenetion by means of enzyms dehydrogenases. A few types of electron carries like NADH, NADPH, FMN and FAD acts as carriers of electrons in following manner :

$$NAD^+ + 2e^- + 2H^+ \longrightarrow NADH + H^+$$

$$NADP^+ + 2e^- \; 2H^+ \longrightarrow NADPH + H^+$$



The flavin nucleotide in most flavoproteins is bound rather tightly to the protein, and in some enzymes, such as succinate dehydrogenase, it is bound covalently. Such tightly bound coenzymes are properly called prosthetic groups. They do not transfer electrons by diffusing from one enzyme to another, rather, they provide a means by which the flavoprotein can temporarily hold electrons while it catalyzes electron transfer from a reduced substrate to an electron acceptor. One important feature of the flavoproteins is the variability in the standard reduction potential (E°) of the bound flavin nucleotide. Tight association between the enzyme and prosthetic group confers on the flavin ring a reduction potential typical of that particular flavoprotein, sometimes quite different from that of the free flavin nucleotide. FAD bound to succinate dehydrogenase, for example, has an E° close to 0.0 V, compared with –0.219 V for free FAD. Flavoproteins are often very complex; some have, in addition to a flavin nucleotlde, tightly bound inorganic ions (iron or molybdenum, for example) capable of participating in electron transfers.

$$E = \Delta E^\circ \frac{RT}{nf} ln \frac{[\text{electron accepter}]}{[e^- \text{ donar}]}$$

or

$$\Delta E = E^\circ + \frac{0.026V}{nf} ln \frac{[e^- \text{ accepter}]}{[e^- \text{ donar}]}$$

$$\Delta G = -nf\Delta E$$

where, f = faraday constant = 96.5 kJ/v mol.

The Relationship between E_0 and E

The free energy change for the general redox reaction occurring in solution

$$A_{ox} + B_{red} \rightarrow A_{red} + B_{ox}$$

is

$$\Delta G = \Delta G^\circ + RT \ln \left(\frac{[A_{red}][B_{ox}]}{[A_{red}]^\circ[B_{ox}]^\circ} \Big/ \frac{[A_{ox}][B_{red}]}{[A_{ox}]^\circ[B_{red}]^\circ} \right) \quad ...(1)$$

$$\Delta G = -nFE$$

and recalling that, for solutes, $[A_{red}]^\circ = [B_{ox}]^\circ = 1$ mol L^{-1}, etc. we have

$$E = E_0 - \frac{RT}{nF} ln \left(\frac{[A_{red}][B_{ox}]}{[B_{red}][A_{ox}]} \right) \quad ...(2)$$

This equation, the *Nernst redox equation*, provides a way of relating E and E_0 for any redox reaction or half-cell reaction.

METABOLISM

The metabolism of an organism (or of a cell) may be defined as 'the sum total of all the enzyme-catalyzed reactions that occur in an organism (or in a cell)'. The large number of reactions in a cell are organized into a relatively small number of sequences or pathways. It is a highly coordinated and purposeful cell activity, in which multienzyme systems cooperate.

Terminology of Metabolism

The various processes constituting metabolism may be divided, somewhat arbitrarily, into catabolism and anabolism. Those processes, whose major function is the generation of chemical energy in forms suitable for the mechanical and chemical processes of the cells, are termed as catabolism (cata^G = down : ballein^G = to throw) ; whereas those processes, which utilize the energy generated by catabolism for the biosynthesis of cell components, are termed as anabolism (ana^G = up). The various activities powered by catabolism include mechanical movement, growth, reproduction, accumulation of foods, elimination of wastes generation of electricity, maintenance of temperature etc. The various anabolic activities may be exemplified by food manufacture etc.

The terms catabolism and anabolism were coined by the physiologist Gasket in 1886.

Some processes can be either catabolic or anabolic, depending on the energy conditions in the cell. These are referred to as **amphibolism**.

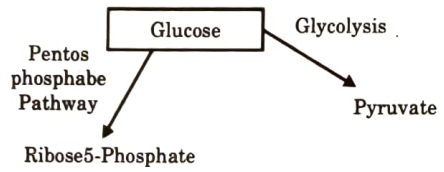
$$\text{Fuels (carbohydrates, fats)} \xrightarrow{\text{Catabolism}} CO_2 + H_2O + \text{Useful energy}$$

$$\text{Useful energy + Small molecules} \xrightarrow{\text{Anabolism}} \text{Complex molecules}$$

Biosynthetic reaction sequences or pathways in the cell require an input of energy and this will normally be in the form of ATP or a reduced coenzyme (NADH, NADPH). It is noteworthy that the usefulness of ATP as an energy source is not a consequence of any special or 'high energy' form of the phosphate bond, but is merely a function of how far the hydrolysis of ATP is displaced from equilibrium.

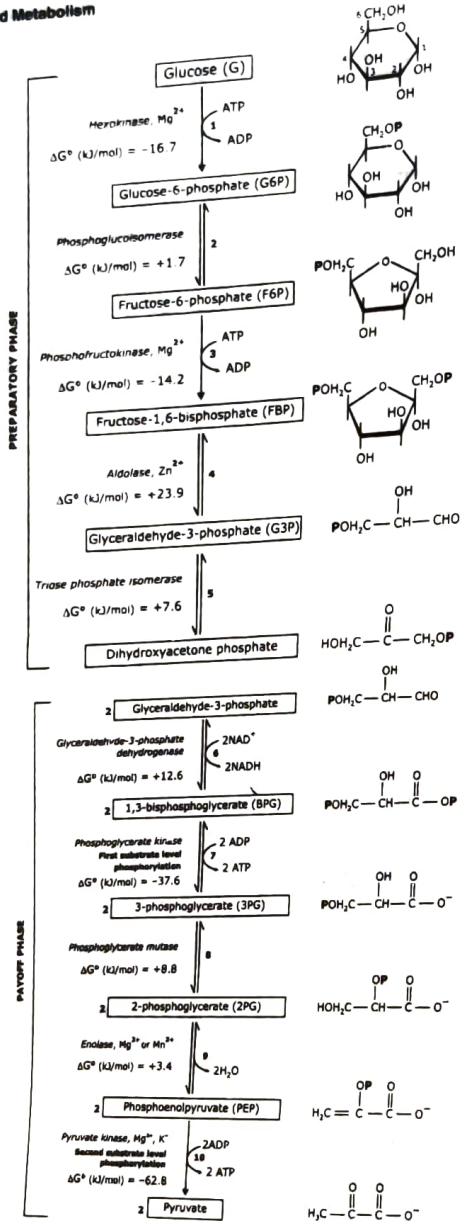
Catabolism of Carbohydrate

The carbohydrate play role as energy production as well as from structural constituents of the individual. Here we will give move emphasis upon the rate of carbohydrate for energy production after its oxidation by means of several pathways. Basically in these pathways the hexose sagar are central to all and in these hexoses the glucose is the main hexose and it is the major fuel of most organisms. There are two major catabolic fate of glucose (hexose) in a living cell, i.e., either it is converted to ribose-5-phosphate after oxidation via pentose phosphate pathway or it is converted to pyruvate via glycolysis.



Glycolysis

Glycolysis is the central energy yielding pathway of same eucaryotic cells and tissues like erythrocytes, Renal medulla, Brain and sperm and also some plants like tubers as well as aquatic plants along with many anerobic microbes. It is an almost central pathway of glucose breakdown with largest flux of carbon is most cell. It in a ten step process driven by a series of enzymes.



These 10 steps are catagorised into two Phases:

1. Preparatory Phase

A preperation of glucose for its breakage. It includes I to V step.

Step I: Phosphorylation of glucose :

Glucose is a very stable molecule as such, so it resists its cleavage, therefor it should be phosphorylated for facilitating cleavage. The phosphoryl group donor is ATP. The reaction is driven by hexokinase engyme which requires Mg^{2+} as cofactor because the substrate for hexokinase is not ATP^{4-} but ATP^{2-} Mg^{2+} complex.

- Hexokinase has four types of Isozymes viz. hexokinase I, II, III and IV.
- The hexokinase IV is found in liver and known as Glucokinase.
- All the four isozymes of hexokinase have different affinities for hexose sugar.
- The hexokinase II of the muscle cell have the most affinity for glucose.
- This reaction is highly energonic and irreversible under physiological conditions. It is a type of random Bi-Bi reaction.
- It is the first regulation point of glycolysis because different affinity for hexose of isozymes of kinase enzym and their different inhibition pattern.

Difference between Glucokinase and Hexokinase

Glucokinase	Hexokinase
1. It is exlusively found in liver.	1. It is found in every tissues like muscles, brain etc.
2. It is only specific for glucose.	2. It is specific for all hexoses.
3. It have relatively less affinity for glucose.	3. It have higher affinity for glucose.
4. It is responsible for distribution of glucose to other tissues.	4. It is responsible only for energy production.
5. It is not inhibited by its reaction product, but inhibited by glucokinase regulatory protein.	5. It is inhibited by glucose-6-P, i.e. its reaction product.

Step II : Conversion of gIucose-6-P to fructose-6-P:

It is a reversible reaction requiring Mg^{2+} and necessary for rearrangement of carbonyl and hydroxyl group of the next step of the reaction, i.e. cleavage of fructose-6- P requires carbonyl at C-2 and hydroxyl at C-4 position, which are not present in glucose-6-P.

It is a general acid base catalytic reaction involving his tebline and Lysine residues of the enzyme phosphohexose isomerase requires Mg^{2+}.

Step III: Phosphorylation of fructose 6-Phosphate to fructose 1-6-Bis Phosphate

In this step, the fructose-6-P is further phosphorylated with the help of enzyme PFK-I (Phosphofructokinase-I) gives to Fructose-1-6-Bis phosphate. It also requires Mg^{2+} and highly energonic, so it is irreversible under physiological conditions and it is the second and main regulatory step of the glycolysis.

Step IV : Cleavage of fructose 1-6-Bis-Phosphate

It is a retro aldol condensation reaction and proceeds through, enolase intermediate. The C_1, C_2 and C_3 carbons of glucose goes to di-hydroxy acetone phosphate and C_4, C_5 and C_6 goes to glyceraldehyde-3-phosphate. The enzyme catalysis reaction is Aldolase which requires Zn^{2+} in some microorganisms.

Step V : Isomerization of Triose Phosphates :

This reaction is a simple isomerization reaction catalyzed by Triose **Phosphate isomerase**. After isomerization, one can not differentiate between the carbons 4-3, 5-2 and 1-6 of the starting fructose molecule in the 2 moles of glyceraldehyde 3-phosphate.

2. Payoff Phase

Production of energy as ATP and reducing equivalent NADH and formation of 2 moles of pyruvate which can be further oxidized to give huge amount of energy by certain proceeding pathways. It includes from step VI to step X.

Step VI: Oxidation of Glyceraldehyde 3-phosphate to 1-3-Bis Phosphoglycerate :

This reaction is catalized by the enzyme, glyceraldehyde-3-phosphate dehydrogenase producing 1-3-Bis phosphoglycerate from glyceraldehyde 3-phosphate.

- In this step the formation of high energy phosphorus hydride bond takes place. This is known as Acyl phosphate having free energy of hydrolysis - 49.3 kJ/mole.
- NO^+ is hybrid. H acceptor.
- This reaction is held if there is no rapid reoxidation of NADH by ETC.
- This reaction is inhibited by iodoacetate by inactivating the enzyme. Therefore, for the production of fructo se-2-6-Bis phosphate, iodoacetate should be used. Because after the PFK-I all the reactions till this stage are reversible.
- The steps VI and VII are energetically controlled and energy conserved in this reaction produced ATP in the next step.

Step VII : Phosphoryl transfer from 1, 3, bisphosphoglycerate to ADP or I ATP generation:

The phosphoryl group from the anhydride cinhage is transferred to form ATP. It is a substrate level phosphorylation by the help of enzyme phosphogylcerate kinase a Mg^{2+} requiring enzyme.

Step VIII: Conversion of 3-Phosphoglycerate to 2-Phosphoglycerate:

It is the two step process. In the first step phosphoryl group of enzyme attached to the position-2 of substrate to form an intermediate that is 2-3-Bis phosphoglycerate.

In the next step, the phosphate from 3 position of substrate transfer to the enzyme phosphoglycerate mutase.

Step IX. Dehydration of 2-Phosphoglycerate to phosphoenol Pyruvate.

In this step-3-Phosphoglycerate is converted to PEP by means of removal of one H_2O/ mole by the action of enzyme enalase.

The total free energy of PEP and 2-phosphoglycerate are same but standard free energy of hydrolysis of the phosphate group of the reactant and product are –17.6 kJ/mole and –61.9 kJ/mole respectively because removal of H_2O fascilitates redistribution of energy.

Step X : Transfer of the phosphoryl group from PEP to ATP or second ATP generation :

It is also a substrate level phosphorylation reaction requires K^+, Mg^{2+} or Mn^{2+} for enzyme Pyruvate kinase.

It is irreversible under physiological conditions with large negative free energy, because the products are stablized by tautomerization.

Balance reaction and Net gain of ATP.

$$\text{Glucose} + 2ATP + 2NAD^+ + 4ADP + 2Pi \longrightarrow 2\text{Pyruvate} + 2ADP + 2NADH + 2H^+ + ATP + 2H^+$$

Net gain of ATP

$$\begin{aligned} -2ATP &\longrightarrow \text{Step I and III} \\ +2ATP &\longrightarrow \text{Step VII} \\ +2ATP &\longrightarrow \text{Step X} \\ +3 \text{ or } 5\ ATP &\longrightarrow \text{From 2NADH Step VI} \end{aligned}$$

Total = 5 or 7 ATP

Note:

If the 2NADH produced in step VI goes to mitochondria via glycerol, 3-phosphate shuffle, then it produced 3 ATP and if it goes via malate as portable suffle, then gives 5ATP.

Glycolysis released only a small fraction of the total available energy of the glucose molecule. When glucose is oxidized completely to CO_2 and H_2O, the total standard free energy charge is -2840 kJ/mole.

Glycolytic degradation of glucose to two molecules of pyruvate ($\Delta G^\circ = -146$ kJ/mole), therefore yields only (146/2840) x 100 = 5.2% of the total energy.

Out of –146 kJ/mole the 61 kJ/mole is required to ATP synthesis in glycolysis.

Thus overall free energy charge of glycolysis is –146 kJ/mol + 61 kJ/mol = – 85 kJ/mol.

Regulation of Glycolysis

The 3-irreversible steps, i.e. I, III and last (X), where I and III are the main regulatory points of the glycolysis.

Regulation of step I:

The hexokinase enzyme which catalizes the entry of free glucose into glycalytic pathway is allosterically inhibited by its reaction product and have different isozymes in different tissues which have different affinities for its substrates and also have different inhibition patterns.

The different isozymes of liver and muscle reflects the different rates of these organs in carbohydrate metabolism. The predominate isozyme of liver is hexokinase also called Glucokinase, which differs in two important respects from the hexokinase of muscle.

(*i*) The glucose concentration at which glucohase is half separated (210 mm) is higher than the usual concentration of glucose in the blood. When the blood glucose concentration is high, as it is a after a meal rich in carbohydrate, excess carbohydrate is transported in hepatocytes where glucokinase converts it to glucose 6-phosphate.

(*ii*) Glucokinase is not inhibited by its reaction product but by glucohinase regulatory protein and it is partial inhibition. This protein have also affinity for fructose-1-phosphate, which compiles with fructose 6-phosphate and cancels its inhibitory effect on glucohinase whereas the hexokinase of myocytes are inhibited by its reaction product, i.e. glucose-6-phosphate.

Regulation at step III

Glucogone phosphorylates the PFK-II when there is high ATP in the cell, which leads to inactivate the phosphoryl are activity of PFK-II and activates phosphate activity.

Since fructose 2-6-Bis phosphate it an activator of PFK-I by binding to its site of binding to inhibitors, i.e., ATP and citrate and ultimately facilitates the formation of fructose-1-6-Bis-phosphate, i.e., Glycolysis.

Hence, we can say that inactivation of phosphorylase activity of PFK-II by means of glucagon at low blood glucose and high ATP and citrate reduces the rate of glycolysis and facilitate gluconeogenesis after activation of phosphatose activity of PFK-I.

At high blood glucose that means high ADP and ip, there is no glucagon and no phosphorylation of PFK-II (no inactivation of phosphorylase activity) and formation of fructose 2-6-Bis-phosphate will be going on which facilitate the glycolysis.

Some important points about Glycolysis.

- All the enzymes of glycolysis are soluble in nature and found in cytosol of the cell.
- Glycolysis is an anaerobic breakdown of glucose but it can also proceeds in aerobic conditions.
- As the step IV of glycolysis have a high +ve free energy, this reaction is dependent to the next steps, i.e., fast removal of the products.
- The intermediates of glycolytic pathway are channeled between the enzymes which speed up the process.
- I, II, III, VII and X step of Glycolysis requires Mg^{2+}.
- X step also requires K^+ and it can also proceeds with Mn^{2+} in place of Mg^{2+}.

Fates of Pyruvate/Fermentation

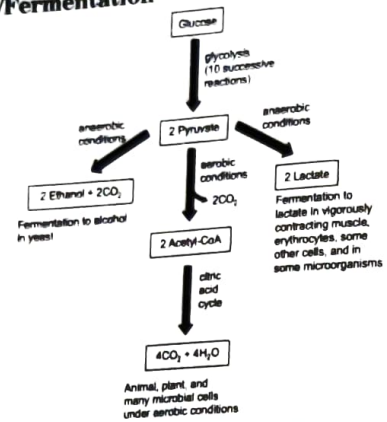
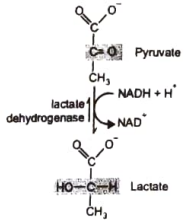


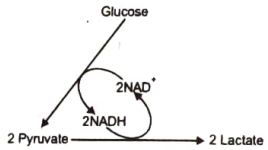
Fig. Three possible catabolic fates of the pyruvate formed in glycolysis.

Lactic acid fermentation

When there is high muscular activity, the skeltal muscle must function under low O_2 content (hyporia), NADH can not be reoxidized to NAD^+ and NAD^+ is required as an electron acceptor for the further oxidation of pyruvate. Under these conditions pyruvate is reduced to lactate, accepting electron from NADH and thereby regenarating the NAD^+ necessary for glycolysis to continue. Certain tissues and cell types (retina, brain and erythrocytes) convert glucose to lactase even under aerobic conditions and lactose is also the product of glycolysis under anarobic conditions in some microorganisms.

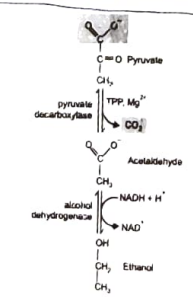


- The dissociation of lactic acid to lactate and H^+ in the fermentation mixture lowers the pH, denaturing casein and other milk products and causing them to precipitate under the correct, controlled conditions the resultant curling produces cheese or yogurt depending on the micro-organism which produces lactate.
- There is no change in oxidation state of carbon or H : C ratio in lactic acid fermentation (from glucose to lactate).



Alcohol or Ethanol fermentation

In some plants tissuses and in some invertebrates, protists and micro-organisms such as Brewer's yeast, pyruvate is converted under hypoxic or anerobic conditions into ethanol and CO_2.



Formation of Acetyl Co-A or Aerobic Breakdown of Pyruvate

Formation of Acetyl Co-A:

In aerobic environment pyruvate is oxidized to acetyl Co-A CO_2 by means of enzyme pyruvate dehydrogenase complex (PDC)

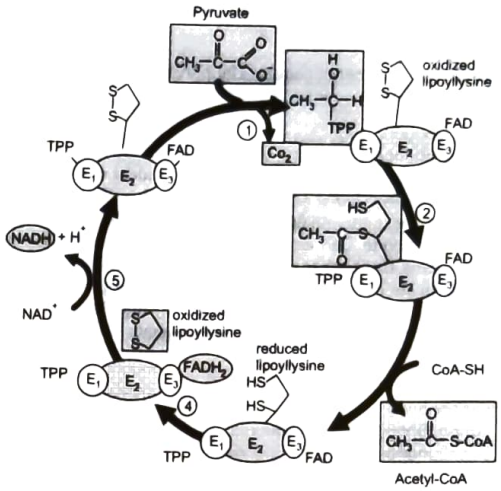


Fig. Oxidative decarboxylation of pyruvate to acetyl-CoA by the pyruvate dehydrogenase complex.

In step (1) pyruvate reacts with the bound thiamine pyrophosphate (TPP) of pyruvate dehydrogenase (E_1), undergoing decarboxylation to the hydroxyethyl derivative. Pyruvate dehydrogenase also carries out step (2), the transfer of two electrons and the acetyl group from TPP to the oxidized form of the lipoylysyl group of the core enzyme, dihydrolipoyl transacetylase (E_2), to form the acetyl thioester of the reduced lipoyl group. Step (3) is a transesterification in which the–SH group of CoA replaces the–SH group of E2 to yield acetyl–CoA and the fully reduced (dithiol) form of the lipoyl group. In step (4) dihydrolipoyl dehydrogenase (E_3) promotes transfer of two hydrogen atoms from the reduced lipoyl groups of E_2 to the FAD prosthetic group of E_3 (shaded yellow). In step (5) the reduced $FADH_2$ of E_3 transfers a hydride ion to NAD^+, forming NADH. The enzyme complex is now ready for another catalytic cycle.

- PDC is cluster of 3-enzymes which are located in the mitochondria of eukaryotes and in the cytosol of 6 prokaryotes.
- There is five co-factors are also required.
- This reaction is oxydative decarboxylation.
- The enzymes and co-enzymes are as follows:

Enzymes :

(1) E_1 → Pyruvate Dehydrogenase

(2) E_2 → Dihydrolipoyl transacetylase

(3) E_3 → Dihydrolipoyl dehydrogenase.

Co-factors :

(1) TPP → Thyamine Pyrophosphate bound with E_1

(2) FAD → Flavin adenine dinucleotide tightly bound with E_3

(3) Lipoate → Bound with E_2

(4) NAD

(5) Co-A-SH.

Regulation of Oxydative Decarboxylation

End-product inhibiton.

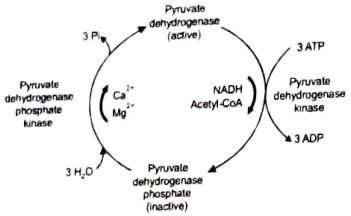
Acetyl-CoA and NADH, both end products of the pyruvate dehydrogenase reaction, are potent allosteric inhibitors of the enzyme. Acetyl-CoA inhibits the transacetylase component whereas NADH inhibits the dihydrolipoyl dehydrogenase component. The inhibitory effects are reversed on addition of coenzyme A and NAD^+ respectively.

Feedback regulation:

The activity of PDC is controlled by the energy charge. The pyruvate dehydrogenase component is specifically inhibited by GTP and activated by AMP.

Covalent modification:

Under conditions of high concentrations of ATP, acetyl-CoA and those of the cycle intermediate, further formation of acetyl-CoA is slowed down. This is accomplished by what is called covalent modification. ATP inactivates pyruvate dehydrogenase complex (PDC) by phosphorylating 3 different serine residues of its pyruvate dehydrogenase component in the presence of an auxiliary enzyme, pyruvate dehydrogenase kinase. NADH, AP and acetyl-CoA stimulate the rate of phosphorylation whereas pyruvate, Ca^{2+} and K^+ inhibit/phosphorylation. However, if the demand for ATP increases causing ATP level to decline, the inactive enzyme complex becomes active again by the hydrolytic removal of phosphate group from its pyruvate dehydrogenase component in the presence of another enzyme, pyruvate dehydrogenase phosphate kinase. Diphosphorylation is enhanced by high levels of pyruvate. Ca^{2+} and Mg^{2+}. Both pyruvate dehydrogenase kinase and pyruvate dehydrogenase phosphate kinase are also present in PDC.



TCA or Citric acid Cycle or Kreb Cycle

TCA cycle has the eight successive steps which ultimatly leads to formation to 2 moles of CO_2 and H_2O.

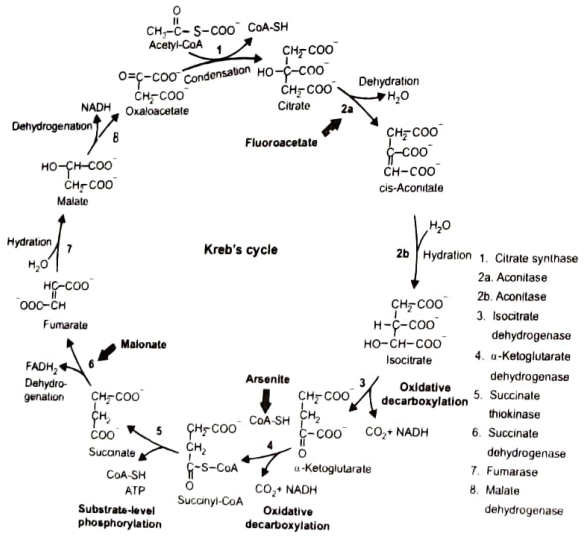


Fig. Kreb cycle

Step I: Formation of citrate.

In this step, condensation of acetyl-Co-A with oxaloacetate takes place producing citrate with the help of enzyme citrate synthase. It proceeds through intermediate, i.e. citroyl-Co-A. The Co-A liberated in this reaction is recycled to participate in the oxidative decarboxylation of another molecule of pyruvate by PDC.

Step II: Formation of Isocitrate.

In this step, dihydration of isocitrate produces isocitrate via an intermediate Cis -Aconitate with the help of enzyme aconitase containing Fe-S-centre.

At physiological pH and tempreture, there is only 10% isocitrate is found but reaction procceeds continuously because of the next step mantains the fear towards right hand side.

Step III: Oxidation of Isocitrate :

In this step, isocitrate is oxidized to α-ketoglutarate (one C-less) and CO_2 with the help of enzyme isocitrate dehydrogenalase.

- Two types of isocitrate dehydrogenases are found in the cell. One of them requires NAD^+ and other $NADP^+$.
- In eukaryotes the NAD^+ dependent enzyme occur in mitochondria whereas $NADP^+$ dependent occurs in both cytosol as well as mitochondria.

Step IV : Oxidation of α-ketoglutarate :

In this step, α-ketoglutarate and CO_2 (one carbon less), by means of enzymetic action of α-keto dehydrogenase complex.

- This reaction is irreversible under physiological conditions.
- This complex is virtually identical to PDC and have homologous E_1, E_2 and E_3 as well as TPP, Lipoate, EAD, NAD and Co-A.

Step V: Formation of succinate :

In this step, conversion of succinage-CoA to succinate takes place with the action of enzyme succinate CoA synthatase.

It is a substrate level phosphorylation reaction.

Step VI: Oxidation of Succinate :

In this step, succinate is oxidized to form fumerate by the enzymetic action of succinate dehydrogenase.

- The enzyme succinate dehydrogenase is only membrane bound enzyme to inner mitochondrial membrane in eukaryotes and in prokaryotes to the plasma membrane.
- The enzyme have 3 different Fe-S-groups, and one bound FAD.
- Malonate is inhibitor of succinate dehydrogenase.

Step VII : Hydration of fumerate :

In this step, the fumerate is hydrated to form malate The enzyme is fumerase.

It is simple hydrolysis reaction and enzyme is highly steriospecific, i.e., it cleaves only trans double bond not cis

Step VIII: Oxidation of Malate :

In this step, malate is oxidized to form oxaloacetate, which goes to accept another molecule of acetyl-Co A. This reaction is catalized by the enzyme malate dehydrogenase.

- The enzyme catalizing this reaction consists of α-subunits and highly steriospecific and requires NAD.
- At high NAD^+/NADH ratio is stimulatory and a low ratio is inhibitory.

Overall Balance sheet and Net gain of ATP:

$$\text{Acetyl-CoA} + 3NAD^+ + FAD + GDP + Pi + 2H_2O \xrightarrow{\text{Kreb's Cycle}} 2CO_2 + CoA\text{-}SH + 3NADH + FADH_2 + GTP + 2H^+$$

Stoichiometry of Coenzyme Reduction and ATP formation in the aerobic oxidation of glucose via glycolysis, the pyruvate dehydrogenase reaction, the citric acid cycle, and oxidative phosphorylation.

Reaction	Number of ATP or reduced coenzymes directly formed	Number of ATP ultimately formed
Glucose ⟶ glucose 6-phosphate	– 1 ATP	– 1
Fructose 6-phosphate ⟶ fructose 1, 6-bisphosphate	– 1 ATP	– 1
2 Glyceraldehyde 3-phosphate ⟶ 2 1,3-bisphasphoglycerate	2 NADH	3 – 5
2 1,3-Bisphosphoglycerate ⟶ 2 3-phosphoglycerate	2 ATP	2
2 Phosphoenolpyruvate ⟶ 2 pyruvate	2 ATP	2
2 Pyruvate ⟶ 2 acetyl-CoA	2 NADH	5
2 Isocitrate ⟶ 2 α-ketoglutrate	2 NADH	5
2 α-Ketoglutarate ⟶ 2 succinyl-CoA	2 NADH	5
2 Succinyl-CoA ⟶ 2 succinate	2 ATP (or 2 GTP)	2
2 Succinate ⟶ 2 fumarate	2 $FADH_2$	3
2 malate ⟶ 2 oxaloacetate	2 NADH	5
Total		30-32

* This is calculated as 2.5 ATP per NADH and 1.5 ATP per FADH2, A negative value indicates consumption.

Regulation of TCA Cycle

Regulation of metabolite flow from pyruvate through the citric acid cycle:

The pyruvate dehydrogenase complex is allosterically inhibited when the ratios of [ATP]/[ADP], [NADH/NAD^+], and [acetyl-CoA]/[CoA] are high, indicating an energy-sufficient metabolic state. When these ratios decrease, allosteric activation of pyruvate oxidation results. The rate of flow through the citric acid cycle can be limited by the availability of the citrate synthase substrates, oxalocetate and acetyl-CoA, or of NAD^+ (depleted by its conversion to NADH), which slows the three NAD-dependent oxidation steps. Feedback inhibition by succinyl-CoA, citrate, and ATP also slows the cycle by inhibiting early steps. In muscle tissue, Ca^{2+} signals contraction and, as shown here, stimulates energy-yielding metabolism to replace the ATP consumed by contraction.

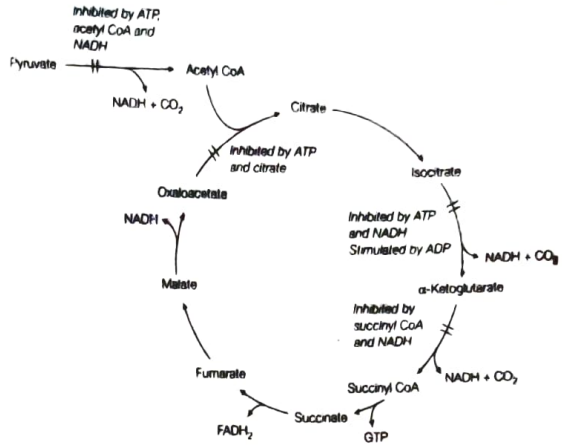


Fig. Regulation points of the citric acid cycle.

Some important points about TCA Cycle :

The citric acid cycle is also called Amphibolic cycle because intermediates of this cycle can be drawn off and used as the precursor of biasynthetic products, e.g. 2kg is a biosynthetic precursor amino acids and acetyl Co-A in lipogenesis.

Difference between Synthase and Synthatase:

- Synthases are the enzymes which catalizes the reaction in which no-nucleotide triphosphates (ATP, GTP etc.) is required.
- Synthatase are the ATP, GTP etc, i.e. nucleotide triphosphate requiring enzyme.
- The complete oxidation of reduced glucose is done through only Kreb's cycle.
- Kreb's cycle is ultimate cycle for the oxidation of all 8-energy producing biomolecules i.e., carbohydrate, lipids and proteins.

Electron Transport System

Redox Potential

The oxidation of a molecule involves the loss of electrons. The reduction of a molecule involves the gain of electrons. Since electrons are not created or destroyed in a chemical reaction, if one molecule is oxidized, another must be reduced (i.e., it is an *oxidation-reduction reaction*). Thus, by definition, oxidation-reduction reactions involve the transfer of electrons. In the oxidation-reduction reaction:

$$NADH + H^+ + \tfrac{1}{2}O_2 \rightleftharpoons NAD^+ + H_2O$$

when the NADH is oxidized to NAD^+, it loses electrons. When the molecular oxygen is reduced to water, it gains electrons. For the reaction:

$$NADH + H^+ + \tfrac{1}{2}O_2 \rightleftharpoons NAD^+ + H_2O$$

$$\Delta E_0' = +1.14V$$

$$\Delta G^{0\prime} = -52.6 \text{ kcal mol}^{-1}$$

Electron transport from NADH

Comparing the energetics of the oxidation of NADH:

$$NADH + H^+ + \tfrac{1}{2}O_2 \rightleftharpoons NAD^+ + H_2O \qquad \Delta G^{0\prime} = -52.6 \text{ kcal mol}^{-1}$$

and the synthesis of ATP:

$$ADP + P_i + H^+ \rightleftharpoons ATP + H_2O \qquad \Delta G^{0\prime} = +7.3 \text{ kcal mol}^{-1}$$

it is clear that the oxidation of NADH releases sufficient energy to drive the synthesis of several molecules of ATP. However, NADH oxidation and ATP synthesis do not occur in a single step. Electrons are not transferred from NADH to oxygen directly. Rather the electrons are transferred from NADH to oxygen along a chain of electron carriers collectively called the *electron transport chain* (also called the *respiratory chain*).

The main part of the electron transport chain consists of three large protein complexes embedded in the inner mitochondrial membrane, called NADH dehydrogenase, the cytochrome bc_1, complex and cytochrome oxidase. Electrons flow from NADH to oxygen through these three complexes as shown. in Fig. Each complex contains several electron carriers (see below) that work sequentially to carry electrons down the chain. Two small electron carriers are also needed to link these large complexes; ubiquinone, which is also called coenzyme Q (abbreviated here as CoQ), and cytochrome c.

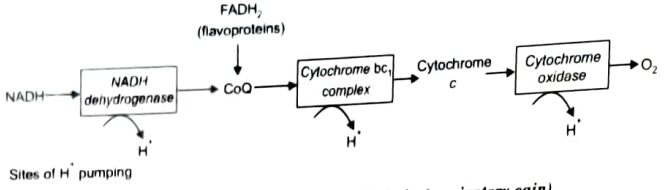


Fig.Overview of the electron transport chain (respiratory cain).

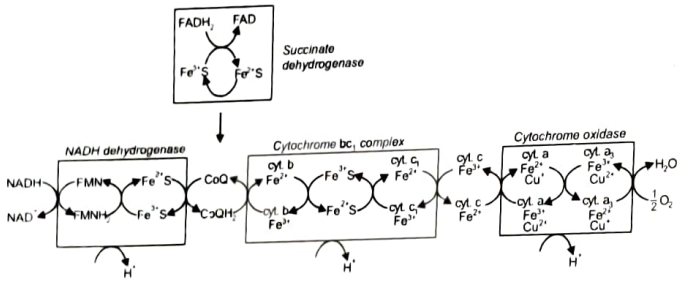


Fig. Details of electron transport.

NADH to NADH dehydrogenase

NADH dehydrogenase (also called NADH-Q reductase or complex I) consists of at least 30 polypeptides. It binds the NADH and re-oxidizes it to NAD^+, passing the two electrons from NADH to a prosthetic group called FMN (flavin mononucleotide) to produce $FMNH_2$. Each electron is accepted together with a hydrogen ion, H^+, such that two electrons and two H^+ are accepted in total. The electrons are then transferred, within the NADH dehydrogenase complex, to iron-sulfur clusters (FeS) in iron-sulfur proteins (also called nonheme iron proteins). Several types of FeS clusters exist but in each case the iron atoms are coordinated to inorganic sulfur atoms and the sulfur of cysteine side chains in the protein. Within an FeS cluster, an electron is carried by the iron atom which, on accepting the electron, changes from the Fe^{2+} (ferric) state to the Fe^{3+} (ferrous) state. As the electron is passed to another electron carrier, the iron atom of the FeS cluster changes back again to the Fe^{3+} state.

NADH dehydrogenase to ubiquinone (CoQ)

Electrons from the FeS clusters of NADH dehydrogenase are passed on to ubiquinone (CoQ), a small lipid-soluble molecule in the inner mitochondrial membrane. This molecule can act as an electron carrier by accepting up to two electrons and two H^+ ions. In so doing, ubiquinone (CoQ) is converted to ubiquinol ($CoQH_2$).

Ubiquinol to cytochrome bc_1 complex

When ubiquinol ($CoQH_2$) donates its two electrons to the next carrier in the chain, the cytochrome bc_1 complex (also called cytochrome reductase or complex III), the H^+ ions are released once more. The cytochrome bc_1 complex contains two types of cytochromes, cytochrome *b* and cytochrome c_1, as well as an FeS protein (Fig.). A

cytochrome is a protein with a bound heme group that contains an iron atom (Fig.). Different cytochromes have different heme groups, but all cytochromes have the ability to act as electron carriers. As the electron is accepted, the iron atom of the heme group change from the Fe^{3+} (ferric) state to the Fe^{2+} (ferrous) state. Figure 2 shows the electrons passing from ubiquinol (QH_2) through the cytochrome *b*, FeS and cytochrome c_1 components of the cytochrome bc_1 complex to the next electron carrier, cytochrome *c*. Since ubiquinol is a two-electron carrier whereas cytochromes are one-electron carriers, the pathway of electron transfer within the cytochrome bc_1 complex is complicated and involves ubiquinol ($CoQH_2$) releasing first one electron and an H^+ ion to become ubisemiquinone (CoQH) and then the second electron and H ion to become ubiquinone (CoQ).

Cytochrome bc_1 complex to cytochrome c to cytochrome oxidase

Cytochrome *c* is a peripheral membrane protein that is loosely bound to the outer surface of the inner mitochondrial membrane. It binds to the cytochrome bc_1 complex and accepts an electron via an Fe^{3+} to Fe^{2+} transition. Then it binds to cytochrome oxidase and donates the electron, with the iron atom of the heme of cytochrome *c* then reverting to the Fe^{3+} state(Fig.).

Cytochrome oxidase to oxygen

Cytochrome oxidase (also called complex IV) contains two cytochromes (cytochrome *a* and a_3). Cytochrome *a* is paired with a copper atom, Cu_A, and cytochrome a_3 is paired with a different copper atom, Cu_B. During electron transfer, the iron atoms of the cytochromes cycle between the Fe^{3+} and Fe^{2+} states whilst the copper atoms cycle between Cu^{2+} and Cu^+. The cytochrome oxidase reaction is complex; it transfers four electrons from four cytochrome *c* molecules and four H^+ ions to molecular oxygen to form two molecules of water:

$$4 \text{ cyt. } c\ (Fe^{2+}) + 4\ H^+ + O_2 \xrightarrow{\text{Cytochrome oxidase}} 4 \text{ cyt. } c\ (Fe^{3+}) + 2\ H_2O$$

Formation of an H^+ gradient

The change in redox potential along the chain is a measure of the free energy change occurring. The potential falls (i.e. becomes more positive) throughout the chain but mainly in three large steps that correspond to the three main protein complexes: the NADH dehydrogenase complex, the cytochrome bc_1 complex and the cytochrome oxidase complex. The large free energy change at each of these three steps, and only these three steps, is large enough to pump H^+ ions from the mitochondrial matrix across the inner mitochondrial membrane and into the intermembrane space. Thus, each of these three complexes is an H^+ pump driven by electron transport. Overall, therefore, electron transport along the chain from NADH releases energy that is used to create an H^+ gradient.

Electron transport from $FADH_2$

Succinate dehydrogenase catalyzes the oxidation of succinate to fumarate in the citric acid cycle. The succinate dehydrogenase contains bound FAD that is reduced to $FADH_2$ in the reaction. The re-oxidation of the $FADH_2$ occurs via succinate-coenzyme Q reductase (also called complex II), an integral protein of the inner mitochondrial membrane. Succinate dehydrogenase is part of this complex but it also contains FeS clusters. During re-oxidation of $FADH_2$, the two electrons pass from the $FADH_2$ to the FeS clusters and are then passed on to ubiquinone (CoQ; Fig.). They then enter the main electron transport chain and cause H^+ ions to be pumped out of the mitochondrion as for the oxidation of NADH. However, succinate-CoQ reductase itself is not an H^+ pump because the free energy change of the overall reaction is too small. The $FADH_2$ of other flavoproteins, such as mitochondrial glycerol 3-phosphate dehydrogenase in the glycerol 3-phosphate shuttle and fatty acyl CoA dehydrogenase in fatty acid oxidation also feed their electrons into the electron transport chain at ubiquinone.

Electron Transport Inhibitors

Several inhibitors of specific electron carriers are known and were used in the original studies to determine the order of the components in the respiratory chain. For example:

- *rotenone* and *amytal* inhibit electron transport at NADH dehydrogenase and so prevent NADH oxidation but the oxidation of $FADH_2$ can still occur since this feeds electrons into the chain at CoQ (i.e. past the point of inhibition);
- *antimycin A* inhibits electron transport at the cytochrome bc_1, complex;
- *cyanide* (CN^-), *azide* (N_3^-) and *carbon monoxide* (CO) all inhibit cytochrome oxidase.

Oxidative phosphorylation

Oxidative phosphorylation is the name given to the synthesis of ATP (*phosphorylation*) that occurs when NADH and $FADH_2$ are oxidized (hence *oxidative*) by electron transport through the respiratory chain. Unlike substrate level phosphorylation it does not involve phosphorylated chemical-intermediates. Rather, a very different mechanism was proposed by Peter Mitchell in 1961, the **chemiosmotic** hypothesis. This proposes that energy liberated by electron transport is used to create a proton gradient across the mitochondrial inner membrane and that it is this that is used to drive ATP synthesis. Thus the proton gradient couples electron transport and ATP synthesis, not a chemical intermediate. The evidence is overwhelming that this is indeed the way that oxidative phosphorylation works. The actual synthesis of ATP is carried out by an enzyme called ATP synthase located in the inner mitochondrial membrane (Fig.).

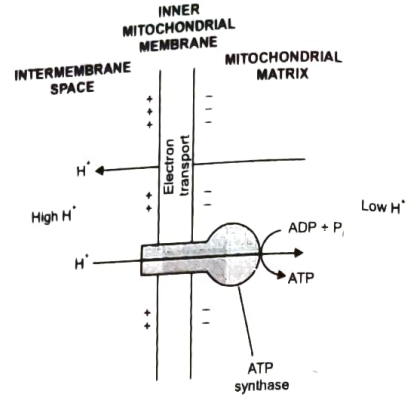


Fig. The mechanism of oxidative phosphorylation

In summary, the process is as follows. Electron transport down the respiratory chain from NADH oxidation causes H^+ ions to be pumped out of the mitochondrial matrix across the inner mitochondrial membrane into the intermembrane space by the three H^+ pumps; NADH dehydrogenase, the cytochrome bc_1 complex and cytochrome oxidase. [Because $FADH_2$ is reoxidized via ubiquinone, its oxidation causes H^+ ions to be pumped-out only by the cytochrome bc_1 complex and cytochrome oxidase and so the amount of ATP made from $FADH_2$ is less than from $NADH_2$.] The free energy change in transporting an electrically charged ion across a membrane is related both to its electrical charge and the concentration of the species. The pumping out of the H^+ ions generates a higher concentration of H^+ ions in the intermembrane space and an electrical potential, with the side of the inner mitochondrial membrane facing the intermembrane space being positive (Fig.). Thus, overall, an electrochemical proton gradient is formed. The protons flow back into the mitochondrial matrix through the ATP synthase and this drives ATP synthesis. The ATP synthase is driven by proton-motive force, which is the sum of the pH gradient (i.e. the chemical gradient of H^+ ions) and the membrane potential (i.e., the electrical charge potential across the inner mitochondria membrane). There is some debate over the exact stoichiometry of ATP production; in past years it was believed that 3 ATP were generated per NADH and 2 ATP per $FADH_2$ but some recent measurements have indicated that the numbers of ATP molecules generated may be 2.5 and 1.5, respectively.

ATP Synthase as a Rotatory Engine

The ATP synthase can be seen as spherical projections from the inner membrane (Fig.). If mitochondria are subjected to sonic disruption, submitochondrial vesicles are formed in which the spheres of the ATP synthase

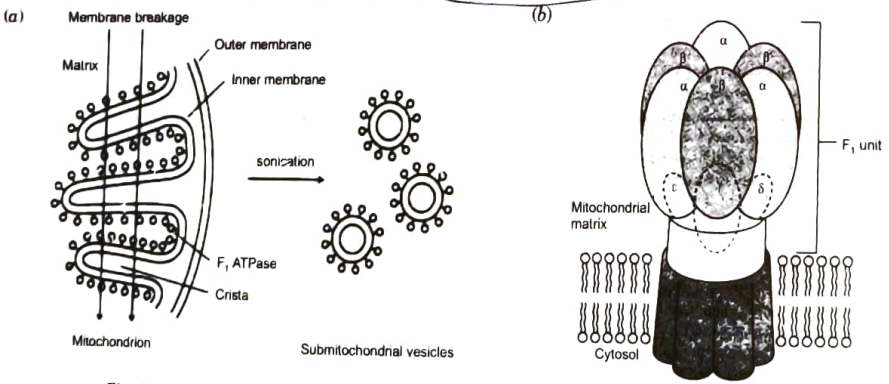


Fig. (a) Sonic disruption (sonication) of mitochondria produces submitochondrial vesicles, (b) schematic representation of the ATP synthase complex.

point outward (Fig. *a*). In 1960, Racker showed that the spheres can be removed and that the isolated spheres *hydrolyze* ATP, that is, the spheres have ATPase activity (called F_1 ATPase; Fig. *b*). F_1 ATPase contains five types of polypeptide in the ratio $(\alpha\beta)_3\gamma\delta\varepsilon$. The stripped submitochondrial vesicles, devoid of the F_1 ATPase, can still transport electrons along the electron transport chain but cannot synthesize ATP. These stripped submitochondrial vesicles contain the other major part of the ATP synthase, called F_0 (coupling factor 0) which is a proton channel that spans the inner mitochondrial membrane (Fig. *b*). Since it is composed of these two major component parts, ATP synthase is also known as F_0F_1 ATPase. The stalk between F_0 and F_1 (Fig. *b*) contains several additional polypeptides. The complete complex harnesses the energy released by electron transport to drive ATP synthesis whereas alone, without coupling to electron transport, the F_1 component hydrolyzes ATP.

Amazingly it has recently been shown that the F_1 portion of ATP synthase behaves as a rotatory engine: during ATP hydrolysis (and presumably also during ATP synthesis) submit γ of the F_1 ATPase rotates relative to $(\alpha\beta)_3$. In fact, this is the smallest rotatory engine so far discovered in nature!

Coupling and Respiratory Control

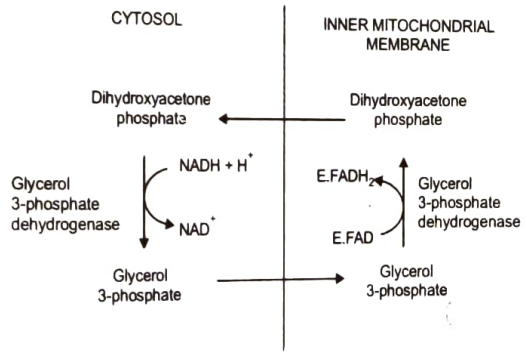
If ADP is added to mitochondria, the rate of oxygen consumption rises as electrons flow down the chain and then the rate of oxygen utilization falls when all the ADP has been phosphorylated to ATP; a process called respiratory control. This mechanism ensures that electrons flow down the chain only when ATP synthesis is needed. If the level of ATP is high and the ADP level is low, no electron transport occurs, NADH and $FADH_2$ build up, as does excess citrate, and the citric acid cycle and glycolysis are inhibited.

Uncouplers

Some chemicals, such as 2,4-dinitrophenol (DNP), act as uncoupling agents, that is, when added to cells, they stop ATP synthesis but electron transport still continues and so oxygen is still consumed. The reason is that DNP and other uncoupling agents are lipid-soluble small molecules that can bind H^+ ions and transport them across membranes (i.e. they are H^+ ionophores). Electron transport occurs and pumps out H^+ ions across the inner mitochondrial membrane but DNP in the same membrane carries the H^+ ions back into the mitochondrion, preventing formation of a proton gradient. Since no proton gradient forms, no ATP can be made by oxidative phosphorylation. Rather the energy derived from electron transport is released as heat.

The production of heat by uncoupling is called nonshivering thermogenesis. It is important in certain biological situations. For example, uncoupling occurs naturally in brown adipose tissue. This tissue is rich in mitochondria, the inner mitochondrial membranes of which contain a protein called thermogenin (or uncoupling protein). Thermogenin allows H^+ ions to flow back into mitochondria without having to enter via the ATP synthase and so uncouples electron transport and oxidative phosphorylation, generating heat. The importance of this natural phenomenon is that brown adipose tissue is found in sensitive body areas of some newborn animals (including humans) where the heat production provides protection from cold conditions. In addition, thermgenesis by brown adipose tissue plays a role in maintaining body temperature in hibernating animals.

Reoxidation of Cytosolic NADH or Shuttle System



The inner mitochondrial membrane is imper- meable to NADH. Therefore NADH produced in the cytoplasm during glycolysis must be reoxidized via a *membrane shuttle*, a combination of enzyme reactions that bypass this impermeability barrier. Figure shows the *glycerol 3-phosphate* shuttle. Dihydroxyacetone phosphate in the cytosol is reduced to glycerol 3-phosphate, and NADH reoxidized to NAD^+, by cytosolic glycerol 3-phosphate dehydrogenase. The glycerol 3-phosphate diffuses across the inner mitochondrial membrane where it is converted back to dihydroxyacetone phosphate by mitochondrial glycerol 3-phosphate dehydrogenase, a transmembrane protein of the inner mitochondrial membrane. The dihydroxyacetone phosphate then diffuses back to the cytosol. The mitochondrial glycerol 3-phosphate dehydrogenase does not use NAD^+ but instead uses FAD. The enzyme-linked $FADH_2$ ($E.FADH_2$) is then reoxidized by transferring its electrons to ubiquinone in the same inner mitochondrial membrane. Note that the shuttle does not allow cytoplasmic NADH to enter the mitochondrion but its operation effectively transports the two electrons from the NADH into the mitochondrion and feeds them into the electron transport chain. Since the electrons from cytoplasmic NADH actually enter the electron transport chain from $FADH_2$, only about two ATPs are synthesized instead of approximately three ATPs from each NADH that arises inside the mitochondrion from the citric acid cycle and fatty acid oxidation.

A similar shuttle, the *malate-aspartate* shuttle, operates in heart and liver (Fig.). Oxaloacetate in the cytosol is converted to malate by cytoplasmic malate dehydrogenase, reoxidizing NADH to NAD^+ in the process. The malate enters the mitochondrion via a malate-α-ketoglutarate carrier in the inner mitochondrial membrane. In the matrix the malate is reoxidized to oxaloacetate by NAD^+ to form NADH. Oxaloacetate does not easily cross the inner mitochondrial membrane and so is transaminated to form aspartate which then exits from the mitochondrion and is reconverted to oxaloacetate in the cytosol, again by transamination. The net result of this cycle of reactions is to transfer the electrons from NADH in the cytosol to NADH in the mitochondrial matrix which is then reoxidized by the electron transport chain.

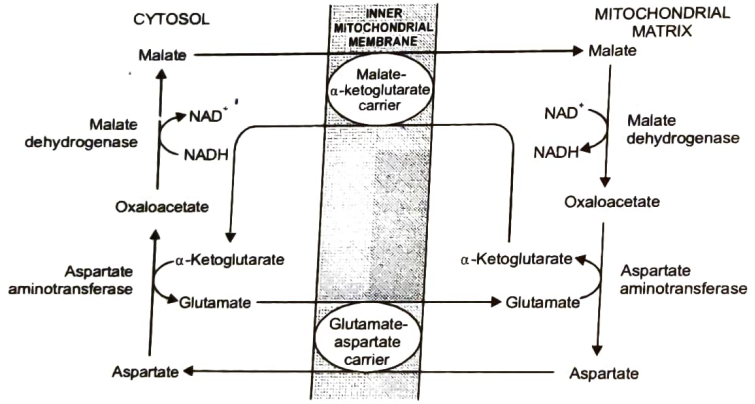


Fig. The malate-aspartate suttle.

The Enter-doudoroff Pathway

Another pathway of glucose catabolism is called the Enter-Doudoroff pathway. It is found in both aerobic and anaerobic prokaryotes but not in eukaryotes. It is fairly widespread, particularly among Gram-negative bacteria, glucose is phosphorylated to glucose-6-phosphate. It is then oxidized to 6-phosphogluconic acid. A dehydration step follows to yield 2-keto-3deoxy-6-phosphogluconic acid (KDPG); the latter is cleaved to pyruvic acid and glyceraldehyde-3-phosphate, which is metablized via some Embden-Meyerhof pathway enzymes to produced a second molecules of pyruvic acid. In the aerobic pseudomonas the catabolism is completed via acetyl-CoA and the tricarboxylic acid cycle.

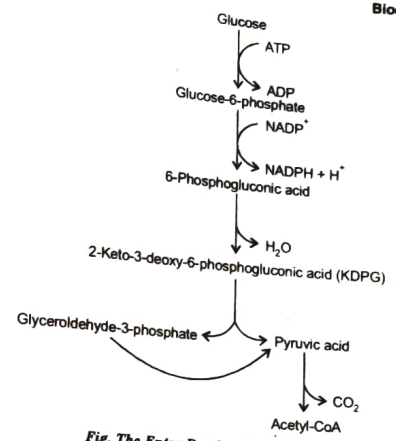


Fig. The Enter Doudoroff pathway

Modification of the Citric Acid Cycle : Glyoxylate Cycle

The task (of converting fats into carbohydrates) in organism is, however accomplised by means of a cyclic set of reactions called glyoxylate cycle or Krebs-Kornberg cycle, the latter nomenclature based on its two principal investigators, Hans A. Krebs and H. L. Kornberg, both of whom discovered this cycle in 1957. The glyoxylate cycle utilizes five enzymes, of which two namely isocitrate lyase and malate synthase are absent in animals. Hence, the glyoxylate phathway does not occur in animals. However, plant cells contain all the 5 enzymes required for the cycle, in subcellular organelles called glyoxysomes, hence this cycle is operative in them. It is interesting to note that the glyoxysomes appear in the cotyledons of lipid-rich seeds (e.g., groundnut, castor, bean) shortly after germination begins and at a time when lipids are being utilized as the major source of carbon for carbohydrate synthesis.

In effect, the glyoxylate cycle bypasses steps 2 to 7 of the citric acid cycle, thereby omitting the two oxidative decarboxylation reaction (Steps 3 and 4) in which CO_2 is produced. The bypass consists of 2 reactions, namely

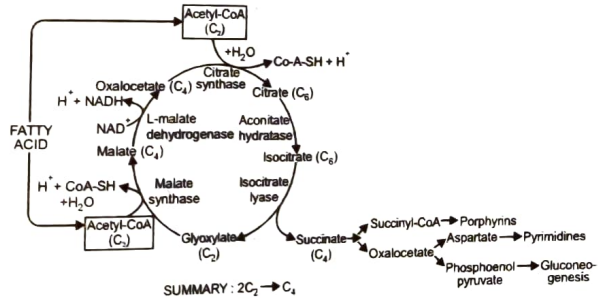


Fig. The glyoxylate cycle Oxidation of fatty provide acetyl-CoA, which enters the cycle at two points. Each turn of the cycle results in generation of one mole of succinate from 2 mole of acetyl-CoA. The net result of the pypass is the formation of C—C bond between C_4 subunits.

(a) Splitting of isocitrate into succinate and glyoxylate, and

(b) Conversion of glyoxylate into malate.

Step 1 : Condensation of acetyl-CoA with axaloacetate

This reaction is catalyzed by the enzyme, citrate synthase.

Setp 2 : Isomerization of citrate into isocitrate

This reaction is catalyzed by the enzyme, aconitate hydratase which is also known as aconitase

Setp 3 : Cleavage of isocitrate into succinate and glyoxylate

Isocitrate, instead of being oxidized *via* the CAC, is rather split into succinate and glyoxylate by the enzyme, isocirate lyase (- isocitrase or isocitratase). Isocitrate lyase from Escherichia coli is, however, inhibited by phosphoenolpyruvate.

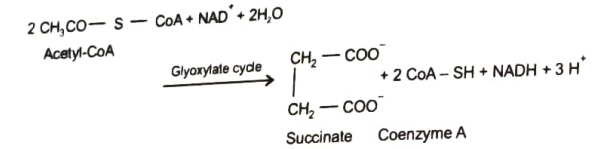
Setp 4 : Conversion of glyoxylate into malate

The glyoxylate, so formed then condenses with a mole of acetyl-CoA to produce L-malic acid by the enzyme, malate synthase.

Setp 5 : Dehydrogenation of malic to oxaloacetate

L-malate is then dehydrogenated, in the presence of L-malate dehydrogenase, to oxaloacetate in the citric acid cycle the convession of isocitrate to malate is an aerobic process, in glyoxylate cycle the conversion takes place anerobically.

The overall reaction for the glyoxylate cycle may be written as



Thus, succinate may be anaplerotic, giving rise to oxaloacetate, or gluconeogenic, giving rise to phosphoenolpyruvate. In this way, net carbohydrate synthesis form fatty acids via acetyl CoA, which is not accomplished in animals, is achieved in the higher plants and microorganisms.

The glyoxylate pathway makes use of both mitcohondrial and glyoxysomal enzymes

Pentose Phosphate Pathway of Glucose Oxidation

In most animal tissues, the major fate of glucose is glycolytic breakdown to pyruvate:"; most of which is oxidized via the citric acid cycle the main function being formation of ATP. Glucose does have other catabolic fates, however, which lead to specialized products needed by the cell. Of particular importance is the synthesis of pentose phosphates.

The **pentose phosphate pathway**, also called the phosphogluconate pathway, produces NADPH and ribose 5-phosphate.

NADPH carries chemical energy in the form of reducing power and is used almost universally as the reductant in anabolic pathway.

In mammals this role of NADPH—and thus the activity of the pentose phosphate pathway—is especially prominent in tissues actively synthesizing fatty acids and steroids, such as the mammary gland, adrenal cortex, liver, and adipose tissue.

These tissues use NADPH to reduce the double bonds and carbonyl groups of intermediates in the synthetic process. Tissues less active in synthesizing fatty acids, such as skeletal muslce, are virtually lacking in the pentose phosphate pathway.

A second function of the pentose phosphate pathway is to generate pentoses, particularly D-ribose, necessary for the biosynthesis of nucleic acids.

Nucleic acid biosynthesis occurs at high rates in growing and regenerating tissues and in tumors.

The first reaction of the pentose phosphate pathway is the dehydrogenation of glucose 6-phosphate by glucose 6-phosphate dehydrogenase to form 6-phosphoglucono-8-lactone, an intramolecular ester. $NADP^+$ is the electron acceptor, and the overall equilibrium lies far in the direction of NADPH formation. The lactone is hydrolyzed to the free acid 6-phosphogluconate by a specific lactonase, then 6-phosphogluconate undergoes dehydrogenation and decarboxylation by 6-phosphogluconate dehydrogenase to form the ketopentose D-ribulose 5-phosphate. This reaction generates a second molecule of NADPH. Phosphopentose isomerase converts ribulose 5-phosphate to its aldose isomer, D-ribose 5-Phosphate. In some tissues, the pentose phosphate pathway ends at this point, and its overall equation is

Glucose 6-phosphate + 2NADP+ + H_2O

→ ribose 5-phosphate + CO_2 + 2NADPH + $2H^+$

The net result is the production of NADPH, a reductant for biosynthetic reactions, and ribose 5-phosphate, a precursor for nucleotide synthesis.

In tissues that require primarily NADPH rather than ribose 5-phosphate, pentose phosphates are recycled into glucose 6-phosphate in a series of reaction. First, ribulose 5-phosphate is epimerized to xylulose 5-phosphate. Then, in a series of rearrangements of the carbon skeletons, six five-carbon sugar phospates are converted to five six-carbon sugar phosphates, completing the cycle and allowing continued oxidation of glucose 6-phosphate with production of NADPH. In erythrocytes, the NADPH produced by this pathway is essential in protecting the cell crom oxidative damage, a genetic defect in glucose 6-phosphate dehydrogenase can have serious consequences.

In the nonoxidative part of the pentose phosphate pathway transketolase, a TPP-dependent enzyme, catalyzes the transfer of a two-carbon fragment (C-1 and C-2) of xylulose 5-phosphate to ribose 5-phosphate, forming the seven-carbon product sedoheptulose 7-phosphate. The remaining three-carbon fragment of xylulose is glyceraldehyde 3-phosphate. Transaldolase then catalyzes a reaction similar to the aldolase reation of glycolysis : a three-carbon fragment is removed from sedoheptulose 7-phosphate and condensed with glyceraldehyde 3-phosphate, forming fructose 6-phosphate. The remaining four-carbon fragment of sedoheptulose is erythrose 4-phosphate. Now transketolase acts again, forming fructose 6-phosphate and glyceraldehyde 3-phosphate from erythrose 4-phosphate formed by two interations of these reactions can be converted into a molecule of fructose 1,6-bisphosphate. The cycle is then complete: six pentose phosphates have been converted to five hexose phosphates.

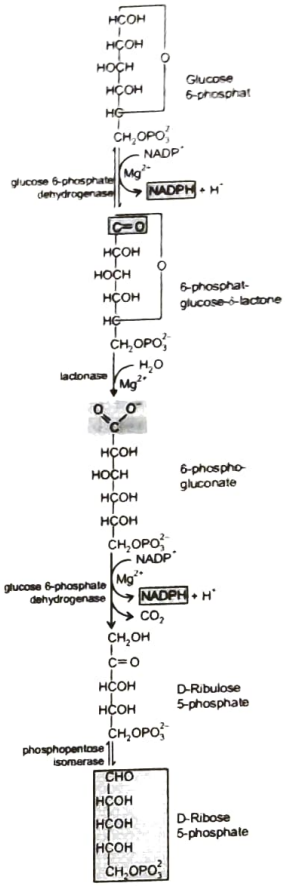


Fig. Oxidative reactions of the pentose phosphate pathway.
The end products are D-ribose 5-phosphate and NADPH.

All reactions of the nonoxidative part of the pentose phosphate pathway are readily reversible and thus also provide a means of converting hexose phosphates to pentose phosphates. This is essential in the photosynthetic fixation of CO_2 by plants.

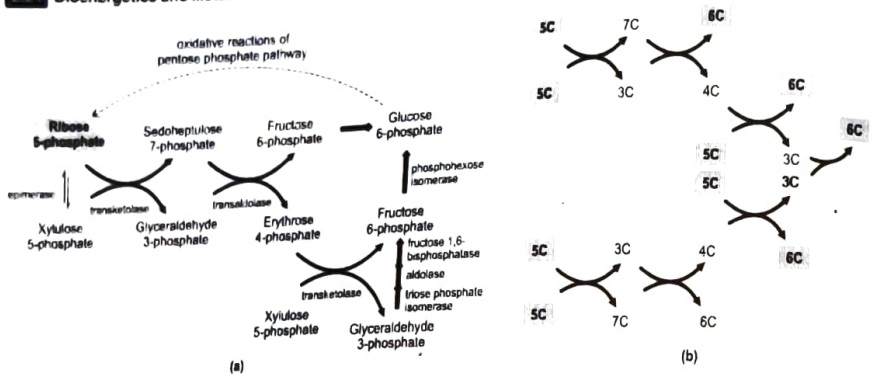
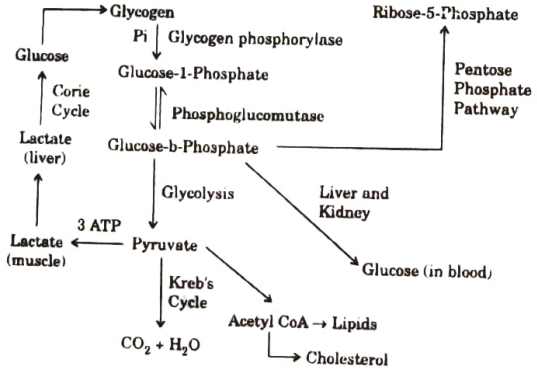


Fig. Nonoxidative reactions of the pentose phosphate pathway.

These reaction (a) convert pentose phosphates back to hexose phosphates, allowing the oxidative reaction to continue. The enzymes transketolase and transaldolase are specific to this pathway; the other enzymes also serve in the glycolytic or gluconeogenic pathway, (b) a schematic diagram showing the pathway from six pentoses (5C) to five hexoses (6C). Note that this involves two sets of the interconversions shown in (a). Every reaction shown here is reversible; unidirectional arrows are used only to make clear the direction of the reactions during continuous oxidation of glucose 6-phosphate. In the light-independent reactions of photosynthesis, the direction of these reactions is reversed.

Glycogen Catabolism

Glycogenin animal tissues and in micro-organisms can be mobilized further within the same cell by phosphorylation reaction catalized by glycogen phosphorylase. This enzyme catalized and attach by iP on the α (a → 4) glycosidic linkage and generates glucose-1-phosphate. Phosphorolysis preserve some of the energy of the glycosidic bond. Glucose-1-Phosphate is converted to glucose-6-P. by the help of enzyme phosphoglucomutase. Three glucose-6-P have 3-major fates. The overview of the whole process is given below :



Note: The similar pathway also operates for starch.

Steps in Glycogen Phosphorylation.

Glycogen phosphorylase (or starch phosphorylase) acts repetitively on the nonreducing ends of glycogen (or amylopectin) branches until it reaches a point four glucose residues away from an (α1 → 6) branch point, where its action stops. Further degradation can occur only after the debranching enzyme, formerly known as oligo (α1 → 6) to (α1 → 4) glucantransferase, catalyzes two successive reactions that remove branches. Glucose 1-phosphate, the end product of the glycogen and starch phosphorylase reactions, is converted to glucose 6-phosphate by phosphoglucomutase, which catalyzes the reversible reaction.

$$\text{Glucose 1-phosphate} \rightleftharpoons \text{glucose 6-phosphate}$$

Phosphoglucomutase requires as a cofactor glucose 1,6-bisphosphate, which has a role analogous to that of 2,3-bisphosphoglycerate in the phosphoglycerate mutase reaction.

Cori Cycle

Because skeletal muscle -stores relatively little glycogen (about 1% of total weight), there is a limit to the amount of glycolytic energy available during all-out exertion. Moreover, the accumulation of lactate and the consequent decrease in pH that occurs in maximally active muscles reduces their efficiency.

After a period of intense musclular activity, heavy breathing continues for some time. Much of the O_2 thus obtained is used for the production of ATP by oxidative phosphorylation in the liver. This ATP is used for gluconeogenesis from lactate, carried in the blood from the muscles to the liver. The glucose thus formed returns to the muscles to replenish their glycogen, completing the Cori cycle.

Skeletal muscle contains 10-30 mm phosphocreatine which can rapidly regenerate ATP from ADP by the creatine kinase reaction. During periods of active contraction and glycolysis, this reaction proceeds predominantly in the direction of ATP synthesis during recovery from exertion, the same enzyme is used to resynthesize phosphocreatine from creatine at the expense of ATP.

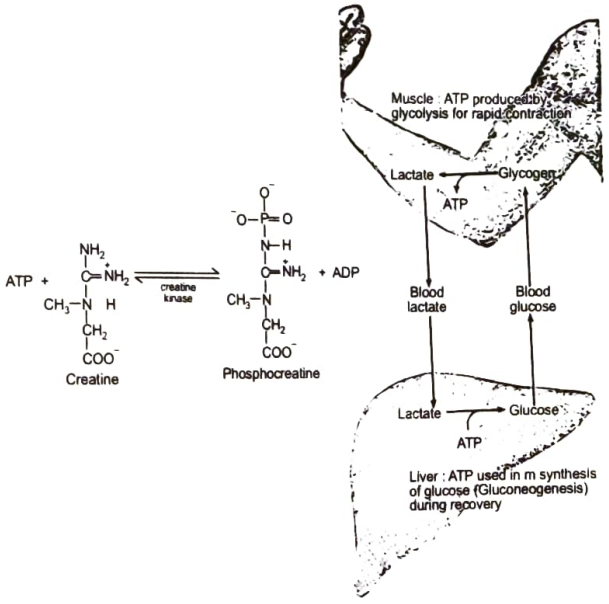


Fig. Matabolic cooperation between skeletal muscle and the liver.

Extremely active musclules use glycogen as an energy source, generating lactate via glycolysis. During recovery, some of this lactate is transported to the liver and used to form glucose via gluconeogenesis. This glucose is released to the blood and returned to the muscles to replenish their glycogen stores. The overall pathway (glucose → lactate → glucooe) constitutes the Cori cycle.

PHOTOSYNTHESIS

Photosynthetic organisms trap solar energy and form ATP and NADPH, which they use as energy sources to make carbohydrates and other organic compounds from CO_2 and H_2O; simultaneously, they release O_2 into the atmosphere. Aerobic heterotrophs (humans, for example) use the O_2 so formed to degrade the energy-rich organic products of photosynthesis to CO_2 and H_2O, generating ATP for their own activities.

The overall equation for photosynthesis in higher plants describes and oxidation-reduction reaction in which H_2O donates electrons (as hydrogen) for the reduction of CO_2 to carbohydrate (CH_2O) :

$$CO_2 + H_2O \xrightarrow{\text{light}} O_2 + (CH_2O)$$

Photochemical Reaction (Light Reaction)

The reaction takes place in grana of chloroplast and includes those activities for which light is essential

Robin Hill (1939) first of all showed that if chloroplasts extracted from leaves of Stellaria media and Lamium album are suspended in a test tube containing suitable electron acceptors, e.g., potassium ferrooxalate (some plants require only this chemical) and potassium ferricyanide, oxygen is released due to photochemical splitting of water. Under these conditions, no CO_2 was consumed and no carbohydrate was produced, but light-driven reduction of the electron acceptors was accompanied by O_2 evolution.

$$\underset{\text{Electron acceptor}}{4Fe^{3+}} + \underset{\text{Electron donor}}{2H_2O} \longleftrightarrow \underset{\text{Reduced product}}{4Fe^{2+}} + 4H^+ + O_2\uparrow$$

The splitting of water during photosynthesis is called photolysis. This reaction is known as Hill reaction.

Hill reaction proves that

(*i*) In photosynthesis oxygen is released from water.

(*ii*) Electrons for the reaction of CO_2 are obtained from water [i.e., a reduced substance (hydrogen donor) is produced which later reduces CO_2].

According to **Arnon** (1954), in this process light energy is converted to chemical energy. This energy is stored in ATP (this process of ATP formation in chloroplasts is known as **photophosphorylation**) and from electron accepter $NADP^+$, a substance which is found in all living beings NADP*H is formed as hydrogen donor. Formation of hydrogen donor NADPH from electron acceptor $NADP^+$ is known as **photoreduction** or production of **reducing power** NADPH.

Note: NADP = Nicotinamide ademine dinucleotide phosphate is coenzyme II (previously called TPN = Triphospho pyridine) ATP is Adenosine triphosphate.

Pigment System I

In pigment system I, different types of molecules of chlorophyll-*a* take part. On the basis of type of wavelengths of light absorbed by these chlorophyll-*a* molecules they are known as Chl-*a*-660, Chl-*a*-670. Chl-*a*-680, Chl-*a*-690. At the end of this energy gathering pigment system, a special type of chlorophyll "*a*" molecule called P700 is present. It absorbs 700 nm wavelength of light and functions as reaction center where photochemical reaction takes place, i.e., from where high energy electrons are released. The other pigments mentioned above, function as antenna in harvesting solar energy.

The energy trapped by a single chlorophyll molecule is not enough to start the first chemical reaction that would occur in light. Chlorophyll and accessory pigments help to capture light over a larger area and pass it onto the reaction centers. Some physiologist also include carotenes and chlorophyll-*b* in pigment system I. Thus a photon absorbed anywhere in the harvesting zone of P700 forms photosystem I. This pigment system is involved in both cyclic and noncyclic photophosphorylation. It is the light of longer wavelengths which is active in photosystem I. PS-I is active in both red and far-red lights.

Brother NADA FAD cooked 442 dishes with water & fi

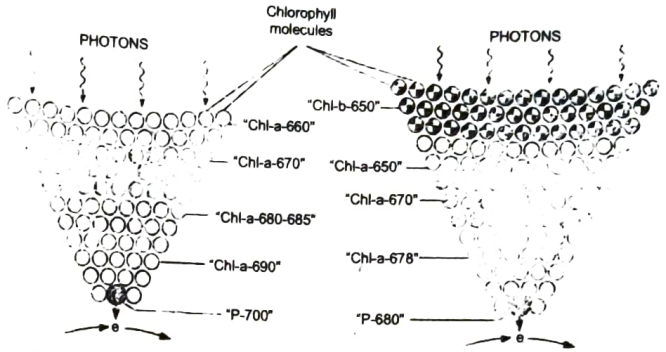


Fig. (a) Reaction center of photosyhstem I, (b). Reaction center of photosystem II

Pigment System II

In pigment system II, Chlorophyll *a*-680, Chlorophyll *a*-660, Chlorophyll *a*-670, Chlorophyll *a*-678 and Chlorophyll *b*-650 are involved. In this pigment system Chlorophyll *a*-660 Chlorophyll *a*-670, Chlorophyll *a*-678 and Chlorophyll *b*-650 function as antenna (according to some physiologists xanthophyll also functions as antenna in this system). **P680** functions as reaction center in this photoact.

In, pigment system II photons of lower wavelength of light are absorbed. It is inactive in far-red light. Pigment system II is involved only in **noncyclic photophosphorylation**.

Red Drop and Emerson's Enhancement Effect

Red Drop Effect (Emerson's first experiment) :

R. Emerson and C.M. Lewis exposed Chlorella plants to monochromatic light, *i.e.*, only one wavelength of light, at a time and measured the quantum yield, *i.e.*, at which wavelength of light the photosynthesis is mot vigorous. As the result of experiment they observed a sharp decrease in quantum yield at wavelengths greater than 680 nm. This decrease in quantum yield took place in the far-red part of the spectrum, *i.e.*, the curve shows yield drops dramatically in the region above 680 nm (red region) hence it is called **"red drop effect"**.

Emerson's Enhancement Effect (Emerson's second experiment) :

Emerson and his coworkers found that the inefficient far-red light in Chlorella beyond 680 nm can be made fully efficient if supplemented with light of shorter wavelengths (red light). The quantum yield from the two combined light beams was found to be greater than the sum effect of both the beams used separately, *e.g.*, if X is the rate of photosynthesis when far-red wavelength is applied, Y is the rate when red wavelength is applied and Z is the rate when both are superimposed. Z is greater than the sum of X + Y. This enhancement of photosynthesis is called as Emerson's enhancement effect.

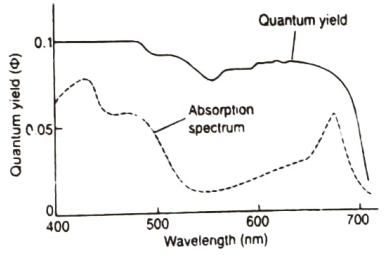


Fig. The Emerson "red drop" effect in the green alga Chlorella.

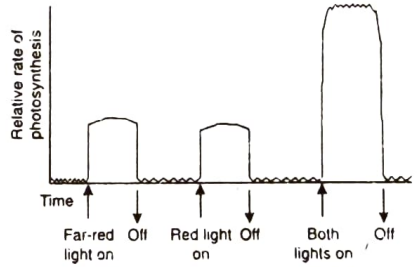


Fig. Emerson " Enhancement effect"

Thus, light with a wavelength greater than 680 nm is much less efficient than light of shorter wavelengths. The rate of photosynthesis when red and far-red light are givn together is greater than the sum of the rates when they are given apart

Z Scheme of Light reactions

When sunlight strikes the thylakoid membrane, the energy is absorbed simultaneously by the antenna pigments of both PS I and PS II and passed onto the reaction centers of both photosystems. Electrons of both reaction center pigments are boosted to an outer orbital and each photoexcited electron is transferred to a primary electron acceptor. The transfer of electrons out of the photosystems leaves the two reaction center pigments missing an electron and thus, positively charged. After losing their electrons, the reaction centers of PS I and PS II can be denoted as $P700^+$ and $P\,680^+$ respectively. Positively charged reaction centers act as attractants for electrons, which sets the stage for the flow of electrons between carriers.

In oxygenic photosynthesis, in which two photosystems act in series, electron flow occus along three legs- between water and PS II, between PS II and PS I and between PS I and $NADP^+$, an arrangement which is described as the Z scheme.

The Z scheme as originally proposed by Hill and Bendall, 1960.

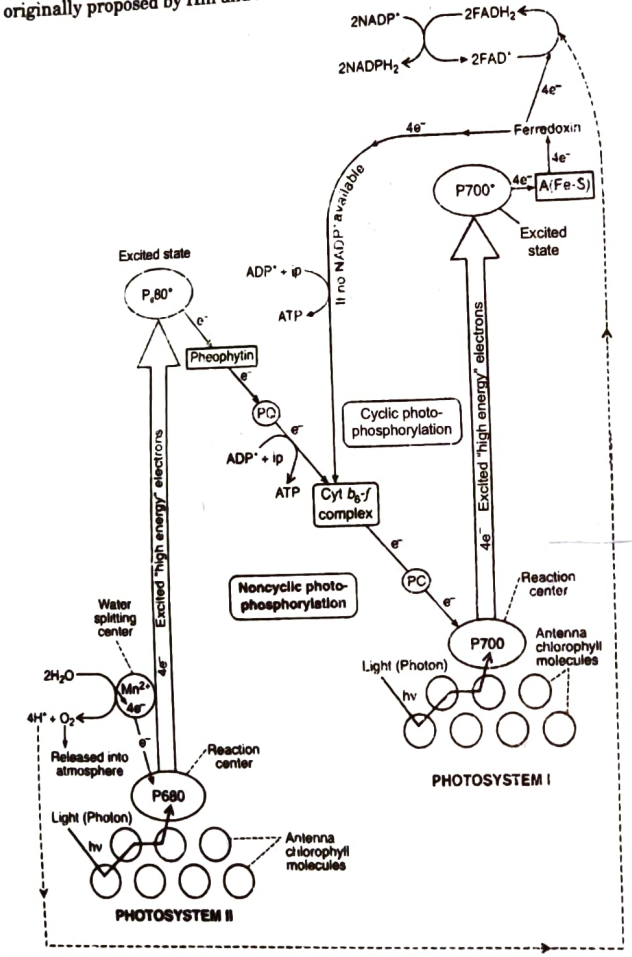


Fig. The Z-scheme of photosynthesis. Simplified diagram of the electron flow from H_2O to $NADP^+$,

Photosystem II

The main characteristics of photosystem II is photolysis or splitting of water, releasing oxygen and reduction of $NADP^+$.

Water is oxidized to oxygen according to following chemical reaction,

$$2H_2O \xrightarrow{Mn^{+2}} 4H^+ + 4e^- + O_2$$

This equation indicates that four electron are removed from two molecules of water, generating one molecule of oxygen (O_2) and four hydrogen ions ($4H^+$). Oxygen-evolving complex is localized on the interior surface of the thylakoid, i.e., towards the lumen side.

To understand the route of electrons in PS II let us start with a **P380 chlorophyll** a reaction center of photosystem II. In the presence of light when energy of a photon reaches to P680 via antenna chlorophyll molecules, it get excited. The excited form of P680, designated as P680*, is rapidly photooxidised as it passes an electron to pheophytin (Pheo) which is a primary electron acceptor in PS II. Pheophytin is a form of chlorophyll-a in which the magnesium ion (Mg^{2+}) has been replaced by two hydrogen atoms. These two molecules of reduced pheophytin then hand on an electron each (one at a time) to reduce **plastoquinone** the lipid soluble electron carrier between PS-II and the cytochrome b_6-f complex. The cytochrome b_6-f complex contains two cytochromes and an iron sulphur (Fe-S) protein. From reduced plastoquinone, *i.e.*, plastohydroquinone, electrons are accepted by cytochrome b_6-f complex, cytochrome b_6-f complex gets reduced in this process. From reduced cytochrome b_6-f complex electrons are taken by a water-soluble, copper containing protein called **plastocyanin** (PC), which is thus reduced. From reduced plastocyanin, electrons are accepted by P700, reaction center of PS-I.

Photosystem I

In photoact I photons of light hit 'pigment-700'. By absorbing quantum energy of photons of light 4 electrons of P700 attain an excited state (*i.e.*, these electrons have more energy). The excited form of P700 is designated as P700*.

High energy electrons released from P700, reaction center of pigment system I are accepted by the substance iron-sulphur-protein complex, designated as A(Fe-S), which acts as an electrons acceptor in photoact I. A(Fe-S) is thus reduced. From reduced A(Fe-S) these electrons are accepted by oxidised ferredoxin (Fd). Ferredoxin is another a low molecular weight (12000 for plant ferredoxins) non-haem iron protein containing labile sulphur, red in colour that is soluble in the stroma. By accepting these electrons Fd is reduced. From reduced Fd the electrons are taken by FAD+ which is reduced to $FADH_2$ (hydrogen comes from photolysis of water), from FADH electrons and hydrogen are accepted by $NADP^+$ and it is reduced to NADPH.

I this way excited electrons released from 'P700' reduce $NADP^+$ to NADPH and with this photosystem I ends.

Table. Comparison of photosystem I and photosystem II

I	II
1. PS I lies on the outer surface of the thylakoid	1. PS II lies on the inner surface of the thylakoid
2. In this system molecular oxygen is not evolve.	2. As the result of photolysis of water, molecular oxygen is evolved
3. Its reaction center is P700.	3. Its reaction center is P680.
4. NADPH is formed in this reaction	4. NADPH is not formed in this reaction.
5. It participates both in cyclic and noncyclic photophosphorylation	5. It participates only in noncyclic photophosphorylation.
6. It receives electrons from photosystem II	6. It receives electrons from photolytic dissociation of water.
7. It is not related with photolysis of water	7. It is related with photolysis of water.

Photophosphorylation

Arnon and his coworkers made important contribution by showing as to how light energy is converted to chemical energy of ATP. This formation of ATP in chloroplasts in the presence of light is known as Photophosphorylation. takes place in two ways :

1. Cyclic photophosphorylation
2. Noncyclic photophosphorylation

Cyclic Photophosphorylation

In photoact I high energy electrons move from P 700 to $NADP^+$ through ferredoxin (Fd). When all of the $NADP^+$ has been reduced, the ferredoxin transfers electrons back to P700 by cyclic route. The electrons released from P700 move through A (Fe-S) to Fd, from Fd to cytochrome b_6-f complex and from b_6-f complex to plastocyanin (PC) and than returns to 'P700' Fig. (a). In this process of return of electrons, energy is stored in ATP in between Fd and cytochrome b_6-f complex.

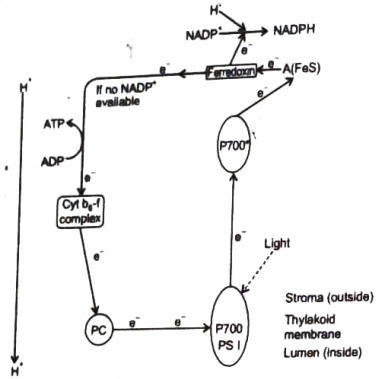


Fig. (a) Cyclic Photophosphorylation. When all of the NADP⁺ is reduced, ferredoxin transfers electrons back to the cytochrome b_6-f complex. Flow of electrons through this leads to increased proton pumping and hence increased ATP synthesis

Because in this photophosphorylation, electrons released from P700 return back to P700 in cyclic manner, ATP being synthesized during the electron flow, is called cyclic **photophosphorylation**. In cyclic photophosphorylation neither oxygen is librated nor NADPH is produced but the energy can be used to produce ATP. Fig. (c)

Noncyclic photophosphorylation

Another electron transport chain starts with the release of electrons from reaction center of photosystem II. The formation of ATP during the process of oxygen releasing photosynthesis is called noncyclic photophosphorylation because electrons move in a linear path from H_2O to $NADP^+$ (Fig. *b*). In this chain high energy electrons released from 'P-680' do not return to 'P-680' but pass Through pheophytin, plastoquinone, cytochrome b_6-f complex, plastocyanin and then enter P-700. In this transfer of electrons from plastoquinone (PQ) to cytochrome b_6-f complex, ATP is synthesized. Because in this process high energy electrons released from 'P-680' do not return to 'P-680' and ATP is formed, this is called **Noncyclic photophosphorylation**. (12 ATP molecules are formed).

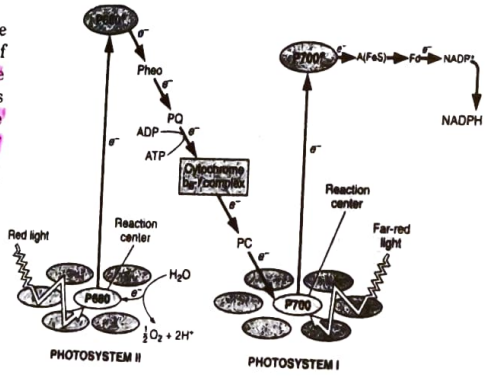


Fig. (b) Noncyclic Photophosphorylation.

In this way during the photoact II, in photochemical reaction photolysis of water takes place, O_2 is released, ATP and NADP. 2H are synthesized.

ATP and NADPH molecules function as vehicles for transfer of energy of sunlight into dark reaction, leading to the formation of high energy carbon containing compounds mainly sugars. ATP and NADPH together are called assimilatory power of the plant and NADPH alone is called reducing power.

In this electron transport chain high energy electrons released from P680 do not return to "P680" but pass through pheophytin (Pheo), plastoquinone (PQ),

Table. Comparison of Cyclic and Non-cyclic Photophosphorylation

Cyclic Photophosphorylation	Non-cyclic Photophosphorylation
1. No oxygen is given off (Anoxygenic)	1. Oxygen is given off (Oxygenic).
2. No water is consumed.	2. Water is used up
3. Only one light-trapping system (Photosystem 1) is involved.	3. Two light-trapping systems (Photosystem I and II) are involved.
4. No NADPH synthesized	4. NADPH synthesized.
5. Last electron acceptor is P700	5. Last electron acceptor is NADP.

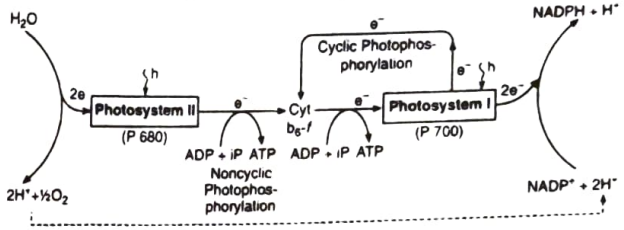


Fig. (c) Two photosystems involved in non-cyclic and cyclic photophosphorylation

Generation and Utilization of ATP:

A proton gradient drives the production of ATP in photosynthesis. The machinery for ATP synthesis in a chloroplast is ATP synthase (ATPase). The ATP synthase consists of a headpiece called F_1 in chloroplasts, which contains the catalytic site of the enzyme and a stalk called CF° which remains embedded in the thylakoid membrane, it forms a proton channel.

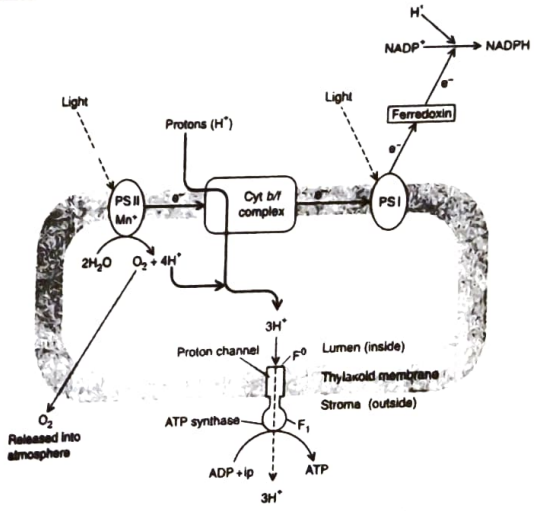


Fig. Proton (H^+) gradient and ATP formation

ATP synthase complexes are not uniformly dispersed throughout the thylakoid membrane, but are preferentially localized in the stroma lamellae and in the exposed region of the grana stacks. The CF_1 headpieces project outward into the stroma in keeping with the orientation of the photon gradient, which has its higher concentration within the thylakoid lumen. Thus, protons (H^+) move from higher concentration in the lumen through the ATP synthase into the stroma thereby driving the phosphorylation process. ATP synthase, bound to the stroma side of the thylakoid membrane in the presence of the proton gradient (H^+ gradients) forms ATP from inorganic phosphate (ip) and ADP. When isolated chloroplasts are engaged in photosynthesis the stroma becomes alkaline as protons (H^+) released by photolysis of water are translocated across the thylakoid membrane into the lumen of the thylakoids. Measurements made during periods of maximal ATP synthesis suggests that difference in H^+ concentrations of 1,000 to 2,000-fold exist across the thylakoid membrane, corresponding to a pH gradient (ΔpH) of 3 to 4 units. The best estimate at the present time is three protons for each ATP synthesized.

Calvin Cycle Photosynthetic Carbon Reduction Cycle (Dark Reaction)

In 1954, Calvin, Benson and their coworkers worked out the path of carbon (M. Calvin was awarded Nobel Prize for this work in 1961).

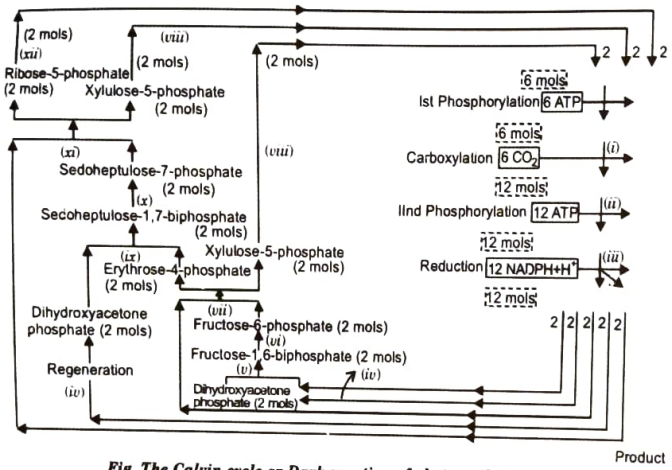


Fig. The Calvin cycle or Dark reaction of photosynthesis

Enzymes : (i) Carboxydismutase, (ii) Phosphoglyceric kinase, (iii) Triose phosphate dehydrogenase, (iv) Phosphotriose isomerase, (v) Aldolase, (vi) Phosphatease, (vii) Transketolase, (viii) Phosphopentose epimerase, (ix) Transldolase, (x) Phosphatase, (xi) Transketolase, (xii) Transketolase, (xii) Phosphopentose isomerase

This path of carbon was studied with the help of radioactive tracer technique using **Chlorella**, a unicellular green alga and CO_2 with C^{14} atomic weight ($C^{14}O_2$).

By provinding radioactive carbon one is able to trace the sequence of steps through which this carbon passes, *i.e.*, what are the intermediate steps of dark fixation of $C^{14}O_2$. The outline of Calvin cycle or PCR cycle is as follws :

First Phosphorylation :

CO_2 reduction starts with a 5-carbon sugar, ribulose-5-phosphate. 6 molecules of this sugar react with 6 molecules of ATP (produced in light reactions) forming 6 molecules of ribulose-1, 5-biphosphate (RuBP) and 6 molecules of ADP (equation I).

$$\underset{(3\text{ mols})}{\text{Ribulose-5-phosphate} + 6\text{ATP}} \xrightarrow{\text{Phosphopentokinase}} \underset{(6\text{ mol})}{\text{Ribulose-1, 5-biphosphate} + 6\text{ADP}} \quad ..(1)$$

Carboxylation :

Ribulose-1, 5-biphosphate (RuBP) (= Ribulose diphosphate) acts as CO_2 acceptor and 6 mols of RuBP react with 6 mols of CO_2 and 6 mols of water giving rise to 12 mols of 3-phosphoglyceric acid (a 3 carbon compound) (equation 2).

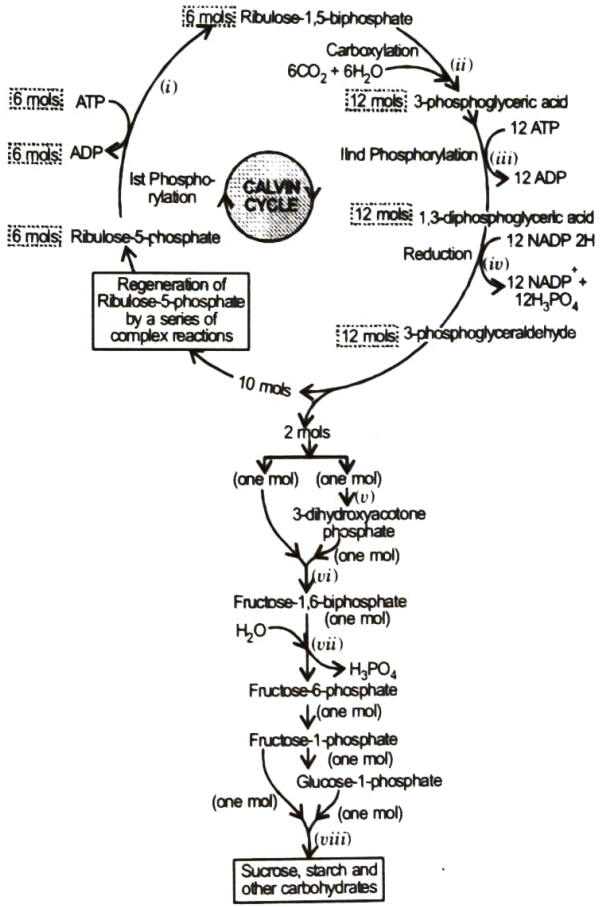


Fig. Simplified diagram of Calvin cycle

Enzymes :

(*i*) Phosphopentokinase, (*ii*) Carboxydisumutase, (*iii*) Phosphoglyceric kinase, (*iv*) Triose phosphate dehydrogenase, (*v*) Phosphotriose isomerase, (*vi*) Aldolase, (*vii*) Phosphatase

$$\underset{(6\text{ mols})}{\text{Ribulose-1, 5-biphosphate} + 6CO_2 + 6H_2O} \xrightarrow[\text{Ribulose-1,5-biphosphate carboxylase*}]{\text{Carboxydismutase or}} \underset{(12\text{mols})}{\text{3-phosphoglyceric acid}} \quad ...(2)$$

* This enzyme is also called Ribulose-1, 5 – biphosphate carboxylase oxygenase-Rubisco (most abundant protein in the world) accounting for nearly 50% of the soluble protein in most leaves.

Second Phosphorylation :

12 mols of 3-phosphoglyceric acid react with 12 mols of ATP (produced in light reactions) giving rise to 12 mols reach of 1, 3-diphosphoglyceric acid + ADP (equation 3).

3-phosphoglyceric acid + 12 ATP $\xrightarrow{\text{Phosphoglyceric kinase}}$ 1, 3-diphosphoglyceric acid + 12 ADP ...(3)

(12 mols) (12 mols)

Reduction :

12 mols of NADP. 2H formed in light reactions are used to reduce 12 mols of 1, 3-diphosphoglyceric acid leading to the formation of 12 mols of 3-phosphoglyceraldehyde, 12 mols of NADP and 12 mols of phosphoric acid (equation 3).

1, 3-diphosphoglyceric acid + 12NADP.2H $\xrightarrow{\text{Triose Phosphate dehydrogenase}}$ 3-phosphoglyceraldehyde + $12NADP^+ + 12H_3PO_4$ ...(4)

(12 mols)

In this way by the reduction of CO_2, 12 molecules of 3-phosphoglyceraldehyde are formed. Out of these 12 molecules, two molecules go to synthesize sugar, starch and other carbohydrates as given below in section A (equation 5-9) and remaining 10 molecules are recycled to regenerate 6 molecules of Ribulose-5-phosphate after a series of complex reactions and intermediates known as Calvin cycle or Calvin-Benson cycle or Reductive Pentose Phosphate Pathway as given in section B.

Section A :

Synthesis of sucrose :

Out of two mols of 3-phosphoglyceraldehyde one mol is converted to its isomer 3-dihydroxyacetone phosphate (equation 5).

3-phosphoglyceraldehyde $\xrightarrow[\text{isomerase}]{\text{Phosphotriose}}$ 3-dihydroxyacetone phosphate ...(5)

(1 mol) (1 mol)

One mol of 3-dihydroxyacetone phosphate reacts with 1 mol of 3-phosphoglyceraldehyde to form one molecule of fructose –1, 6-biphosphate (equation 6).

– 3 Phosphoglyceraldehyde + 3 – Dihydroxyacetone phosphate $\xrightarrow{\text{Aldolase}}$ Fructose- 1, 6-biphosphate .....(6)

(1 mol) (1 mol) (1 mol)

One mol of fructose – 6 – phosphate and one mol of phosphoric acid is released from one mol of fructose -1, 6-biphosphate with the help of the enzyme phosphatase with tuilizations of one mol of H_2O (equation 7).

Fructose-1, 6-biphosphate $\xrightarrow[H_2O]{\text{Phosphatriose}}$ Fructose-6-phosphate + H_3PO_4 ...(7)

Fructose-6-phosphate is converted to fructose-1-phosphate which forms glucose-1-phosphate (equation 8).

Fructose-6-phosphate ⟶ Fructose-1-phosphate ⟶ Glucose-1-phosphate .....(8)

(1 mol) (1 mol) (1 mol)

Fructose-6-phosphate and Glucose-1-phosphate $\xrightarrow{\text{Phosphatase}}$ Sucrose ...(9)

(1 mol) (1 mol) (1 mol)

From sucrose are produced other carbohydrates, e.g., starch etc.

Section B :

Regeneration of Ribulose-5-phosphate (Reductive Pentose Phosphate Pathway)

(*i*) 3-Phosphoglyceraldehyde $\xrightarrow[\text{isomerase}]{\text{Triose phosphate}}$ Dihydroxyacetone phosphate

(PGAL) (DHAP)

[4 mols] [4 mols]

(*ii*) Dihydroxyacetone phosphate + 3-phosphoglyceraldehyde $\xrightarrow{\text{Aldolase}}$ Fructose-1, 6-diphosphate

[2 mols] [2 mols] [2 mols]

(*iii*) Fructose –1, 6-diphosphate + $2H_2O$ $\xrightarrow{\text{Phosphatase}}$ Fructose -6- phosphate + $2H_3PO_4$

[2 mol] [2 mols]

(*iv*) Fructose-6-diphosphate + 3-phosphoglyceraldehyde $\xrightarrow{\text{Transketolase}}$

[2 mols] [2-mols]

Xylulose-5-phosphate + Erythrose-4-phosphate

[2 mols]

(*v*) Erythrose-4-phosphate + Dihydroxyacetone phosphate $\xrightarrow{\text{Transaldolase}}$

[2 mols] [2 mols]

(Produced in reaction iv)of the 4 mols produced in reaction (i) Sedoheptulose-1, 7– diphosphate

[2 mols]

(*vi*) Sedoheptulose-1, 7-diphosphate $\xrightarrow{\text{Phosphatase}}$ Sedoheptulose-7-phosphate + $2H_3PO_4$

[2 mols] [2 mols]

(*vii*) Sedoheptulose-7-phosphate + 3-phosphoglyceraldehyde $\xrightarrow{\text{Transketolase}}$

[2 mols] [2 mols]

Ribose-5-phosphate + Xylulose-5-phosphate

[2 mols] [2 mols]

(*viii*) Ribose-5-phosphate $\xrightarrow{\text{Phosphopentose isomerase}}$ Ribulose-5-phosphate

[2 mols] [2 mols]

(produced in reaction *vii*)

(*ix*) Xylulose-5-phosphate $\xrightarrow{\text{Phosphopentose epimerase}}$ Ribulose-5-phosphate

[4 mols] [4 mols]

2 mols produced in reaction *vii*

2 mols produced in reaction *iv*

SOME IMPORTANT POINTS

1. **Light reaction ..... in thylakoid membrane:**

$$24H_2O \longrightarrow 24OH^- + 24H^+$$

$$18\ ADP + 18Pi \longrightarrow 18ATP$$

$$12\ NADP^+ + 24H^+ \longrightarrow 12NADP.\ 2H$$

$$24\ OH^- - 24e^- \longrightarrow 24OH$$

$$24OH \longrightarrow 12H_2O + 6O_2$$

2. **Chemical dark reaction ..... in stroma**

$$6CO_2 + 18ATP + 12NADP.2H \rightarrow C_6H_{12}O_6 6H_2O + 18ADP + 18Pi + 12NADP^+$$

$$(I + II)6CO_2 + 24H_2O \xrightarrow[\text{Chloroplast}]{\text{Light}} \underset{\text{Carbohydrate}}{C_6H_{12}O_6} + 18H_2O + 6O_2$$

C_3 and C_4 Plants

C_3 plants:

Those plants in which in the dark reaction of photosynthesis, the first stable compound formed is phosphoglyceric acid (PGA) (3C), are known as C_3 plants such as sunflower, pumpkin, mango and mostly dicot plants, rarely in monocots.

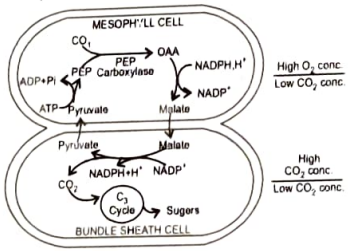
C_4 Plants:

Those plants in which in the reaction of photosynthesis, the first stable compound formed is oxaloacetic acid (OM) (4C) are known as C_4 plants such as sugarcane, maize, Cyperus, Rotundus, Panicum maximum, Chloris gayana, Atriplex spongiosa, Digitaria samguinalis, Amaranthus thlipa gesnariana etc.

C_4 cycles is characteristic of family graminae, cyperaceae and some dicot plants.

Hatch and Slack Cycle or C_4 Cycle

By 1965 it was known that the synthesis of carbohydrates only takes place by light reaction and C_3 pathway, but in 1967 M.D. Hatch and C.R. Slack proposed another process of CO_2 fixation, in which first stable product instead of a three carbon compound (PGA), there takes place formation of a four carbon compound (Oxaloacetic acid). Hence these plants are known as C_4 plant.



Hatch and Slack pathway (C_4 plants).

- This type of carbon assimilation process has been observed in several members of cyperaceae, amarentaceae chenopodiaceae, portulacaceae, graminae etc.
- Leaves of C_4 plants have Krantz anatomy because of the presence of radially arranged cells of bundle sheath around V.B. These cells lack grana in their chloroplasts.
- On the other hand mesophyll cells are smaller and contain grana.
- As photorespiration is absent in C_4 plants so they are more efficient than C_3 plants.
- In C_4 plants, CO_2 taken from atmosphere is accepted by phosphoenol pyruvic acid present in chloroplast of mesophyll cells of these leaves. This results in the formation of 4-carbon compound oxaloacetic acid.
- Oxaloacetic acid is converted to another 4-carbon compound malic acid, which enters in the chloroplasts of bundle sheath cells.
- Malic acid gets dissociated here into pyruvic acid and carbon dioxide.This carbon dioxide reacts with ribulose 1, 5 biphosphate (RuBP) which is already present in chloroplasts of bundle sheath cells.
- Pyruvic acid re-enters in mesophyll cells and regenerates phosphoenol pyruvic acid.
- CO_2 after reacting with RuBP forms sugars and carbohydrates in bundle sheath cells according to Calvin cycle.

CAM Plants:

- In some succulent plants, there is found a biochemical process, the crassulacean acid metabolism (CAM).Thus these plants are referred as CAM plants.
- This phenomenon is characteristic of family crassulaceae. Kalanchoe, Sedum' Cactus, Orchid, Pineapple and some other members of cactaceae, orchidaceae, bromeliaceae, liliaceae, asclepiadaceae, vitaceae and euphorbiaceae also possess this phenomenon.
- Most CAM plants possess succulent habit.

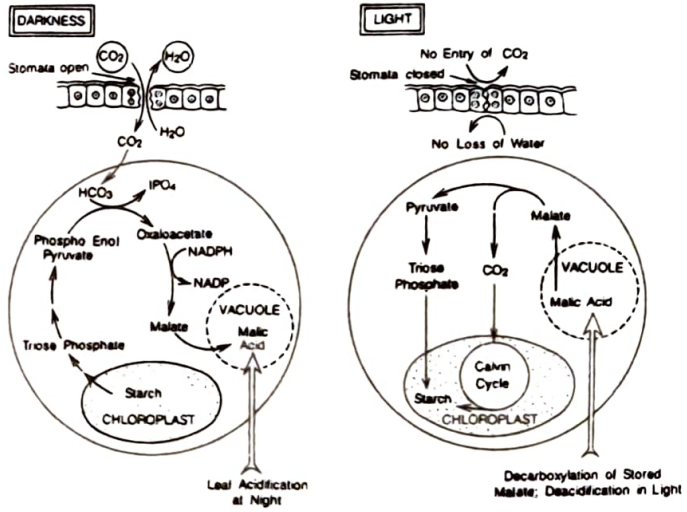
CAM Cycle:

Formation of organic acid takes place in two steps :

(*i*) Formation of malic acid in the night when stomata remains open.

(*ii*) Formation of carbohydrate by malic acid in the day time when stomata remains closed (deacidification).

- The two cycles (CAM and Calvin) occurs in the mesophyll cells and there is no differentiation between the type of cell as found in C_4 plants.
- During night when stomata are open, CO_2 is fixed through the action of PEP carboxylase to malic acid. This phase is completed in two steps. In the first, PEP fixes CO_2 and is converted into OAA.
- In the second, OAA is subsequently converted into malic acid by the enzyme malic dehydrogenase.



Crassulacean acid metabolism showing night uptake and fixation of CO_2 and decarboxylation and refixation of CO_2 in the day.

The CO_2 fixed in maleic acid during dark period through PEP carboxylase ultimately is converted to hexose or other carbohydrates.

Peculiarities of CAM Plants

- They are generally succulent plants.
- Stomata opens in the night and closes in the day time.
- Malic acid is formed in the night and carbohydrate formed in the day time.

Energy Relationship of Photosynthesis:

- Only 1% of the total energy falling on leaves is used in photosynthesis.
- For the reduction of one molecule of CO_2 with water more than 110 kcal is required.
- One mole quanta of red light has 43 kcal energy which is about 1/3 of the total requirement means about 3 photons would be required. One quanta of blue light has 63.4 kcal. Here 8-10 photons of blue light are required to reduce one molecule of CO_2.

Quantum Number:

- Number of photons of light required to reduce one molecule of carbon dioxide is referred as quantum number.
- The life time of all excited chlorophyll molecule is little, less than 0.01 second.

Quantum Yield:

- The number of oxygen molecules released per photons (quanta) of light is known as quantum yield.
- Emerson and Lewis (1943) found that average maximum quantum yield in photosynthesis is 12% which drops sharply near the far red end of spectrum.

Quantum Requirement :

- The number of photons required to release one molecule of oxygen by chlorophyll molecule is known as quantum requirement.
- Probably quantum requirement is 8 photons or 8 quanta to release one molecule oxygen.

Factors Affecting Photosynthesis

Two types of factors affect this process :

(*i*) External factors, and

(*ii*) Internal factors.

- For study of these factors it is essential to have a knowledge of *Blackman's law of limiting factors.*
- Before 1905, it was customary to study the effect of individual factors on the rate of photosynthesis in terms of minimum (at which photosynthesis will start), optimum (at which photosynthesis is at its best) and maximum (above which photosynthesis does not occur).
- Many workers obtained different values of these cardinal points under different conditions, *e.g.*, they found optimum carbon dioxide concentration to be greater at high intensity than at low intensity.
- F.F. Blackman (1905) criticised the above said concept and established law of limiting factors.
- He stated that when a process is conditioned as to its rapidity by a number of separate factors, the rate of the process is limited by the pace of the slowest factor, (*i.e.*, factors present in minimum amount).

Example:

- Suppose 1 unit of light intensity is capable of fixing 5 cc of carbon dioxide.
- If 1 cc of carbon dioxide is provided to the plant, photosynthesis takes place (at zero concentration of carbon dioxide there is no photosynthesis) but soon becomes constant as concentration of carbon dioxide is limiting the process.
- If we go on increasing the amount of carbon dioxide up to 5 cc the rate of photosynthesis also increases.
- But if we increase the amount of carbon dioxide concentration above 5 cc the rate will not increase as now light works as a limiting factor.
- However if we double the light intensity the rate of photosynthesis will increase until carbon dioxide concentration again becomes limiting.

(*i*) External Factors:

(1) Carbon dioxide:

- If percentage of CO_2 increases and becomes high than rate of photosynthesis increases.
- Very high concentration of CO_2 may have toxic effect on plants resulting in paralysing its metabolic activities.

(2) Light:

- Red and blue light have maximum rate of photosynthesis.
- UV, IR. green and yellow light have no or very low rate of photosynthesis.
- If light intensity increase upto 100 ft candle to 300 ft candle, then rate of photosynthesis increases.
- When the intensity of light is very low (in the morning) the rate of photosynthesis is very low and it just equals the rate of respiration.
- Generally, longer period of day light results in more photosynthesis.

(3) Temperature :

- Normally 0 - 50°C temperature is required for photosynthesis.
- For tropical plants 18-30°C temperature is required for process.
- For temperate plants 9-10°C temperature is required for the process.
- For desert plants 55°C maximum temperature is required for photosynthesis.
- For hot spring algae 75°C maximum temperature is required for this process.

(4) Water:

- Only 1% of the total absorbed water is utilized for photosynthesis.
- When percentage of water increases, then rate of photosynthesis also increases.
- If water supply is extremely meagre to as to cause wilting of leaves, the rate of photosynthesis may be reduced even to the value of zero.

(5) Oxygen:

Normally, it gains no effect but if oxygen percentage increases, then rate of photosynthesis is decreases.

(6) Effect of chemical compounds

In presence of hydrocyanic acid, hydroxylamine, ether etc. rate of photosynthesis is decreased.

(*ii*) Internal Factors:

(1) Chlorophyll:

It is most essential for the process of photosynthesis because it is photoreceptive pigment and directly involved in trapping the light energy.

(2) Protoplasmic factor:

Rate of photosynthesis in young leaves is higher than the old leaves because young leaves have protoplasm which is more active.

(3) Accumulation of photosynthetic products :

Accumulation of food substances cause the low rate of photosynthesis.

(4) Hormones:

Presence of cytokinins and gibberellic acid increase the rate of photosynthesis.

(5) Photosynthetic quotient (P.Q.)

Photosynthetic quotient gives an idea of the ratio of the volume of O_2 liberated to the volume of CO_2 utilized during photosynthesis.

$$\text{P.Q.} = \frac{\text{Vol of } O_2 \text{ liberated}}{\text{Vol of } CO_2 \text{ used}} = \frac{6O_2}{6CO_2} = 1$$

The value of P.Q. is unity in glucose.

Photorespiration

Otto Warburg made an observation that O_2 inhibits photosynthesis in C_3 plants. This phenomenon, originally known as the Warburg effect, was later recognized as the light-dependent release of CO_2 due to oxygenase activity of Rubisco. Rubisco is a bifunctional enzyme. It catalyzes both the carboxylation and the oxygenation of RuBP. Oxygenation of RuBP leads to the production of one molecule of glycerate-3-phosphate and one of glycolate-2-Phosphate. The principal factors influencing the enzyme's oxygenase activity are the ambient concentrations of O_2, CO_2 and temperature. High temperatures promote oxygenation, and hence photorespiration, because the solubility of CO_2 in water declines more rapidly than that of O_2 as the temperature is increased. Second the specificity factor of Rubisco decreases with increasing temperature.

$$\text{Ribulose-1,5-bisphosphate} \xrightarrow[\text{Rubisco}]{+O_2} \text{2-phosphoglycolate} + \text{3-phosphoglycerate}$$

Photorespiration pathway (glycolate pathway)

Glycolate-2-Phosphate produced by Rubisco when it oxygenates RuBP cannot be utilized within the Calvin cycle. The pathway by which glycolate-2-Phosphate is further metabolized is described as glycolate pathway.

This pathway involves three subcellular compartments, the chloroplasts, peroxisomes and mitochondria. The key features of this pathway are the conversion of the two-carbon molecule, glycolate-2-Phosphate, to glycine and decarboxylation of two molecules of glycine to serine, CO_2 and NH_3. The three-carbon molecule, serine, is then converted into glycerate-3-Phosphate, which re-enters the Benson—Calvin cycle.

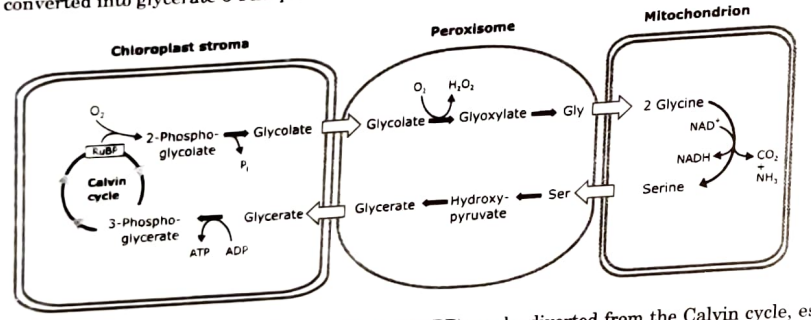


Figure : Photorespiration. Ribulose-1,5-bisphosphate (RuBP) can be diverted from the Calvin cycle, especially when the concentration of CO_2 is low. RuBP carboxylase/oxygenase catalyzes the oxidation of RuBP to form phosphoglycolate. In the reactions that follow, more O_2 is used, CO_2 is generated, and ATP is hydrolyzed, as metabolites pass from the chloroplast to near by peroxisomes and mitochondria and then back into the chloroplast.

Nitrogen Fixation and Assimilation

Nitrogen Cycle

The nitrogen cycle refers to the movement of nitrogen through the food chain of living organisms (Fig.). This complex cycle involves bacteria, plants and animals. All organisms can convert ammonia (NH_3) to organic nitrogen compounds, that is compounds containing C-N bonds. However, only a few microorganisms can synthesize ammonia from nitrogen gas (N_2). Although N_2 gas makes up about 80% of the earth's atmosphere, it is a chemically unreactive compound. The first stage in the nitrogen cycle is the reduction of N_2 gas to ammonia, a process called nitrogen fixation. Ammonia can also be obtained by reduction of nitrate ion (NO_3^-) that is present in the soil. Nitrate reduction can be carried out by most plants and microorganisms. The ammonia resulting from these two processes can then be assimilated by all organisms. Within the biosphere there is a balance between total inorganic and total organic forms of nitrogen. The conversion of organic to inorganic nitrogen comes about through catabolism, denitrification and decay (Fig.).

Nitrogen Fixation

The process of converting atmospheric N_2 gas into ammonia (nitrogen fixation) is carried out by only a few microorganisms, termed diazatrophs. These are some free-living soil bacteria such as Klebsiella and Azotobacter, cyanobacteria (blue-green algae), and the symbiotic bacteria (mainly Rhizobium). The symbiotic Rhizobium bacteria invade the roots of leguminous green plants (plants belonging to the pea family, e.g., beans, clover, alfalfa) and form root nodules where nitrogen fixation takes place. The amount of N_2 fixed by these

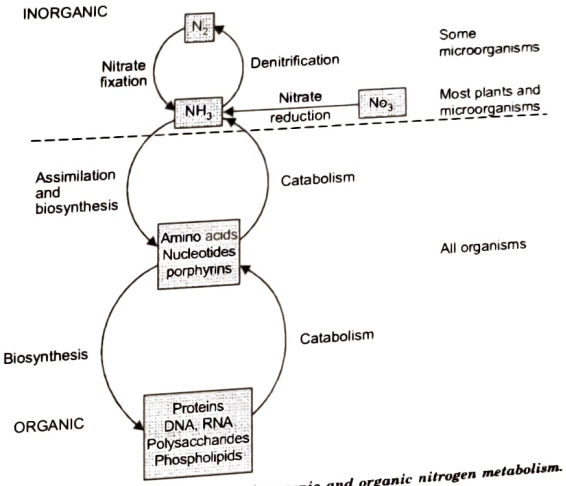


Fig. The interrelationships between inorganic and organic nitrogen metabolism.

diazatrophic microorganisms has been estimated to be in the order of 10^{11} kg per year, about 60% of the earth's newly fixed nitrogen. Lightning and ultraviolet radiation fix another 15%, with the remainder coming from industrial processes.

The chemical unreactivity of the N ≡ N bond is clearly seen when one considers the industrial process of nitrogen fixation. This process, devised by Fritz Haber in 1910 and still used today in fertilizer factories, involves the reduction of N_2 in the presence of H_2 gas over an iron catalyst at a temperature of 500°C and a pressure of 300 atmospheres.

$$N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$$

Nitrogenase Complex

Biological nitrogen fixation is carried out by the nitrogenase complex which .consists of two proteins: a reductase, which provides electrons with high reducing power, and a nitrogenase, which uses these electrons to reduce N_2 to NH3 (Fig.). The reductase is a 64 kDa dimer of identical subunits that contains one iron-sulfur cluster and two ATP binding sites. The nitrogenase is a larger protein of 220 kDa that consists of two α- and two β-subunits ($\alpha_2\beta_2$) and contains an iron-molybdenum complex. The transfer of electrons from the reductase to the nitrogenase protein is coupled to the hydrolysis of ATP by the reductase. Although the reduction of N_2 to NH_3 is only a six-electron process:

$$N_2 + 6e^- + 6\,H^+ \rightarrow 2\,NH_3$$

The reductase is imperfect and H_2 is also formed. Thus two additional electrons are also required:

$$N_2 + 8e^- + 8\,H^+ \rightarrow 2\,NH_3 + H_2$$

The eight high-potential electrons come from reduced ferredoxin that is produced either in chloroplasts by the action of photosystem I or in oxidative electron transport (Fig.). The overall reaction of biological nitrogen fixation:

$$N_2 + 8e^- + 16\ ATP + 16\ H_2O \rightarrow 2\ NH_3 + H_2 + 16\ ADP + 16\ P_i + 8\ H^+$$

highlights that it is energetically very costly, with at least 16 ATP molecules being hydrolyzed.

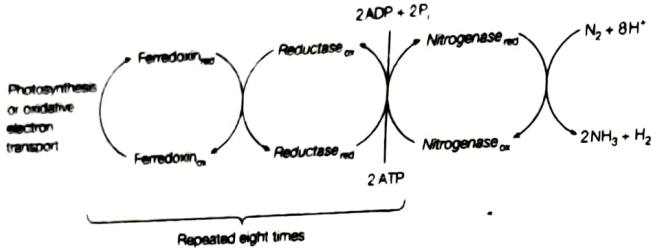


Fig. The flow of electrons in the nitrogenase-catalyzed reduction of N_2

Leghemoglobin

The nitrogenase complex is extremely sensitive to inactivation by O_2, so the enzyme must be protected from this reactive substance. In the root nodules of leguminous plants, protection is afforded by the symbiotic synthesis of leg-hemoglobin. The globin part of this monomeric oxygen-binding protein is synthesized by the plant, whereas the heme group is synthesized by the Rhizobium. The leghemoglobin has a very high affinity for O_2, so maintaining a low enough concentration to protect the nitrogenase.

Nitrogen Assimilation

The next step in the nitrogen cycle is the assimilation of inorganic nitrogen, in the form of ammonia, into organic nitrogen-containing compounds. All organisms assimilate ammonia via two main reactions catalyzed by glutamate dehydrogenase and glutamine synthetase giving rise to the amino acids glutamate (Glu) and glutamine (Gln), respectively. The amino nitrogen in Glu and the amide nitrogen in Gln are then used in further biosynthetic reactions to give rise to other compounds.

Glutamate dehydrogenase

Glutamate dehydrogenase catalyzes the reductive amination of the citric acid cycle intermediate α-ketoglutarate (Fig.). Although the reaction is reversible, the reductant used in the biosynthetic reaction is NADPH. This enzyme is also involved in the catabolism of amino acids.

Glutamine synthetase

Glutamine synthetase catalyzes the incorporation of ammonia into glutamine, deriving energy from the hydrolysis of ATP (Fig.). This enzyme is named a synthetase, rather than a synthase, because the reaction couples bond formation with the hydrolysis of ATP. In contrast, a synthase does not require ATP.

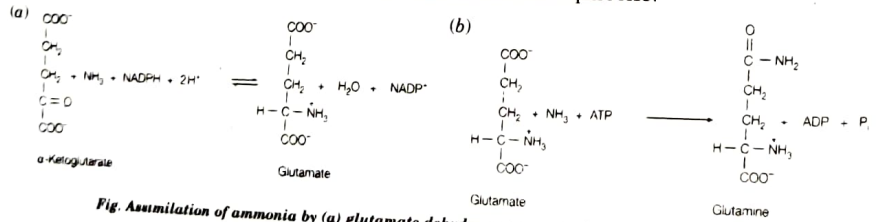


Fig. Assimilation of ammonia by (a) glutamate dehydrogenase and (b) glutamine synthetase.

EXERCISE – I

MCQ TYPE QUESTIONS

1. During anaerobic conditions, the rate of glycolysis increases is called as:
   (a) Compensation point (b) Extinction point
   (c) Warburg effect (d) Pasteur effect
2. The incomplete breakdown of sugars in anaerobic respiration results in the formation of:
   (a) Fructose and water
   (b) Glucose and CO_2
   (c) Alcohol and CO_2
   (d) Water and CO_2
3. Cyanide resistant pathway is:
   (a) Anaerobic respiration
   (b) Aerobic respiration
   (c) Both (a) and (b)
   (d) None of these
4. Fermentation is
   (a) anaerobic respiration
   (b) incomplete oxidation of carbohydrates
   (c) complete oxidation of carbohydrates
   (d) none of the above.
5. Conversion of pyruvic acid into ethyl-alcohol is facilitated by the enzymes
   (a) carboxylase
   (b) dehydrogenase
   (c) carboxylase and dehyrdogenase
   (d) phosphatise
6. During anaerobic respiration in yeast:
   (a) Water and CO_2 are end products
   (b) CO_2, C_2H_5OH and energy are end products
   (c) H_2S, $C_6H_{12}O_6$ and energy are end products
   (d) H_2O, CO_2 and energy are only end products
7. Anaerobic respiration takes place in the
   (a) mitochondria
   (b) cytoplasm
   (c) lysosomes
   (d) ER
8. Decarboxylation is not involved in
   (a) Electron Transport System
   (b) Glycolysis
   (c) Kreb's cycle
   (d) Alcoholic fermentation
9. Which one of the following is the first step of glycolysis?
   (a) Breakdown of glucose
   (b) Phosphorylation of glucose
   (c) Conversion of glucose into fructose
   (d) Dehydration of glucose
10. The product formed by malic dehydrogenase is
    (a) Malic acid (b) Fumaric acid
    (c) Oxaloacetic acid (d) Succinic acid
11. Which intermediate compound is involved in the synthesis of amino acids?
    (a) Malic acid (b) Citric acid
    (c) a- ketoglutaric acid (d) Isocitric aicd
12. Kreb's cycle was discovered by Krebs in pigeon muscles in 1940. Which step is called gateway step-link reaction/transition reaction in respiration
    (a) Glycolysis
    (b) Formation of acetyl CoA
    (c) Citric acid formation
    (d) ETS terminal oxidation
13. The universal hydrogen acceptor is
    (a) NAD (b) ATP
    (c) CoA (d) FMS
14. ATP synthesis occurs on the
    (a) Outer membrane of mitochondrion
    (b) Inner membrane of mitochondrion
    (c) Matrix
    (d) None of the above
15. Largest amount of phosphate bond energy is produced in the process of respiration during
    (a) Anaerobic respiration
    (b) Glycolysis
    (c) Kreb's cycle
    (d) None of the above
16. In Kreb's cycle the hydrogen atoms removed at succinate level are accepted by
    (a) FAD (b) ADP
    (c) ATP (d) NAD
17. In how many steps, CO_2 is released in aerobic respiration of pyruvic acid?
    (a) One
    (b) Six
    (c) Three
    (d) Twelve

18. By the reaction of α-ketoglutaric acid with ammonia, through which of the process glutamic acid is formed?
(a) Oxidative amination
(b) Reductive amination
(c) Ammonification
(d) Transamination

19. In an electron transport chain in terminal oxidation the cytochrome which donates electrons to O_2 is
(a) Cytochrome b (b) Cytochrome c
(c) Cytochrome a_3 (d) Cytochrome a

20. The correct sequence of electron acceptor in ATP synthesis is
(a) Cyt a_3 a b c (b) Cyt b c a a_3
(c) Cyt b c a_3 a (d) Cyt c b a a_3

21. Which transfer electrons in E.T.S. (electron transport system)?
(a) Phytochrome (b) F1 particles
(c) Fe-S (d) None of these

22. Final electron acceptor in ETS is
(a) H_2O (b) O_2
(c) Cyt a_3 (d) Cyt a

23. Activity of succinic dehydrogenase involves the following in TCA cycle
(a) NAD (b) FAD
(c) GDP (d) ATP

24. HMP shunt is an alternative to
(a) Kreb's cycle (b) Aerobic glycolysis
(c) Calvin cycle (d) C4 pathway

25. Which of the following respiratory material may show the unit value of R.Q.?
(a) Stem of wheat (b) Leaf of barley
(c) Leaf of oat (d) All the above

26. Which causes RQ to vary?
(a) Respiratory substrate
(b) Light and O_2
(c) Respiratory product
(d) Temperature

27. A mixture containing equal quantity of germinating maize and groundnut seeds are taken. The RQ of this mixture would be
(a) One (b) Less than one
(c) More than one (d) Infinity

28. The value of RQ at compensation point is
(a) Unity (b) Infinity
(c) >1 (d) Zero

29. The RQ of a plant organ depends upon the nature of the substrate which is
(a) Reduced (b) Oxidized
(c) Catabolized (d) Metabolized

30. Which of the following pair is a combination of highest and lowest energy molecules
(a) Glucose and pyruvic acid
(b) Palmitic acid and acetyl CoA
(c) Glucose and malic acid
(d) Malic acid and acetyl CoA

31. Which among the following is the most appropriate reason for storing green coloured apples at low temperature (refrigerator)?
(a) The rate of photosynthesis is reduced
(b) Respiration and photosynthesis are completely inhibited
(c) The rate of respiration is reduced
(d) The rate of photosynthesis and respiration is reduced

32. In the process of respiration in plants 180 grams of sugar plus 192 grams of oxygen produce
(a) Large amount of CO_2, no water and no energy
(b) 132 grams of CO_2, 54 grams of water and 337 kcals of energy
(c) 264 grams of CO_2, 108 grams of water and 674 kcals of energy
(d) 528 grams of CO_2, 216 grams of water and 1348 kcals of energy

33. In nitrogen cycle, which of the following plays an important role?
(a) rhizopus (b) nitrobacter
(c) mucor (d) all green alga

34. Maximum contribution of O_2 is from
(a) phytoplankton (b) grasslands
(c) herbs and shrubs (d) dense forest

35. For recycling of material which one is most necessary?
(a) consumer (b) producer
(c) decomposer (d) none of these

36. Which of the following present in maximum amount in atmosphere?
(a) oxygen (b) nitrogen
(c) carbon dioxide (d) hydrogen

37. Which of the following cycle would be affected if decomposers of an ecosystem vanish?
(a) producer's cycle (b) consumer's cycle
(c) decomposers cycle (d) biogeochemical cycle

38. Phosphorus cycle is
(a) gaseous cycle
(b) perfect cycle
(c) imperfect cycle
(d) partly gaseous and partly sedimentary

39. Inorganic elements that food into usable energy is
(a) S (b) P
(c) Ca (d) Mg

40. Which one of the following is not a gaseous biogeochemical?
(a) sulphur cycle (b) phosphorus cycle
(c) nitrogen cycle (d) carbon cycle

41. During light reaction of photosynthesis, which of the following phenomenon is observed during cyclic phosphorylation as well as non-cyclic phosphorylation?
(a) release of O_2
(b) formation of ATP
(c) formation of NADPH
(d) involvement of PS- I and PS - II pigment systems

42. The primary acceptor, during CO_2 fixation in C_3 plants, is
(a) Phosphenol pyruvate(PEP)
(b) ribulose 1,5-diphosphate(RuDP)
(c) phosphogluceric acid(PGA)
(d) ribulose monophophate

43. In C_4 plants, CO_2 combines with
(a) Phosphoenol pyruvate
(b) phosphoglyceraldehyde
(c) phosphoglyceric acid
(d) ribulose diphosphate

44. Photorespiration is favoured by
(a) high temperatur and low O_2
(b) high humidity and temperature
(c) high O_2 and low CO_2
(d) high CO_2 and low O_2

45. Chlorophyll molecule at its carbon atom 3 of the pyrrole ring II has one of the following
(a) carboxylic group (b) magnesium
(c) aldehyde group (d) methyl group

46. The rate of photosynthesis is higher in
(a) very high light
(b) continuous light
(c) red light
(d) green light

47. The first step for initiation of photosynthesis will be
(a) Photolysis of water
(b) Excitement of chlorophyll due to absorption of light
(c) ATP formation
(d) Glucose formation

48. For assimilation of 1 CO_2 molecule, the energy required in form of ATP and $NADPH_2$ in C_3 cycle are
(a) 2 ATP and 2 $NADPH_2$
(b) 5 ATP and 3 $NADPH_2$
(c) 3 ATP and 2 $NADPH_2$
(d) 18 ATP and 12 $NADPH_2$

49. Which pair is wrong?
(a) C_3-maize
(b) C_4-kranz anatomy
(c) Calvin cycle PGA
(d) Hatch and Slack cycle-OAA

50. Which pigment system is inactivated in red drop?
(a) PSI and PSII (b) PSI
(c) PSII (d) None of these

51. Which of the following is wrong in relation to photorespiration?
(a) It occurs in chloroplast
(b) It occurs in day time only
(c) It is a characteristic of C_4 plants
(d) It is characteristics of C_3 plants

52. Stomata of cam plants
(a) Are always open
(b) Open during the day and close at night
(c) Open during the night and close during the day
(d) Never open

53. As compared to a C_3 plant, how many additional molecules of ATP are needed for net production of one molecule of hexose sugar by C_4 plant?
(a) Two (b) Six
(c) Twelve (d) Zero

54. Photosynthesis in C_4 plants is relatively less limited by atmospheric CO_2 levels because
(a) Effective pumping of CO_2 into bundle sheath cells
(b) Rubrics in C 4plants has higher affinity for CO_2
(c) Four carbon acids are the primary initial CO_2 fixation products.
(d) The primary fixation of CO_2 is mediated via PEP carboxylase.

55. During photorespiration, the oxygen consuming reaction occur in
(a) Stroma of chloroplast
(b) Stroma of chloroplast and mitochondria
(c) Stroma of chloroplast and peroxisomes
(d) Grana of chloroplast and peroxisomes

56. In Photosystem I, the first acceptor is
(a) An iron-sulphur protein
(b) Ferredoxin
(c) Cytochrome
(d) Plastocyanin

57. Electrons from exited chlorophyll molecule of photosystem II are accepted first by
(a) Plastoquinone (b) Ferredoxin
(c) Cytochrome-b (d) Cytochrome -f

58. Cyclic photophosphorylation results in the higher plant contains
(a) ATP and NADPH (b) ATP, NADPH, and O_2
(c) ATP (d) NADPH

59. Carbon dioxide is necessary for photosynthesis. The chemical used to remove this gas most effectively from entering a control apparatus is
(a) calcium oxide
(b) distilled water
(c) potassium hydroxide solution
(d) sodium carbonate.

60. During photosynthesis, oxygen in glucose comes from
(a) oxygen in air (b) water
(c) carbon dioxide (d) both (b) and (c)

61. The C_4-plants differ from C_3 plants with reference to the
(a) substrate that accepts CO_2 in carbon assimilation
(b) type of end product
(c) type of pigment involved in photosynthesis
(d) number of ATP that are consumed in preparing sugar.

62. Photosynthetic bacteria contain
(a) Quantasome (b) PS-1 and PS-2
(c) PS2 (d) PS1

63. Chloroplast contains maximum quantity of
(a) Pyruvic carboxylase
(b) Hexokinase
(c) RuDP carboxylase
(d) None of the above

64. What is common between photosynthesis and respiration
(a) Cytochrome (b) Light
(c) H_2O (d) Temperature

65. Photosynthesis is
(a) Oxidative, exergonic, catabolic
(b) Reductive, endergonic,anabolic
(c) Reductive, exergonic, anabolic
(d) Reductive, endergonie, catabolic

66. Which is the following can photosynthesise at low temperature (-24°C)
(a) Bacteria (b) Lichen
(c) Yeast (d) *Batrachospermum*

67. Bacteriochlorophyll 'a' differs in
(a) One pyrroll ring with one hydrogen
(b) One pyrrol ring with two hydrogen
(c) One pyrrol ring with three hydrogen
(d) One pyrro] ring with four hydrogen

68. The First intermediate of photosynthesis is
(a) Fructose-1,6-diphosphate
(b) Ribulose-1,5-biphosphate
(c) Xylulose-5-phosphate
(d) Phosphoglyceraldehyde

69. C4 plants are found among
(a) Gramineae only
(b) Monocots only
(c) Dicots only
(d) Monocots as well as dicots

70. Coenzyme used for conversion of Triacylglycerols to dihydroxyacetone is
(a) Mg^{+2} (b) NAD^+
(c) $FADH_2$ (d) ATP

71. The large negative delta G values is shown by
(a) Hexokinase
(b) phosphofructokinase
(c) Pyruvate kinase
(d) All of these

72. In animal tissues experiencing anaerobic conditions, pyruvate is reduced to
(a) Galactose (b) Fructose
(c) Lactate (d) Mannose

73. The stable form of pyruvate
(a) Enol (b) Eeto
(c) Cis (d) Trans

74. The prevalent form of phosphoglycerate mutase is phosphoenzyme with a phosphoryl group attached to
(a) Serine (b) Lysine
(c) Glycine (d) Histidine

75. In the phosphoglycerate mutase reaction the phosphoryl group is moved from
(a) C3 - C2 (b) C2 - C3
(c) C7 - C5 (d) C2 - C5

76. Fructose bisphosphate aldolase cleaves fructose, 1-6 phosphate between
(a) C3 and C4 (b) C2 and C3
(c) C4 and C5 (d) C1 and C2

77. Glycolysis and TGA cycle is coupled by
(a) Hexokinase
(b) Phosphofructokinase
(c) Lactate dehydrogenase
(d) Phosphoglycerate kinase

78. The deltaG^0 is 1.67kJ/mol and deltaG under cellular condition is –2.92 kJ/mol this indicates
(a) Reversible
(b) Operates near equilibrium
(c) Both (a) & (b)
(d) Irreversible

79. During the isomerization of glucose – 6 – phosphate, carbonyl oxygen moves from
(a) C3 - C4 (b) C2 - C3
(c) C1 - C2 (d) C2 - C4

80. Glucokinase is an
(a) Inducible enzyme (b) Allosteric enzyme
(c) Lipase (d) Topoisomerase

81. True substrate for hexokinase is
(a) $MgATP^2$ (b) MgATP
(c) ATP (d) ATP^2

82. In the metabolic process, energy is stored in the form of
(a) ATP
(b) NADH
(c) FADH2
(d) All of these

83. TCA cycle reactions takes place in
(a) SER
(b) RER
(c) Mitochondria
(d) Nucleus

84. Pyruvate dehydrogenase complex is composed of
(a) Thiamine pyrophosphate, coenzyme A, lipoic acid, NAD^+, FAD
(b) Thiamine pyrophosphate, coenzyme A, lipoic acid, ATP, FAD
(c) Thiamine pyrophosphate, coenzyme A, lipoic acid, NAD^+, ATP
(d) Thiamine pyrophosphate, coenzyme A, NAD^+, FAD

85. Acetyl CO-A reacts with oxaloacetate in
(a) Aldol condensation (b) Perkin condensation
(c) Kolbe's reaction (d) Hydration

86. Acyl group is activated to acyl co-A by
(a) Carbonyl carbon attack by nucleophiles
(b) Carbonyl carbon attack by electrophiles
(c) Removal of water molecule
(d) Removal of carbondioxide

NUMERICAL TYPE QUESTIONS

1. How many ATP are used in glycolysis or for complete phosphorylation of a glucose molecule, how many ATP molecules are required
2. Total ATP production during EMP pathway is
3. The number of molecules of pyruvic acid formed from one molecule of glucose at the end of glycolysis is
4. Complete oxidation of 1gm molecule of glucose gives rise to
5. What is the value of RQ of castor seeds, if the imaginary values of Ganong's respirometer are as follows?
(i) First rise of saline = 10ml
(ii) Second rise of saline after adding KOH = 30ml
6. RQ in aerobic respiration is
7. For the synthesis of one glucose molecule the Calvin cycle operates for
8. Number of ATP molecules produced at pyruvate stage
9. Number of ATP molecules produced per molecule of glucose
10. In the first five series of reaction number of glyceraldehyde – 3 – phosphate molecules produced by glucose is
11. In enolase catalyzed reaction the deltaG value reaches
12. Number of ATP molecules produced per glucose oxidation in aerobic pathway

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2015

1. Match the compounds in Group I with the correct entries in Group II.

Group I	Group II
P. Cyanide	1. K^+ ionophore
Q. Antimycin A	2. Electron transfer from cytochrome *b* to cytochrome c_1
R. Valinomycin	3. F_1 subunit of ATP synthase
S. Aurovertin	4. Cytochrome oxidase
	5. Adenine nucleotide translocase

(*a*) P-5, Q-2, R-3, S-1 (*b*) P-5, Q-2, R-1, S-3
(*c*) P-4, Q-2, R-1, S-3 (*d*) P-4, Q-5, R-3, S-1

2014

2. The most plausible explanation for a sudden increase of the respiratory quotient (RQ) of a microbial culture is that
(*a*) cells are dying
(*b*) yield of biomass is increasing
(*c*) the fermentation rate is increasing relative to respiration rate
(*d*) the maintenance rate is decreasing

3. The product(s) resulting from the hydrolysis of maltose is/are
(*a*) a mixture of α-D-Glucose and β-D-Glucose
(*b*) a mixture of D-Glucose and L-Glucose
(*c*) α-D-Glucose only
(*d*) β-D-Glucose only

4. The reactions leading to the formation of amino acids from the TCA cycle intermediates are
(*a*) carboxylation
(*b*) isomerization
(*c*) transamination
(*d*) decarboxylation

5. Match the following photoreceptors with their prosthetic groups and spectral specificity

Photoreceptor	Moiety that absorbs light	Absorption (nm)
P. Phototropin	1. Chromobilin	A. 400-500
Q. Cryptochrome	2. FAD	B. 600-800
R. Phytochrome	3. FMN	C. 500-600

	P	Q	R
(*a*)	3-A	1-A	1-B
(*b*)	1-B,	1-A	3-B
(*c*)	3-A	1-A	1-C
(*d*)	2-C	1-C	1-A

2013

6. Match the entries in **Group I** with the enzymes in **Group II.**

Group I	Group II
P. NAD^+	1. Glutathione peroxidase
Q. Selenium	2. Nitrogenase
R. Pyridoxal	3. Lactate dehydrogenase phosphate
S. Molybdenum	4. Glycogen phosphorylase

(*a*) P-3, Q-2, R-4, S-1
(*b*) P-4, Q-1, R-3, S-2
(*c*) P-3, Q-1, R-4, S-2
(*d*) P-3, Q-4, R-2, S-1

Common Data Questions

Common Data for Questions 7 and 8 :

A solution was prepared by dissolving 100 mg of protein X in 100 ml of water.

Molecular weight of protein X is 15,000 Da; Avogadro's number = 6.022×1023.

7. Calculate the molarity (μM) of the resulting solution.
(*a*) 66.6
(*b*) 6.6
(*c*) 0.67
(*d*) 0.067

8. The number of moleculespresent in this solution is
(*a*) 40.15×10^{19}
(*b*) 6.023×10^{19}
(*c*) 4.015×10^{19}
(*d*) 0.08×10^{19}

2012

9. During photorespiration under low CO_2 and high O_2 levels, O_2 reacts with ribulose 1, 5 - bisphosphate to yield
(*a*) one molecule each of 3-phosp-hoglycerate and 2-phosphoglycolate
(*b*) two molecules of 3-phosphoglycerate
(*c*) two molecules of 2-phosphoglycolate
(*d*) one molecule each of 3-phosp- hoglycerate and glyoxylate

10. Match the entries in Group I with the process paranieters in Group II.

Group I	Group II
P. Clark electrode	1. Liquid level
Q. Redox probe	2. Dissolved oxygen concentration
R. Load cell	3. Vessel pressure
S. Diaphragm gauge	4. pH (anaerobic process)

Codes :

	P	Q	R	S
(*a*)	2	1	3	4
(*b*)	4	2	3	1
(*c*)	2	4	1	3
(*d*)	2	1	4	3

11. Match the high energy compounds in Group I with the biosynthetic pathways for the molecules in Group II.

Group I	Group II
P. GTP	1. Fatty acid
Q. UTP	2. Phospholipid
R. CTP	3. Protein
S. Acyl coenzyme A	4. Peptidoglycan

Codes :

	P	Q	R	S
(*a*)	3	2	4	1
(*b*)	2	4	3	1
(*c*)	4	3	1	2
(*d*)	3	4	2	1

12. Match the vitamins in Group I with the processes/ reactions in Group II.

Group I	Group II
P. Pantothenic acid	1. Electron transport
Q. Vitamin B2	2. Transfer of 1-C units
R. Vitamin B6	3. Decarboxylation
S. Folic acid	4. Fatty acid metabolism
	5. Hydrolysis

Codes :

	P	Q	R	S
(*a*)	5	2	4	1
(*b*)	4	1	3	2
(*c*)	4	2	1	3
(*d*)	2	1	3	5

COMMON DATA QUESTIONS

Directions (Q. 13 – 14) : *In a muscle, the extracellular and intracellular concen-trations of Na^+ are 150 mM and 12 mM and those of K^+ are 2.7 mM and 140 mM respectively. Assume that the temperature is 25 °C and that the membnine potential is – 60 mV, with the interior more negatively charged than the exterior.*

($R = 8.314$ J mol^{-1} K^{-1}; $F = 96.45$ kJ mol^{-1} V^{-1})

13. The free energy change for the transport of three Na^+ out of the cell is
(*a*) + 1.5 kJ/mol (*b*) + 17.4 kJ/mol
(*c*) + 18.9 kJ/mol (*d*) + 36.3 kJ/mol

14. The free energy change for the transport of two K^+ into the cell is
(*a*) +8.0 kJ/mol (*b*) +11.6 kJ/mol
(*c*) +19.6 kJ/mol (*d*) +31.2 kJ/mol

2011

15. An alternative to glycolysis pathway is
(*a*) glyoxylate pathway
(*b*) pentose phosphate pathway
(*c*) citric acid cycle
(*d*) gluconeogenesis

16. In N-linked glycosylation, the oligosaccharide chain is attached to protein by
(*a*) asparagine
(*b*) arginine
(*c*) serine
(*d*) threonine

17. Match the products in **Group I** with their respective organisms in **Group II.**

Group I
P. Glycerol
Q. Glutamic acid
R. Curdlan
S. Amphotericin B

Group II
1. *Corynebacterium glutamicum*
2. *Alcaligenes faecalis*
3. *Dunaliella salina*
4. *Streptomyces nodosus*

	P	Q	R	S
(a)	2	1	3	4
(b)	4	2	1	3
(c)	3	1	2	4
(d)	2	1	4	3

18. Match items in **Group I** with **Group II.**

Group I

P. Glycolytic pathway

Q. Eukaryotic oxidative metabolism

R. Glyoxylate cycle

S. Calvin cycle

Group II

1. Chloroplast
2. Glyoxysomes
3. Mitochondria
4. Cytosol

	P	Q	R	S
(a)	1	2	3	4
(b)	2	3	4	1
(c)	4	3	2	1
(d)	3	4	1	2

2010

19. During lactic acid fermentation, net yield of ATP and NADH per mole of glucose is

(a) 2 ATP and 2 NADH

(b) 2 ATP and 0 NADH

(c) 4 ATP and 2 NADH

(d) 4 ATP and 0 NADH

20. Oxidation reduction reactions with positive standard redox potential (ΔE^0) have

(a) positive ΔG^0 (b) negative ΔG^0

(c) positive $\Delta E'$ (d) negative $\Delta E'$

Linked Answer Questions

Statement for Linked Answer Questions (21 and 22):

The standard redox potential values for two half-reactions are given below. The value for Faraday's constant is 96.48 kJ V^{-1} mol^{-1} and Gas constant R is 8.31 JK^{-1} mol^{-1}.

$NAD^+ + H^+ + 2e^- \leftrightarrow NADH$ -0.315 V

$FAD + 2H^+ + 2e^- \leftrightarrow FADH_2$ -0.219 V

21. The ΔG for the oxidation of NADH by FAD is

(a) -9.25 kJ mol^{-1} (b) -103.04 kJ mol^{-1}

(c) $+51.52$ kJ mol^{-1} (d) -18.5 kJ mol^{-1}

22. The value of $\Delta G'$, given K_{eq} as 1.7, at 23°C will be

(a) -17.19 kJ mol^{-1} (b) -19.8 kJ mol^{-1}

(c) $+52.82$ kJ mol^{-1} (d) -117.07 kJ mol^{-1}

2007

23. A protein binds to phosphocellulose column at pH 7.0 and elutes at pH 8.0. If the protein has to be further purified on a DEAE Sephacel column, the binding buffer should have a pH of

(a) 5 (b) 6

(c) 7 (d) 8

24. Match items in group 1 with correct examples from those in group 2

Group 1	Group 2
P. Catabolic product	1. Griseofulvin
Q. Bioconversion	2. Bakers yeast
R. Biosynthetic product	3. 6-Aminopenicillanic acid
S. Cell mass	4. Ethanol

(a) P-4, Q-3, R-2, S-1

(b) P-3, Q-4, R-1, S-2

(c) P-4, Q-3, R-1, S-2

(d) P-1, Q-4, R-3, S-2

25. The respiratory coefficient for the reaction

$aCH_mO_n + bO_2 + cNH_3 \rightarrow dCH_aO_\beta N_\gamma + eH_2O + fCO_2$

is defined as

(a) f/a (b) e/b

(c) b/f (d) f/b

NUMERICAL TYPE QUESTIONS

2016

1. The equilibrium potential of a biological membrane for Na^+ is 55 mV at 37°C. Concentration of Na^+ inside the cell is 20 mM. Assuming the membrane is permeable to Na^+ only, the Na^+ concentration outside the membrane will be _____ mM.

(Faraday constant: 23062 $cal.V^{-1}.mol^{-1}$, Gas constant: 1.98 $cal.mol^{-1}.K^{-1}$)

2015

2. The standard free energy change ($\Delta G'^{\circ}$) for ATP hydrolysis is -30 $kJ.mole^{-1}$. The *in vivo* concentrations of ATP, ADP and P_i in *E. coli* are 7.90, 1.04 and 7.90 *m*M, respectively. When *E. coli* cells are cultured at 37°C, the free energy change (ΔG) for ATP hydrolysis *in vivo* is _____ kJ. $mole^{-1}$.

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (d)	2. (c)	3. (a)	4. (a)	5. (c)	6. (b)	7. (b)	8. (b)	9. (b)	10. (c)
11. (c)	12. (b)	13. (a)	14. (b)	15. (c)	16. (a)	17. (c)	18. (b)	19. (c)	20. (b)
21. (c)	22. (b)	23. (b)	24. (b)	25. (d)	26. (a)	27. (b)	28. (d)	29. (b)	30. (b)
31. (c)	32. (c)	33. (b)	34. (a)	35. (c)	36. (b)	37. (d)	38. (c)	39. (b)	40. (b)
41. (b)	42. (b)	43. (a)	44. (c)	45. (d)	46. (c)	47. (b)	48. (c)	49. (a)	50. (c)
51. (c)	52. (c)	53. (c)	54. (d)	55. (c)	56. (a)	57. (a)	58. (c)	59. (c)	60. (c)
61. (a,b, d)	62. (d)	63. (c)	64. (a)	65. (b)	66. (b)	67. (b)	68. (d)	69. (b)	70. (b)
71. (d)	72. (c)	73. (b)	74. (d)	75. (a)	76. (a)	77. (b)	78. (c)	79. (c)	80. (a)
81. (a)	82. (d)	83. (c)	84. (a)	85. (b)	86. (a)				

Numerical Type Questions

1. 2	2. 8 ATP molecules	3. 2	4. 6,86,000 cal	5. 0.75
6. Infinity	7. 6 times	8. 2	9. 2	10. 2
11. 0	12. 30 - 38			

EXERCISE – II

MCQ Type questions

1. (c)	2. (c)	3. (a)	4. (c)	5. (a)	6. (c)	7. (a)	8. (Marks to All)		9. (a)
10. (c)	11. (d)	12. (b)	13. (d)	14. (d)	15. (b)	16. (a)	17. (c)	18. (c)	19. (b)
20. (b)	21. (d)	22. (a)	23. (c)	24. (c)	25. (d)				

Numerical Type Questions

1. (147.0 : 170.0) 2. (–48 to – 46)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. The increase of rate of glycolysis in absence of O_2 is due to the non-availability of those enzymes which are easily available in presence of O_2.
2. $C_6H_{12}O_6 \rightarrow 2\,C_2H_5OH + 2\,CO_2 + 21$ Kcal
3. Cyanide is deadly poison which inhibit the activity of cytochrome oxidase enzyme of electron transport chain. As in case of aerobic respiration there is no electron transport chain involved, therefore, anaerobic respiration is cyanide resistant pathway.
4. In the absence of O_2, fermentation or anaerobic respiration occurs.
5. Pyruvic acid → (Decarboxylase, CO_2) Acetaldehyde → (Dehydrogenase, + 2 $NADH_2$) ethyl alcohol + 2 NAD
6. During anaerobic respiration in yeast glucose formed CO_2, C_2H_5OH and energy.
7. In anaerobic respiration only glycolysis is found, in which there is no need of mitochondria so that the above mechanism happens in cytoplasm.
8. Decarboxylation is the removal of carbon from a compound by using carbon to make carbon dioxide. During Kreb's cycle the acetyl group pass round the cycle and the two carbon atoms are lost in the forms of carbon dioxide in two decarboxylation reactions.
9. In the first step, glucose is phosphorylated to glucose 6-phosphate.
10. Malate is dehydrogenated by malic dehydrogenase to form oxaloacetic acid.
11. Amino acid like aspartate and glutamate are synthesised on deamination from organic acids like OAA and α- ketoglutaric acid respectively which enter into Kreb's cycle.
12. In anaerobic conditions, pyruvic acid undergoes anaerobic respiration/ fermentation. Whereas, under aerobic conditions the pyruvic acid enters into mitochondria and is converted to acetyl-coA. Acetyl-coA enters the Kreb's cycle along with oxaloacetic acid. So, it acts as a connecting link between glycolysis and Kreb's cycle.
13. $NAD + 2e^- + H^+ \rightarrow NADH$. So, it is effectively accepts a hydrogen ion and gets converted into NADH.

14. ATP synthesis is an endothermic and oxidative process and thus it occurs on the on the inner membrane of the mitochondria. The cristae/ mesosome are also involved in the synthesis in eukaryotes/prokaryotes.
15. Largest amount of phosphate bond energy is formed in aerobic respiration i.e., Kreb's cycle. In total 38 ATP are produced, out of which 8 ATP are formed during glycolysis and 30 ATP in Kreb's cyle.
16. Dehydrogenation converts succinate into 4-carbon fumerate with the aid of an enzyme, succinate dehydrogenase and liberates a pair of hydrogen atoms. The latter then passes to FAD forming $FADH_2$.
17. The three steps in which CO_2 is produced during aerobic respiration are:
    1. During acetyl CoA formation.
    2. During formation of a α-ketoglutaric acid in Kreb's cycle.
    3. During succinic acid formation in Kreb's cycle
18. Reductive amination is the method of amino acid synthesis in plants. In this method, α-ketoglutaric acid reacts with ammonia (NH_3) and forms the amino acid called glutamic acid. The enzyme glutamate dehydrogenase enhances the reaction.
19. Cytochrome a_3 it is terminal cytochrome of ETC. It has Fe^{+++} and Cu^{++} both. With Fe^{+++}, it picks up electrons and through Cu^{+} it passes electrons to oxygen.
20. These are arranged in order of their increasing redox potential and electron flow through the chain in step wise manner from the more electronegative compound to the more electropositive O_2. On the basis of redox potential cytochrome acceptor order is cyt b c a a_3.
21. Fe-S complex is responsible for passing electrons (also protons) from mitochondria NADH/NADPH to ubiquinone.
22. The oxygen is the terminal acceptor of electrons which forms a water molecule in presence of protons in ETS.

    $\frac{1}{2} O_2 + 2e^- \longrightarrow O^{2-}$

    $2H^+ + O^{2-} \longrightarrow H_2O$
23. During the sixth step succinate is oxidized into fumarate by the enzyme succinate dehydrogenase and two hydrogens are transferred to FAD to form $FADH_2$.
24. HMP is an alternative method of aerobic respiration and occurs in cytoplasm.
25. In the above green organs the value of RQ is always one because in them for respiration hexose molecule is needed, in which CO_2 and O_2 molecules are equal.
26. The value of RQ always depends upon chemical nature of the substrate.
27. Maize contains carbohydrates and groundnut seed contains fatty acid. RQ of these substances are respectively less than 1. This total RQ will be less than one.
28. At compensation point, volume of O_2 absorbed is zero and volume of CO_2 evolved is zero.

$$\text{R.Q.} = \frac{\text{volume of carbon dioxide evolved}}{\text{volume of oxygen absorbed}}$$

$$= \frac{0}{0} = 0$$

29. Respiratory quotient depends upon the type of substrate participating in the respiration.
30. Palmitic acid (one mole) on complete oxidation will produce 129 ATP molecules. Complete oxidation of one mole of acetyl CoA in TCA results in production of 12 ATP molecules CO_2 and H_2O.
31. At low temperature in green apple, the galactose sugar consumption reduces in respiration due to which they remain as such fresh for long time.
32. Reaction of respiration is

    $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + 674$ kcal

    In this reaction considering molecular weight of each element, we get

    $C_6H_{12}O_6$ = 6 times 12 + 12 times 1 + 6 times 16

    = 180 gm

    O_2 = 6 times 32 = 192gm

    Now we consider products of this reaction i.e. $6CO_2$ AND $6H_2O$

    $6CO_2 = 6(12 + 32) = 264$gm

    $6H_2O = 6(2+16) = 108$ gm
33. Nitrobacter is a N2 fixing organism,it fixes atmospheric free N2 into soluble salts like nitrites and nitrates. The fixed N2 is absorbed by plants when herbivores feed on these plants, the N2 flows on the carnivores through food chain hence it is important for the N2 cycle.
34. Major source of liberation photosynthesis and about 90% of total photosynthesis is done by Phytoplankton.
35. The decomposers are heterotrophic organisms that break up the dead bodies of plant and there waste products into smaller bits or molecules. The reducers release molecules to the environment as chemical to be used again by the producers.

36. 78.03% of the atmosphere consist of nitrogen some is being added to it through volcanic eruption erosion of sedimentary rocks and denitrification.
37. Recycling of material is carried out by decomposers .If decomposers of an ecosystem vanish recycling of material are stopped.
38. Because atmosphere or gaseous cycle is absent.
39. Usable energy is ATP in which phosphorous is present.
40. Phosphorous cycle - purely sedimentary cycle.
41. Photolysis of water, release of O_2 and formation of NADPH takes place during non-cyclic phosphorylation, while ATP is synthesized in both the cyclic as well as non-cyclic phosphorylations.in cyclic phosphorylations, while passing between ferredoxin and plastiquinone and or over the cytochrome complex, the electron loses sufficient energy to form ATP and ADP and inorganic phosphate .In non-cyclic phosphorylation, while passing over cyctochrome complex, the electrons loses sufficient energy for the synthesis of ATP.
42. The carbon dioxide in C_3 plants combines with ribulose 1,5-diphosphate (RuDP) (now called RuBP-Ribulose 1,5-bisphophate) and gives rise to a 3-C compound, phophoglyceric acid. It is first stable product of C_3 cycle. The process can be described as Calvin cycle.
43. In C_4 plants (sugarcane, maize) CO_2 combines with a 3-carbon compound phosphenol pyruvate (PEP) to form a 4-carbon compound oxaloaxetic acid. This reaction is catalyzed by PEP carboxylase and takes place in mesophyll cells.
44. In the course of studies on plant productivity it was discovered that a considerable proportion of the photosynthetic products was being oxidized to CO_2 and water without the production of metabolically useful energy. This was traced to a type of respiration which occurs in small organelles called peroxisomes, and because it occurs only in light, is called photorespiration. Under conditions of low CO_2 and high O_2 concentration which are normal atmospheric conditions, as much as 30-50% of the photosynthetically fixed carbon may be lost as carbon dioxide by photorespiration and the energy that was expended to fix it is wasted to the plant. So, it is called oxygenation.
45. Chlorophyll structure was first studied by Wilstatter Stoll and Fischer in 1912.It has a tadpole like configuration with a head called poryphrin and tail made up of long chain alcohol called phytol. Poryphrin head is made up of 4 pyrolle rings. Carbon atoms 1, 3, 5 and 8 have methyl groups (CH_3).
46. Blue and red regions of light spectrum are most effective in photosynthesis. Blue wavelength of light carries more energy while red wavelength of lesser energy. Therefore the most efficient wavelength's of light are those of red light.
47. The process of photosynthesis involves two distinct steps. They are light reaction or photochemical phase and dark reaction or biosynthetic phase. In photochemical phase (occurs in thylakoids of chloroplast) the light energy is absorbed by chlorophyll and carotenoid pigments and is then converted into chemical energy which is stored in the energy rich ATP molecules. This is the first step of photosynthesis.
48. In Calvin cycle assimilation of 1 CO_2 molecule require energy in the form of 3 ATP and 2 $NADPH_2$ molecules.
49. In maize C_4 cycle takes place so it is called C_4 plant. Kranz anatomy is present in C_4 plants. PGA is related to Calvin cycle and OAA to hatch and slack cycles as both are the first stable products in these corresponding cycles.
50. Emerson and Lewis while working on chlorella and calculating the quantum yield for different wavelengths, observe a sharp decline in photosynthetic rate at wavelength higher than 680 nm. This is called red drop. As the pigment system II is driven by light and have component P_{680}, it remains inactive during red drop.
51. A considerable amount of photosynthetic products get oxidized to CO_2 and H_2O without the production of useful energy in C_3 plants. This respiration occurs in chloroplast peroxisomes and mitochondria and because it occurs in light as known as photorespiration.
52. In most of the plants stomata open during daytime close in night. But in CAM plants stomata open during night and are closed in daytime light.
53. In C_3 cycle, for formation of one mole of hexose sugar, 18 ATP and 12 NADPH are used. In C_4 plants, for formation of one mole of hexose , 30ATP and 12NADPH are required. Thus, an additional 12 ATP s are required in C_4 cycle compared to C_3 cycle.
54. In C_4 plants there are two carboxylation reactions, first in mesophyll chloroplast and second in bundle sheath chloroplast. CO_2 acceptor molecule in C_4 plants is phosphoenol pyruvate (PEP) and PEP carboxylase (PEPcase) is the key enzyme. RuBP is present in bundle sheath chloroplast where C_3 cycle takes place.

55. Photorespiration is an alternate pathway for rubsico, the main enzyme of photosynthesis (specifically, the Calvin cycle). It is the energetically-wasteful substitution of oxygen for carbon dioxide in the dark reactions of photosynthesis, which occurs when plant stomata close and carbon dioxide concentration declines. It is light dependent respiration. It produces glycolic acid in chloroplasts in the light. The glycolic acid may be oxidized in presence of enzymes of peroxisomes.

56. High energy electrons released from P_{700} reaction center of pigment system I are accepted by the substance iron-sulphur-protein complex, designated as A (Fe-S), which acts as an electron acceptor in photoact-1. A (Fe-S) is thus reduced. From reduced A (Fe-S) these electrons are accepted by oxidized ferredoxin (Fd). Ferredoxin is another low molecular weight non-haeme iron protein containing labile sulphur.

57. The electrons released during photolysis of water are picked up by P_{680} photocentre of photosystem II. The electron extruded by the photocentre of PS II picked up by the quencher phaeophytin. From here the electron passes over a series of carriers a downhill journey losing its energy at every step. The major carriers are cytochrome-f complex and plastoquinone. Electron is first accepted by plastoquinone PQ which further transfer it to cytochrome complex. While passing over cytochrome complex, the electron less sufficient energy for the creation of proton gradient and synthesis of ATP from ADP and inorganic phosphate. The process is called photo-phosphorylation (non-cyclic).

58. Photophosphorylation is the light driven or light energized synthesis of ATP. It was discovered by Amon et al in 1954. It is of two- main types- cyclic and non cyclic. In cyclic photophosphorylation PSI takes part and only ATP be formed. In non- cyclic PSI and PSI1 both take part in which ATP and $NADPH_2$ formed.

59. Like other alkali metals, potassium hydroxide combines with CO_2 as such:-

$$2\,KOH + CO_2 \rightarrow K_2CO_3 + H_2$$

60. Ruben and Kamen (1941) and Ruben et al (1941) suspended Chlorella in water having nonradioactive isotope of oxygen, ^{18}O, instead of natural oxygen ^{16}O. The suspension was illuminated. Oxygen evolved was tested by means of mass spectrophotometer. It was found to be heavy isotope, ^{18}O. This is possible only if oxygen evolved during photosynthesis comes from splitting of water. But oxygen in glucose comes from CO_2 as shown in analyzed equation

$12\,H_2O + 6\,H_2O \rightarrow C_6H_{12}O_6 + 6\,O2 + 6\,H_2O$

(In presence of Light and chlorophyll)

The equation is analyzed as follows

$6\,CO_2 \longrightarrow 6\,C + 6\,O + 6\,O$

$12\,H_2O \longrightarrow 12\,H + 12\,H + 12\,O$

61. Plants that utilize primarily RuBP to fix CO_2, which results in the formation of the three - carbon compound (3-PGA), are called C_3 plants. Hatch and Slack proposed a new pathway of CO_2 fixation via the carboxylation of Phosphoenol Pyruvate (PEP). Because the products are four carbon compounds (oxaloacetic, malic and aspartic acids), plants exhibiting this pathway are referred as C_4 plants. Ribulose diphosphate is the first acceptor of CO_2 in C_3 plants and phosphoenol pyruvate is the first acceptor of CO2 in C4 plants. In C_3 plants, fixation of one molecule of CO_2 uses 3 ATP and 2 NADPH, whereas in C_4 plants fixation of one molecule of CO_2 uses 5 ATP and 2 NADPH.

62. Unlike in the case of plant photosynthesis, cyclic phosphorylation takes place in bacterial photosynthesis, with ATP being generated in each cycle. Light driven electron transfer is cyclic in *Rhodobacter sphaeroides* and other purple bacteria. It involves PS I only.

63. RuDP carboxylase constitutes 16% of chloroplast protein.

64. Cytochrornes- Hence proteins (iron containing proteins) serving as electron carriers in respiration photosynthesis and other oxidative reduction reactions.

65. Reduction of carbon, anabolised organic compound will give ATP during respiration.

66. Lichen's can photosynthesize at very low temperature i.e., - 24°C.

67. Purple sulphur bacteria and non-sulphur bacteria contain bacterio-chlorophyll, which is having 2H-atoms more than chlorophyl a and it absorbs 850-950 nm (infra-red) wavelength of light.

68. 3 molecules of CO2 combine with 3 molecules of RuBPto produce 3 molecules of an unstable 6-C compound which immediately breaks down into molecules of 3-phosphoglyceric acid.

69. C4 are found mostly in monocots only

70. Coenzyme used for conversion of Triacylglycerols to dihydroxyacetone is NAD+

71. The large negative delta G values is shown by Hexokinase, pyruvate kinase, phosphofructokinase

72. In animal tissues experiencing anaerobic conditions, pyruvate is reduced to lactate

73. The stable form of pyruvate is keto

74. The prevalent form of phosphoglycerate mutase is phosphoenzyme with a phosphoryl group attached to Histidine

75. In the phosphoglycerate mutase reaction the phosphoryl group is moved from C3 – C2

76. Fructose bisphosphate aldolase cleaves fructose, 1-6 phosphate between C3 and C4

77. Glycolysis and TCA cycle is coupled by phosphofructokinase

78. The deltaG0′ is 1.67kJ/mol and delta G under cellular condition is –2.92 kJ/mol, this indicates Reversible, operates near equilibrium of cell

79. During the isomerisation of glucose – 6 – phosphate, carbonyl oxygen moves from C1 – C2

80. Glucokinase is an inducible enzyme

81. True substrate for hexokinase is MgATP-2

82. In the metabolic process, energy is stored in the form of ATP, FADH2, and NADH

83. TCA cycle reactions takes place in mitochondria

84. Pyruvate dehydrogenase complex is composed of thiamine pyrophosphate, coenzyme A, lipoic acid, NAD+, FAD

85. Acetyl CO-A reacts with oxaloacetate in Perkin condensation

86. Acyl group is activated to acyl co-A by carbonyl carbon attack by nucleophiles

NUMERICAL TYPE QUESTIONS

1. Two molecules of ATP are consumed in glycolyis during phosphorylation of glucose to fructose 1, 6- diphosphate.

2. The net reaction of glycolysis would be:

$C_6H_{12}O_6 + 2\,ADP + 2Pi + 2NAD + (glucose) \rightarrow 2\,C_3H_4O_3 + 2\,NADH\,(+H^+) + 2\,ATP + 2\,H_2O$

$Glucose + 8\,ADP + 8Pi \rightarrow pyruvate + 8ATP + 2H_2O$

3. One glucose molecule undergoes glycolysis and forms 3 phosphoglyceraldehyde and 1, 3-bisphosphoglycerate, each of which forms 1 pyruvate molecule. Thus in total 2 molecules of pyruvate are formed from 1 glucose molecule.

4. 1 molecule of glucose yields 56 kcal/ 2 ATP in anaerobic respiration and 686000 calories/total of 38 ATP in aerobic respiration.

5. $\dfrac{\text{volume of carbon dioxide}}{\text{volume of oxygen}} = \dfrac{V2cc}{V1cc+V2cc}$

$R.Q. = \dfrac{30cc}{10cc + 30cc} = 0.75$

6. During anaerobic respiration, due to the absence of O_2 the value of RQ is infinite (or more than one).

$C_6H_{12}O_6 \longrightarrow 2CO_2 + 2C_2H_5OH$

$R.Q. = \dfrac{2}{0} = INFINITE$

7. As glucose is a 6 carbon compound, and each Calvin cycle contributes one carbon atom so 6 turns of Calvin cycle are required to synthesize 1 molecule of glucose.

8. Number of ATP molecules produced at pyruvate stage is 2

9. Number of ATP molecules produced per molecule of glucose is 2

10. In the first five series of reaction number of glyceraldehyde – 3 – phosphate molecules produced by glucose is 2

11. In enolase catalyzed reaction the deltaG value reaches 0

12. Number of ATP molecules produced per glucose oxidation in aerobic pathway is 30 – 38

EXERCISE – II

MCQ TYPE QUESTIONS

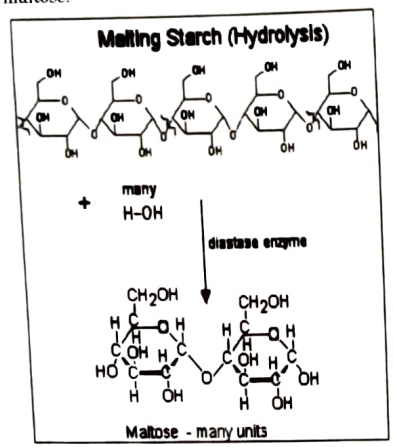
1.

Group I	Group II
P. Cyanide	4. Cytochrome oxidase
Q. Antimycin A	2. Electron transfer from cytochrome *b* to cytochrome c_1
R. Valinomycin	1. K^+ ionophore
S. Aurovertin	3. F_1 subunit of ATP synthase

2. High density culture is dependent upon high levels of available dissolved oxygen at times of maximum growth (using cascades of process parameter such as stirrer speed, gas flow and oxygen supplementation) or control of feed to match the quantity of substrate which is already present in the bulk culture. In a microbial culture, microbial metabolism is controlled on the basis of respiratory quotient (R.Q.). If, however, there occurs a sudden increase in the R.Q. of the microbial culture, it implies that rate of fermentation is increasing more than that of the rate of respiration.

3. Maltose or malt sugar is the least common disaccharide in nature. It is present in germinating grain, in a small proportion in corn syrup, and forms on the partial hydrolysis of starch. It is a reducing sugar. In the hydrolysis of

any di- or polysaccharide, a water molecule helps to break the acetal bond. The acetal bond is broken; the H from the water is added to the oxygen glucose. The -OH is then added to the carbon on the other glucose. In this case the diastase enzyme is very specific and cuts the starch chain in two units of glucose which is maltose.



4. Transamination refers to the transfer of an amine group from one molecule to another.This reaction is catalyzed by a family of enzymes called transaminases. Actually, the transamination reaction results in the exchange of an amine group on one acid with a ketone group on another acid. It is analogous to a double replacement reaction. The most usual and major keto acid involved with transamination reactions is alpha-ketoglutaric acid, an intermediate in the citric acid cycle. A specific example is the transamination of alanine to make pyruvic acid and glutamic acid.

5. Phototropin have flavin mononucleotide (FAD) molecules present in them which absorb light in the range of 400 to 500 nm of wavelength. Cryptochrome are comprised of flavin dinucleotide moieties which give light absorbance at the same range of 400-500 nm because of the same flavin molecule. Phytochrome has chromobilin present in it which is excited and shows absorbance within the range of 600-800 nm of wavelength.

6. Lactate dehydrogenase (NAD+ oxidoreductase) is a ubiquitous enzyme among vertebrates and carries out the reaction: NAD+ + lactate <=> NADH + Pyruvate. Glycogen Phosphorylase catalyzes phosphorolytic cleavage of α (1, 4) glycosidic linkages of glycogen, releasing glucose-1-phosphate as the reaction product. Pyridoxal phosphate (PLP), a derivative of vitamin B_6, serves as prosthetic group for Glycogen Phosphorylase. Nitrogenase is a two-protein complex. One component, called Nitrogenase Reductase (NR) is an iron-containing protein that accepts electrons from ferredoxin, a strong reductant, and then delivers them to the other component, called Nitrogenase, or Iron-Molybdenum protein. There are several proteins in mammalian cells that can metabolize hydrogen peroxide and lipid hydroperoxides. These proteins include four selenium-containing glutathione peroxidases that are found in different cell fractions and tissues of the body. These proteins are called glutathione peroxidases.

7. Given, Molecular weight of the Protein X, m = 15,000 Da = 15,000 g/mol = 15 mg/mol

   Weight of the Protein in solution, W = 100 mg

   Volume of the solution, V = 100 mL

   Therefore, molarity of the solution

   = (W/m) × (1000/V) M = (100/15) × (1000/100)

   = 66.67 M

8. Number of moles present in the solution of protein = (100/15) = 6.667 moles

   Number of molecules present in the solution

   = no. of moles × Avogadro's number

   = $6.667 \times 6.022 \times 10^{22}$

   = 40.146×10^{22} molecules

9. During photo respiration under low CO_2 and high O_2 reacts with ribulose– 1, 5 bisphosphate to yield-one molecule each of 3-phosphoglycerate and 2-phosphoglycolate.

10. The Clark electrode is an electrode that measures oxygen on a catalytic platinum surface using the net reaction:

    $O_2 + 4e^- + 2H_2O \rightarrow 4OH^-$

    These are also known as dissolved oxygen electrodes.

    Redox probe also known as a pH probe is used for the measurement of pH.

    A load cell is a transducer that is used to create an electrical signal whose magnitude is directly proportional to the force being measured. It can be used to measure the liquid levels. To measure level, the load cell must be incorporated into the vessel's support structure.

The Diaphragm Pressure Gage uses the elastic deformation of a diaphragm (i.e. membrane) instead of a liquid level to measure the difference between an unknown pressure and a reference pressure.

11. Guanosine-5′–triphosphate (GTP) is used as a source of energy for protein synthesis and gluconeogenesis. GTP is essential to signal transduction, in particular with G-proteins in second-messenger mechanisms where it is converted to guanosine diphosphate (GDP) through the action of GTPases.

    Peptidoglycan, also known as murein, is a polymer consisting of sugars and amino acids that form a mesh-like layer outside the plasma membrane of all bacteria (except Mycoplasma), forming the cell wall. Uridine triphosphate (UTP), which is a pyrimidine nucleotide, has the ability to act as an energy source for the synthesis of peptidoglycan.

    The phosphatidic acid is used to synthesize additional phospholipids utilizing cytidine triphosphate (CTP) as an energy source and creating a CDP-diacylglycerol molecule.

    Fatty-acyl-CoA Synthase, or more commonly known as yeast fatty acid synthase, is an enzyme complex responsible for fatty acid biosynthesis

12. Pantothenic acid, also called pantothenate or vitamin B_5, is a water-soluble vitamin. Pantothenic acid is used in the synthesis of coenzyme A (CoA). CoA is also important in the biosynthesis of many important compounds such as fatty acids, cholesterol, and acetylcholine.

    Riboflavin is a water-soluble B vitamin, also known as vitamin B_2. In the body, riboflavin is primarily found as an integral component of the coenzymes, flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN). FAD is part of the electron transport (respiratory) chain, which is central to energy production.

    Vitamin B_6 is a water-soluble vitamin and is part of the vitamin B complex group. Several forms of the vitamin are known, but pyridoxal phosphate (PLP) is the active form and is a cofactor in many reactions of amino acid metabolism, including transamination, deamination, and decarboxylation.

    Folic acid is a B vitamin. It helps the body make healthy new cells. Transfer of 1-C units is important for many biosynthetic reactions. To carry out the transfer of 1-carbon units, NADPH must reduce folic acid two times in the cell.

15. The pentose phosphate pathway (PPP) is a process of glucose turnover that produce NaDPH as reducing equivalents and pentose as essential part of nucleotides. There are two different phases in a pathway. One is irreversible oxidase phase in which glucose-6-phosphate is converted to ribulose-6-phosphate by oxidative decarboxylation and NaDPH is generated. The other is reversible non- oxidative phase in which phosphorylated sugars are interconverted to generate xylulose - 5p, ribulose-5p and ribose-5p. Phosphoribosyl pyrophosphate formed from ribose-5p is an activated compound used in the biosynthesis of histidine and purine/pyrimidine nucleotides. This pathway main purpose is to regenerate $NaDP^+$ through an oxidation/reduction reaction. NaDPH is used for reductive reaction in anabolism, especially in fatty acid synthesis.

16. All N-linked glycans are derived from a common 14-sugar oligosaccharide synthesized in the ER, which is attached co-translationally to a protein while this is being translated inside the reticulum. The process of the synthesis of this glycan, known as Synthesis of the N-glycan precursor or LLO, constitutes one of the most conserved pathways in eukaryotes, and has been also observed in some eubacteria. The attachment usually happens on an asparagine residue within the consensus sequence asparagine-X-threonine by an complex called oligosaccharyl transferase(OST). After being attached to an unfolded protein, the glycan is used as a label molecule in the folding process (also known as Calnexin/Calreticulin cycle).

17. Glycerol is produced by Dunaliella salina. Glutamic acid is produced by corynebacterium glutamicum. Curdlan is synthesised by Alcaligenes faecalis . AmphotericinB is apolyene antifungal drug, often used intravenously for systemic fungal infections. It was originally extracted from Streptomyces nodosus, a filamentous bacterium.

18. In Plants, Cholroplast is the host for Glycolytic pathway. Oxidative oxidation in eukaryotes takes place at the Glyoxysomes. Mitochondria is the house of glyoxalate cycle in eukaryotes while cytosol host the Calvin cycle.

19. Lactic acid fermentation is caused by some fungi and bacteria. The most important lactic acid producing bacteria is lactobacillus. It is used to convert lactose lactic acid in yogurt production. In lactic acid fermentation, NADH is oxidized into 2 molecules of NAD^+ with a net gain of 2ATP and 2lactate.

    Pyruvate + 2NADH → Lactic Acid

    + $2NAD^+$ + 2ATP.

20. Redox potential E_0 can be calculated by the ratio between Gibbs free energy and product of Faradays Number with number of participating moles. It implies that the value of ΔG° will be negative as Faraday's constant is a positive number.

21. We have, from Gibbs free energy equation:-

$$\Delta G = -nF\Delta E$$

Where,

ΔG = change in Gibbs free energy,

n = number of electrons per mole product,

F = Faraday's constant,

ΔE = change in electrode potential of the reaction.

Now,

$$\Delta G_{OXIDATION} = -nF\Delta(E_{REDUCTION} - E_{OXIDATION})$$

Given, from equation: n = 2 ($2e^-$ are exchanged)

$E_{OXIDATION}$ = –0.315 V [Since NADH is oxidized to NAD^+ by loss of $2e^-$]

$E_{REDUCTION}$ = –0.219 V [Since FAD is reduced to $FADH_2$ by accepting $2e^-$]

Therefore,

$$\Delta G_{OXIDATION} = -2 * 96.48 * [(-0.219) - (-0.315)]$$
$$= -18.524 \text{ kJ/mol}$$

Therefore ΔG for oxidation of NADH by FAD is:

–18.524 kJ/mol

22. We know, $\Delta G' = \Delta G^0 + RT \log K_{eq}$

Where, $\Delta G'$ = change in Gibbs free energy\

ΔG^0 = standard change in Gibbs free energy

R = Universal Gas constant

T = absolute temperature

K_{eq} = equilibrium constant

Given,

ΔG^0 = –18.5 kJ/mol, R= 8.31 J/K/mol

= 8.31 * 10^{-3} kJ/K/mol, K_{eq} = 1.7,

T = 23°C = (273 + 23) K = 296K

Therefore,

$\Delta G'$ = –18.5 + [8.31 * 10^{-3} *296 * log(1.7)]

= –17.19 kJ/mol

Therefore the value of $\Delta G'$ is –17.19 kJ/mol

23. If the protein is most stable at pH's above its pI, then an anion-exchange gel matrix should be used. If it is most stable below its pI, then a cation-exchange gel matrix should be used. Here the protein eluted at pH 8 so it would take pH7 to bind in DEAE column.

24. Bioconversed product is 6 Amiono penicillinic Acid, Grisseofuvin is biosynthetic product, Bakers yeast have definite cell mass ie Biomass and Ethanol is a catabolic product.

25. The respiratory quotient (RQ) is calculated from the ratio RQ = CO_2 eliminated / O_2 consumed where the term "eliminated" refers to carbon dioxide (CO_2) removed ("eliminated") from the body. So, RQ for the equation is f/b.

NUMERICAL TYPE QUESTIONS

1. $$E_{eq\ Na^+} = \frac{RT}{ZF} \ln\left[\frac{[Na^+]_0}{[Na^+]_i}\right]$$

Given $E_{eq\ Na^+}$ = 55 mV

= 55×10^{-3} V

R = 1.98 cal mol^{-1} K^{-1}

T = 273 + 37

= 310 K

F = 23062 cal V^{-1} mol^{-1}

$[Na^+]_i$ = 20 mM.

Z = 1

$$\therefore\ 55 \times 10^{-3} = \frac{1.98 \times 310}{1 \times 23062} \ln\left(\frac{[Na^+]_0}{20}\right)$$

$$\therefore\ [Na^+]_0 = 20 \times 7.897$$

= 157.94 mM

2. $ATP + H_2O \rightarrow ADP + P_i$

ΔG° = – 30 KJ/mol

T = 37°C

$$\Delta G = \Delta G^\circ + RT \ln \frac{[ADP][P_i]}{[ATP]}$$

$$= -30 + 8.314 \times 10^{-3} \times (37 + 273) \ln \frac{(1.04 \times 10^{-3} \times 7.90 \times 10^{-3})}{(7.90 \times 10^{-3})}$$

= –30 – 17.70 = – 47.70 KJ/mol

■■

3 CHAPTER

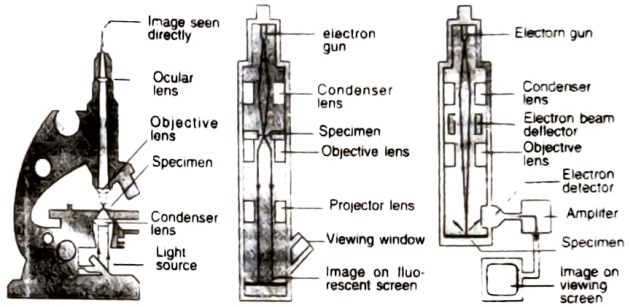
Biophysical and Biochemical Techniques

Microscopy

A microscope may be defined as optical instrument consisting of a lens, or combination of lenses for making enlarged or magnified images of minute objects.

Light Microscope or Compound Microscope

The compound microscope has two set lenses. One is known as objective and the other eye piece. These are mounted in a holder commonly known as body tube. The lens system nearest to the specimen is called objective which magnifies the specimen to a definite number of times. The second lens system is called eye piece which is the nearest to eye. It further magnifies the image formed by the objective. Accurate focusing is attained by a special screw appliance called as fine adjustment.



Types of Microscopes.

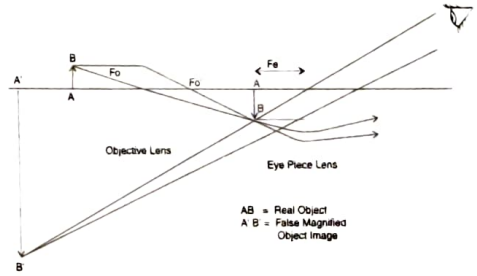


Fig. Image formation in compound microscope.

(i) Parts of Compound Microscope :

A compound microscope consists of several parts as described below:

(a) Objective Lens:

The chief characteristics of the objective lens are: (a) to gather the light rays coming any point to the object, (b) to unite the light in a point of image, (c) to magnify the image.

There are three types of objectives, namely (a) achromatic which are simplest in construction and adequate for most purpose, (b) flourite are those in which the aberrations are largely eliminated by use of these objectives, (c) apochromatic which are most corrected with respect to aberrations.